The Incidence of Diversity: Labor Market Effects of Corporate Diversity Pledges*

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Abstract

Do diversity initiatives create new positions for minorities, or move them between firms? I build a comprehensive dataset of the numerical, dated workforce diversity targets of American companies, and match them to administrative and LinkedIn data on their employees. Relative to never-pledged firms, those with a numerical race pledge raised their share of Black workers by 4.5 percent on average over the four years after the pledge. These Black hires were no more often placed in diversity or staff roles, and fell no further behind the white men hired alongside them in college rank, experience, promotion or retention. Although the pledgers largely fell short of what they promised, they diversified their workforces by tilting the flow of outside hires toward Black candidates. For every 100 positions the pledgers added for Black workers at director level and above, 42 went to Black workers who moved from similar jobs at other firms.

Keywords: Corporate Diversity Pledges, DEI, Labor Market Incidence, Executive Labor Markets, Voluntary Affirmative Action, LinkedIn
JEL Codes: J15, J23, J44, J71, M14, M51

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1 Introduction

Racial gaps in economic outcomes in the United States are large and persistent. Black Americans earn less, hold less wealth, and move up the income distribution less often than white Americans (Bayer and Charles, 2018; Derenoncourt et al., 2024; Chetty et al., 2020). Corporate America plays a role in this. For instance, Black applicants are less likely to be called back than white applicants with the same résumé (Bertrand and Mullainathan, 2004; Kline et al., 2022). In the aftermath of the murder of George Floyd in 2020 and the protests that followed, consumers, employees, and institutional investors pressed companies to disclose their demographics and set targets for minority representation (Balakrishnan et al., 2023; Bourveau et al., 2026). Many of the largest U.S. firms responded with public promises about the racial and female make-up of their workforces. The median race target implied an increase in the share of the targeted group of one percentage point per year for three to five years.

These diversity pledges carry a tension between their aim and their execution. Companies promised fast and visible change, but the aim of the pressure behind them was to give minorities better labor market opportunities in a sustainable fashion. Developing an internal promotion pipeline adds minority executives to the economy, but it requires a long horizon to become effective. A firm in a hurry could also lower its screening standards for minority candidates or create window-dressing roles. Such a lower standard can backfire, because it weakens the targeted group's incentive to invest in skills (Lundberg and Startz, 1983; Coate and Loury, 1993). A third way of accomplishing quick change is to hire minorities who already hold similar positions at other firms. However, such a hire merely moves an executive from one firm to another, potentially adding no minority executive to the economy. In the presence of employment quotas that bind only some employers and with a thin supply of minority workers, Welch (1976) predicted the sorting of these workers towards the covered employers rather than an increase in their total number.

I study the effects of these pledges to answer three questions. Did the firms that set numerical targets actually change the composition of their workforce? Did they lower the bar for the minorities they hired, and were the new positions window dressing? Finally, did the pledges create new opportunities for minorities, or reallocate existing ones between firms?

To answer these questions, I construct a comprehensive dataset of the specific pledges of every S&P 1500 firm, coded from firms' own filings, reports and press releases between

2015 and 2025. I document that 186 firms made numerical, specific commitments to diversify their workforce. To study the effects at the company level, I collect self-disclosed information on their workforce composition for each year (EEO-1 files). To measure the effects at different levels of the company and to follow workers transitioning between jobs, I match workers to companies using LinkedIn data for 23.5 million U.S. workers from 2014 to 2024. Given that LinkedIn data does not provide racial, ethnic or gender information, I predict this information from the profiles' names, locations and colleges. I validate this name-based measure against data from voter registry files that contain self-reported race, and with profile photographs.

Since the pledges mostly addressed the Black population, I compare the Black share at the pledgers with that at never-pledged S&P 1500 firms in an event study design. While pledger and non-pledger firms are not identical, two factors speak for the validity of this design. First, the Black share followed the same trend at pledgers and controls before the pledges. Second, the name-based estimates hold with sector-by-year fixed effects and with other staggered-adoption estimators. Furthermore, to study worker-level outcomes within firms, I use triple-difference designs comparing Black new hires with the white men hired by the same firm in the same year.

To determine the channel through which firms implemented their commitments, I split each pledger's yearly change in the Black share into hires from outside, promotions from below, separations and promotions out of the band. Because my data also covers workers at employers outside the S&P 1500, I can trace back the people the pledgers hired to the firms they left, and the vacancies they left behind to whoever filled them. For each Black director who moved from a director job elsewhere to a pledger after its pledge, I follow the position that she left empty to see whether it was filled within one year, and by whom.

I document the following four findings. First, after a numerical race pledge, the Black share of the pledgers' workforces rose relative to never-pledged firms using my measures of race. Using the firms' official racial make-up data, I estimate a firm-wide rise averaging 0.47 percentage points over the four years after the pledge, or 4.5 percent of the pledgers' 2019 share. This translates to an increase of about 20,000 Black employees. Using the aggregate-level data from Onorato et al. (2026) estimated based on both photographs and surnames, I estimate a firm-wide rise of 0.38 percentage points. Using my name-based measure, I estimate an increase of 0.21 percentage points. Split by seniority, my name-based measure shows increases of 0.20 to 0.30 points in every band. On the firm-years that both sources cover, my firm-wide estimate is still smaller than in the EEO-1 reports,

mostly because predicting race from names is more prone to miss some Black workers. Even the largest estimate is far below the one percentage point a year that the median target promised.

Second, firms accomplished this compositional shift mostly through external hiring rather than internal promotions. Specifically, of every 100 Black workers added at director level and above, 79 were hired from outside and 21 were promoted from lower bands. This was not accompanied by an expansion of total headcount, which, if anything, grew more slowly at pledgers than at controls.

Third, the pledges did not widen the gaps in observable characteristics between the Black hires and the white men hired alongside them. After the pledges, the gaps between Black and white-male hires into director level and above did not widen in college ranking, experience, predicted pay, retention or promotion. The gap in the rank of their previous employers narrowed, indicating that pledging firms did not dip into lower-ranked firms to recruit Black workers. The gaps not widening is consistent with two readings. Either firms had been passing over qualified Black candidates, or the supply of qualified Black candidates was thin and the pledgers met their targets partly by hiring sitting directors away from other firms. I also classify each hire's title by its function and role. I find that the new Black hires were not differentially placed more in diversity, human-resources or staff roles, and no more often in roles new to the firm.

Fourth, about three in five of the added Black directors were new to the economy. Of the 79 outside hires in every 100 added seats, 45 came straight from a director-level job at another firm. Those firms refilled the vacated seats with a Black worker only 7.4 percent of the time, no more often than they usually hire Black directors, so each such move mostly reallocated a Black director from one firm to another. Counting the replacements, the net number of Black directors in the U.S. labor market rose by about 58 for every 100 the pledgers added.

Taken together, the findings imply that looking at firm-level effects of corporate diversity initiatives overstates their aggregate effect. I find no sign that the increase came from a lower bar, the route by which a target can backfire in Coate and Loury (1993). The incidence of a target therefore depends on the margin used to meet it. A target on a firm's stock of directors credits a director hired away from another firm the same as one promoted from within. A target on promotions, and on hires from below the director level, would credit only the margins that add Black directors one for one, which in this episode produced 55 of the 58.

I contribute to three literatures. The first studies labor market discrimination and the

role that policies have in reducing it (Becker, 1957; Phelps, 1972; Arrow, 1973; Coate and Loury, 1993; Giuliano et al., 2009; Lang and Lehmann, 2012; Hsieh et al., 2019; Kline et al., 2022; Onorato et al., 2026). Work on affirmative action, reviewed by Holzer and Neumark (2000a), has mostly studied mandates and quotas, such as federal contractor obligations (Leonard, 1990; Miller, 2017; Kurtulus, 2016; Holzer and Neumark, 1999, 2000b), bans on race-conscious admissions (Hinrichs, 2012; Bleemer, 2022), caste reservations (Bertrand et al., 2010) or board quotas (Bertrand et al., 2019). The closest paper in the literature is Bertrand et al. (2019), who find that Norway's board quota raised the representation and pay of the women appointed, with little spillover to the women below them. In contrast to that, the pledges studied here are voluntary and reach below the board. I contribute to this literature by following workers across large firms over a decade and over pledging and non-pledging firms. This lets me separate growth from reallocation and outside hiring from promotion, and compare the Black hires with the white men hired alongside them.

The second literature studies the displacement and general-equilibrium effects of labor market policies. In Crépon et al. (2013), job-search assistance raised the employment of treated job seekers in France partly at the expense of untreated job seekers in the same labor markets. Related studies find spillovers on non-participants from job-search assistance in Denmark (Gautier et al., 2018), from longer unemployment insurance in Austria (Lalive et al., 2015) and from training in France (Ferracci et al., 2014), and general-equilibrium effects of a public employment program in India (Muralidharan et al., 2023). While most of those papers study supply-side interventions that change the search behavior or skills of workers, I study a demand-side shock on a subset of firms competing for a thin pool of senior talent. I contribute a transparent accounting of the partial versus general equilibrium effects of workforce diversity initiatives, following each vacated seat as the first link of a vacancy chain (White, 1970; Mercan and Schoefer, 2020).

The third literature studies corporate commitments and their consequences (Flammer, 2015; Bhagwat et al., 2020; Edmans et al., 2023; Balakrishnan et al., 2023; Gormley et al., 2023; Raghunandan and Rajgopal, 2024; Cai et al., 2024; Bolton and Kacperczyk, 2026; Chen et al., 2026; Law and Tan, 2026). Baker et al. (2024) find that firms whose diversity language runs ahead of their workforce receive better ESG ratings and more money from ESG funds, but that the language does not predict later increases in diversity. I contribute to this literature by introducing a novel dataset of the specific diversity pledges of American companies, by showing that public commitments of firms can in fact alter real corporate hiring behavior and by bringing general-equilibrium incidence into the study of corporate governance. The dataset separates dated numerical race targets from general

diversity language, and I find that, at director level and above, the Black share rose only at the firms that named a number.

The rest of the paper is structured as follows: Section 2 describes the institutional setting, Section 3 the data, and Section 4 the empirical strategies. Section 5 reports the effects on the workforce, Section 6 reports the worker-level comparisons, Section 7 discusses the aggregate-level effects and Section 8 concludes.

2 Institutional Setting

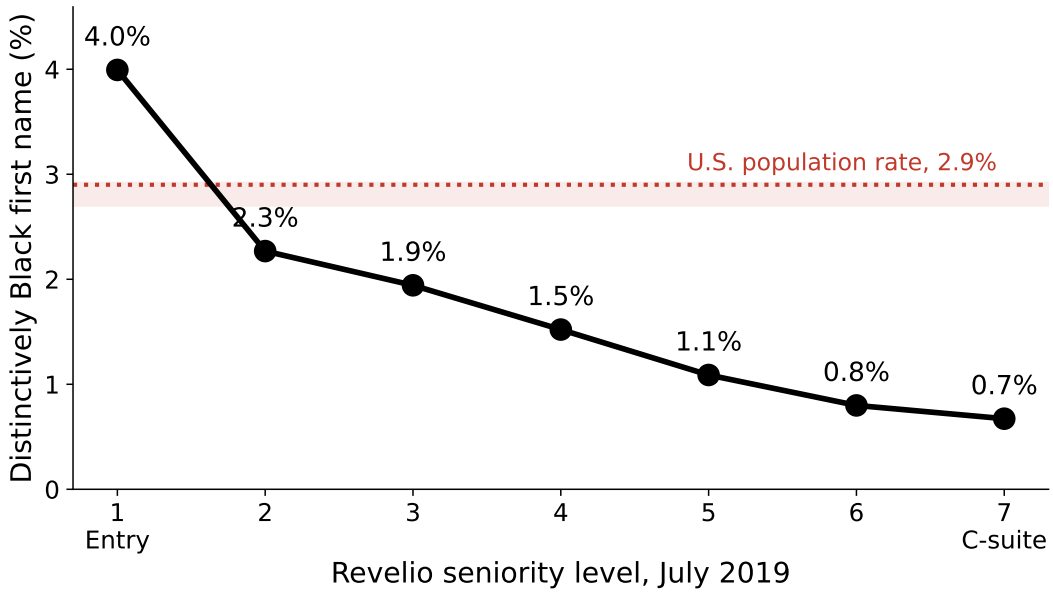
Minorities face steep barriers to advancing up the corporate ladder in the United States (Stainback and Tomaskovic-Devey, 2009; Bourveau et al., 2026). Economy-wide, Black workers in 2018 held 15 percent of all jobs at private employers with 100 or more employees, but only 7.5 percent of first- and mid-level management roles and 3.3 percent of executive positions (U.S. Equal Employment Opportunity Commission, 2018). Within the S&P 1500 firms in my sample, LinkedIn data reveals a monotonic decline across seven seniority rungs. Figure 1 plots the share of workers in July 2019 with a distinctively Black first name, which represents 2.9 percent of the U.S. population (Rosenman et al., 2023) but falls from 4.0 percent at entry level to 1.1 percent among directors and 0.7 percent in the C-suite.

Large firms have run diversity programs for decades (Kalev et al., 2006). They train managers to avoid biased hiring, add diversity criteria to managers' evaluations, and create diversity committees and staff positions responsible for diversity. In 2014, large technology firms also voluntarily began to disclose the racial and gender composition of their workforces.[1] The evidence on these programs is mixed. Diversity training, for instance, seems to change attitudes only temporarily and to have little effect on behavior at work (Kalinoski et al., 2013; Bezrukova et al., 2016; Chang et al., 2019). Adding diversity criteria to managers' evaluations seems to do little to raise the share of women and Black workers in management, while committees and staff responsible for diversity come with the broadest increases in managerial diversity (Kalev et al., 2006).

Discriminatory practices on the basis of race have been illegal in the U.S. since the Civil Rights Act of 1964, which bars race-based hiring and promotion. It only allows racial preferences in temporary voluntary plans that correct a manifest imbalance (Holzer and

[1]Google first published the gender and racial make-up of its workforce in May 2014 (Bock, 2014), and Facebook, Apple and other large technology firms released similar reports within months (Williams, 2014; Nichols, 2014).

FIGURE 1: BLACK REPRESENTATION ACROSS THE CORPORATE HIERARCHY



Notes: The figure plots the share of workers with a distinctively Black first name, by the Revelio seniority level held on July 1, 2019. A first name is distinctively Black if at least 80 percent of the people with that name are Black in the voter-file tables of Rosenman et al. (2023). The sample is the 10.4 million workers whose first name appears in the tables, 95 percent of all 11.0 million at the 1,448 S&P 1500 firms on that date (all 1,449 index firms except Old Dominion Freight Line, which is missing from the panel used for this figure). The dotted line is the U.S. population rate, 2.9 percent: the share of each racial group with a distinctively Black first name in the six-state voter files behind the name tables, weighted by the national racial mix. Sources: Rosenman et al. (2023), Revelio Labs.

Neumark, 2000a). However, there is no law that requires private employers to meet racial targets in their workforces. The closest was the mandate for federal contractors to keep an affirmative action plan. This mandate (Executive Order 11246) forced federal contractors with at least 50 employees and a $50,000 contract to keep written affirmative action plans with placement goals (Miller, 2017; U.S. Department of Labor, 2024).

The murder of George Floyd on May 25, 2020, led to pressure on corporations by consumers, employees, and institutional investors (Balakrishnan et al., 2023; Pajuste et al., 2024). On social media, the #PullUpOrShutUp campaign asked brands that had posted statements of support to publish, within 72 hours, how many Black employees they had in their corporate offices and in leadership roles, and it asked consumers to stop buying from these brands until they did (Callahan, 2020). Employees made similar demands inside their firms. For instance, a group of Amazon employees asked the company to release data on race and promotions and to set diversity targets for its management (Weise, 2020). Institutional investors then turned these demands into governance campaigns. On July 1, 2020, the New York City Comptroller wrote, on behalf of three of the city's pension

funds, to 67 S&P 100 companies that had issued statements in support of racial equality, asking them to commit to publishing their EEO-1 reports (Office of the New York City Comptroller, 2020). Over the following months, BlackRock, State Street and Vanguard asked their portfolio firms to disclose the same data.

Many of the largest publicly traded companies answered with publicly disclosed, numerical targets for the composition of their U.S. workforce. In all, 157 S&P 1500 firms set a numerical race target between 2015 and 2024. About two thirds of the firms with such targets set them in either 2020 or 2021. In contrast to the contractor plans, which were confidential, pledges were public and they were accompanied by proxy statements, sustainability reports and voluntary disclosures of workforce data.

TABLE 1: EXAMPLES OF CORPORATE NUMERICAL RACE PLEDGES

Firm	Pledge
Cisco	"achieve a 25% increase in representation of all employees who self-identify as AA/Black from entry-level through manager level and a 75% increase in representation of employees who self-identify as AA/Black from Director through VP+ level" by 2023
McDonald's	"increase representation of historically underrepresented groups in leadership roles, Senior Director and above in the U.S., from 29% to 35% by the end of 2025."
State Street	"triple our Black and Latinx leadership (senior vice presidents+) and double our percentage of Black and Latinx populations over the next three years."

Notes: The table reports the numerical race pledges of Cisco, McDonald's and State Street. The pledges are quoted from the firms' own documents: Cisco's statement "Social Justice Beliefs and Actions" (September 2020), McDonald's announcement of February 19, 2021, and State Street's statement "10 Actions to Address Racism and Inequality" (July 2020). Source: pledge dataset (Section 3.1).

Table 1 quotes the pledges of Cisco, McDonald's and State Street. To give an example, Cisco aimed to increase Black representation especially at the top of the company, with a goal of a 75% increase in representation from Director through Vice President levels by 2023. McDonald's aimed to raise the share of historically underrepresented groups in U.S. leadership roles, senior director and above, from 29 to 35 percent by the end of 2025, which represents about 1.2 percentage points a year. It also linked executive pay to meeting its diversity goals. State Street, like about one fifth of race pledgers did, mentioned both Black and Latin employees in its pledge.

Since 2023, there have been notable cases of pushback against these initiatives. Examples include the Supreme Court's decision in *Students for Fair Admissions*, which raised the legal risk of race-conscious decisions, and an executive order in January 2025 that revoked the contractor mandate to have affirmative action plans.

3 Data & Measurement

3.1 Corporate Diversity Pledges Dataset

To identify firms that made numerical diversity pledges, I construct a new dataset of corporate commitments that covers every firm in the S&P 1500 from 2015 to 2025, based on publicly available company documents. I code the treatment variable from the firms' proxy statements, annual and current reports, sustainability and diversity reports and press releases.[2]

I start by defining a dictionary of keywords related to racial diversity pledges (e.g., "African American", "people of color", "BIPOC" and "underrepresented minorities"; Appendix A.1). For each firm and with the help of an LLM, I go through 10 years of their documents looking for those keywords and extracting the sections that contain them. I record the verbatim quote, and extract the source, the group, the number, the deadline, the positions covered and the stated announcement date. Every record that could make a firm a pledger is checked by hand against its primary text. Three independent readers then re-read the firms whose status or date is hardest to establish, together with a random sample of the rest. Finally, in a blinded double-coding of 100 random pledges from their quotes alone, the codes agree with the stored group, target and deadline in 89 to 98 percent of cases (Appendix A.4).

I define a numerical race pledge as a public target that the firm itself sets for its U.S. workforce or part of it. The target must name a racial or ethnic group that includes Black or Hispanic workers, and it must set a numerical goal for that group's share or headcount, to be reached by a deadline. I classify vague language, a goal without a number or a date, membership in a coalition and a donation as general statements. To give an example, Walmart's 2021 annual report promises "a diverse, equitable and inclusive company" and "a diverse associate base" but names no group and gives no number or date. Appendix A.2 details the specific definition of numerical race pledgers, gives the specific conditions and explains how I resolve the hard cases.

Of the 386 firms in the S&P 1500 that made some public diversity commitment, 186 made a numerical pledge between 2015 and 2024, and the other 200 made only general statements or set targets outside the definition.[3] Among these numerical pledgers, 157

[2]Proxy statements are forms DEF 14A and DEFA14A, annual reports form 10-K, and current reports form 8-K with their exhibits; sustainability reports include ESG reports.

[3]The general pledgers that serve as a comparison group are the 193 firms with general statements and no numerical race target: the 186 firms that made only general statements and seven firms that made general

set targets for race or ethnicity (of those, 49 pledged exclusively that) and 137 set gender targets (of those, 29 pledged gender only). The main analysis uses 115 of the 136 race pledgers that pledged between 2018 and 2022 (Section 4.1 and Table 2).

Race pledgers came from all eleven GICS sectors, and no sector accounts for more than a fifth of them (Table A.2): consumer discretionary, industrials, financials, information technology and health care each have between 17 and 31 race pledgers. However, relative to each sector's size in the index, consumer-facing firms pledged most often. About a fifth of consumer staples and communication services firms and a sixth of consumer discretionary firms set a numerical race target, against about 2 percent of energy, utilities and real estate firms.

I code a pledger's date as the day of the first public announcement of its earliest qualifying race target, and its cohort as the year of the first July 1 on or after that date. Figure 2 depicts the number of firms making numerical commitments by the year of their first numerical race pledge, or of their first gender pledge if they set no race target. Eighteen firms pledged between 2015 and 2019. Following the pressure from the Black Lives Matter movement, 66 percent of the workforce commitments came in 2020 and 2021, and the wave tailed off through 2024.[4] The dataset also records what each firm later did with its target. By their 2025 filings, 114 of the 115 pledgers in the main sample had softened or withdrawn their targets, which I call a retreat.

To see which positions the targets cover, I map the specific wording of each pledge to the levels of the corporate ladder. For instance, vice presidents map to Revelio level 6, executive, and above. Figure 3 counts the targets that cover each level, and almost every pledger sets a target for the senior levels. Specifically, 89 percent of race pledgers and 86 percent of gender pledgers have a target that covers director-level positions. Forty-four percent of race pledgers have a race target for the whole workforce, against 25 percent of gender pledgers, and 29 percent limit their race targets to director level and above, against 47 percent for gender.[5] Seventy-two race targets of 42 pledgers state the target share, the starting share and both dates. Among these, the median asks for a rise of 1.0 percentage point a year, and a quarter ask for more than 1.5 points.[6]

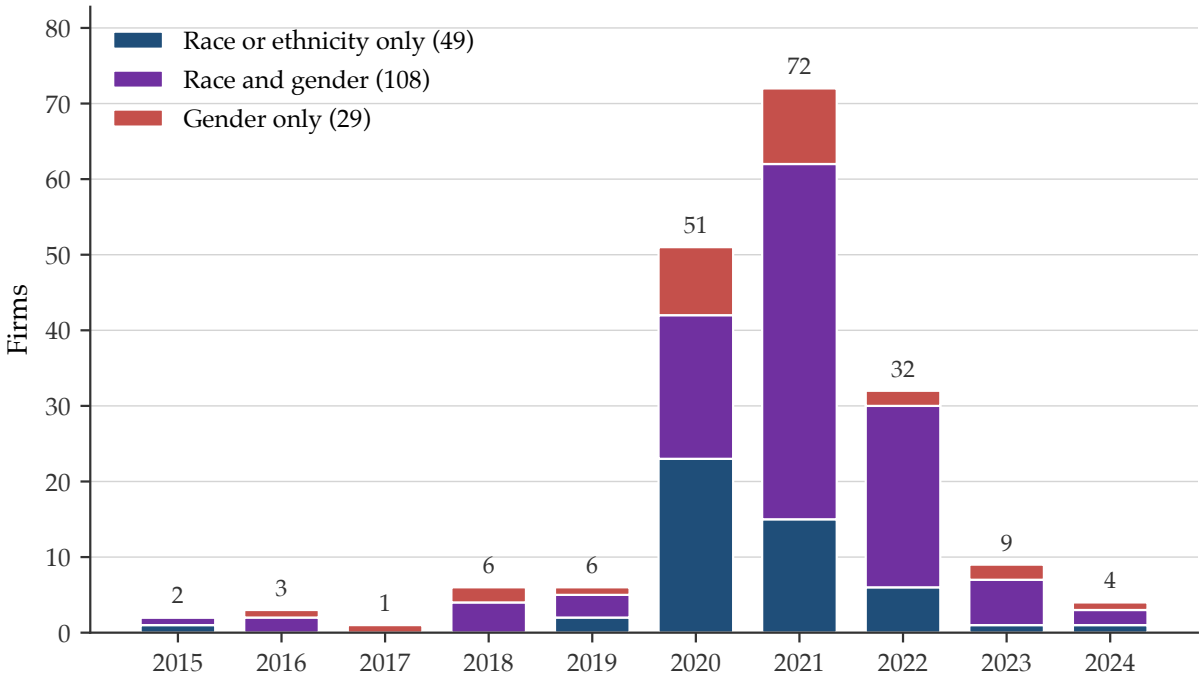
statements about race but set a numerical target only for women.

[4]Among the firms with both race and gender targets, 18 had set their gender target in an earlier year than their race target. While gender targets continued commitments that many firms had already made, race targets were the new commitment of 2020. The race goals were thus the larger shock to what firms had promised.

[5]The race targets name the United States in 82 percent of cases, while seven in ten gender targets are global.

[6]Table A.3 reports the distribution by the lowest level a target covers and by the group it names.

FIGURE 2: ADOPTION OF NUMERICAL DIVERSITY PLEDGES BY TARGET GROUP AND YEAR



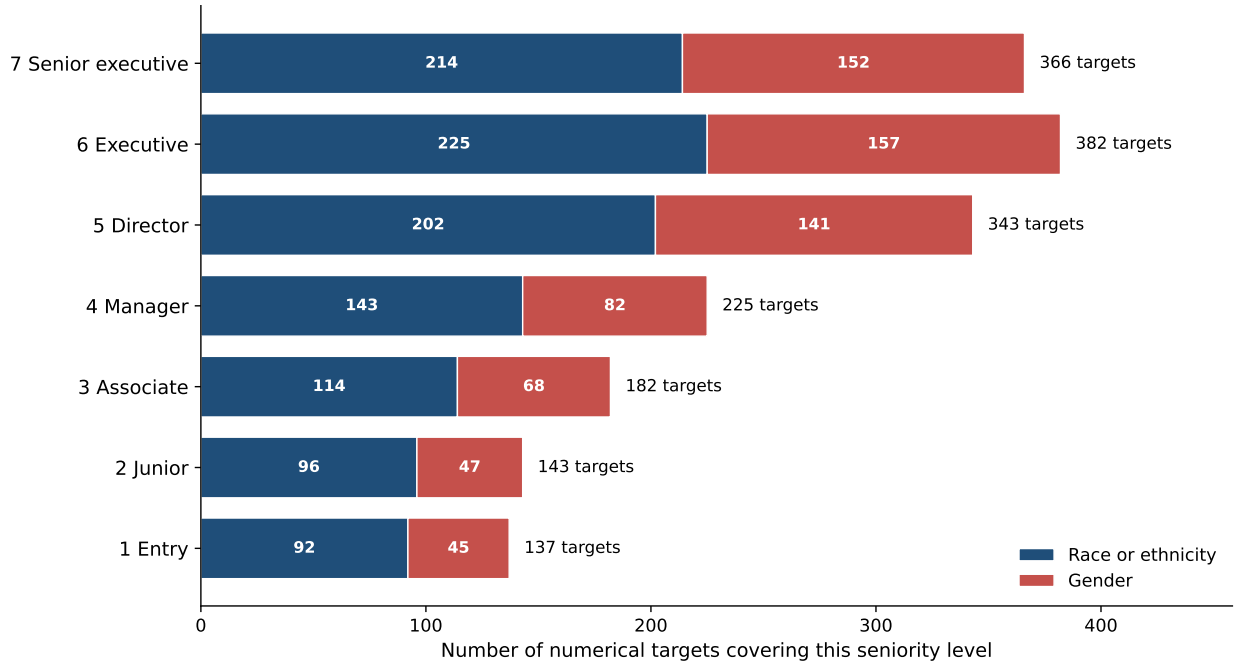
Notes: The bars count the 186 S&P 1500 firms with a numerical pledge covering U.S. workers by calendar year of their first numerical race pledge, or of their first gender pledge if they have no numerical race target. Firms with a numerical race target and no numerical gender target are race or ethnicity only, firms with the reverse are gender only, and firms with both are race and gender. Most race dates are exact to the day, and most gender dates are exact only to the year. Source: pledge dataset.

Since the vast majority of race pledgers cover Black workers, I concentrate the estimation on the Black share. In the main sample, 42 percent of the race pledgers name Black workers explicitly, and the rest use an umbrella group that includes them. About a fifth of race pledgers name Hispanic workers, always alongside Black workers (Table A.4). I therefore use the Black and Hispanic share as the secondary outcome.

3.2 Official and Photo-Based Race Data

EEO-1 reports are documents collected by the government each year that count the firm's U.S. employees by job category, sex and race or ethnicity. This data is, in most cases, self-reported. Every private employer with at least 100 employees and every federal contractor with at least 50 employees and a $50,000 contract files this report. Hispanic workers of any race form one group, so the Black group excludes them. Because the reports are confidential, I combine two sources. For 2016 to 2020, I use the consolidated reports of

FIGURE 3: SCOPE OF NUMERICAL DIVERSITY TARGETS ACROSS SENIORITY LEVELS



Notes: Each bar counts the targets of the 186 numerical pledgers whose scope covers one Revelio level, split into race and gender targets. A target that spans several levels counts at each of them. The 11 joint targets count once as race and once as gender, so the 401 targets with a coded scope give 412 entries. Levels 1 and 2, 3 and 4, and 5 to 7 are the entry, middle and director-and-above bands. Source: pledge dataset.

federal contractors that the Labor Department's Office of Federal Contract Compliance Programs released under the Freedom of Information Act. From 2021 on, I use the reports that firms published themselves. A firm's count jumps when it acquires or sells a business. To deal with this, I cut a firm's series at those events and at known mergers and give each stretch its own fixed effect (Appendix E.5). In the event study of Section 5.1, I keep the 102 pledgers and 685 controls with at least one report.

Since the job categories of the reports do not line up with the seniority bands of Section 3.3, I use them only for the Black share of all employees. After 2020, I only have the reports of firms that chose to publish them.

The second measure comes from Onorato et al. (2026), who predict the race of LinkedIn users at Fortune 1000 firms from profile photographs and surnames, and report each firm's Black share by calendar year. I observe 92 of my pledgers and 491 of my controls in their panel.

I also use three other sources. I get revenue, net income and total assets from the SEC's XBRL company facts to compare pledgers and controls in Table 3. The race predictions

of Section 3.4 use IPEDS counts of graduates by race, and I validate them against self-reported race in the North Carolina voter file (Appendix C.3).

3.3 LinkedIn Profiles

The workforce data come mainly from a November 2025 snapshot of LinkedIn profiles (Revelio Labs, 2025). LinkedIn data has been widely used to measure labor-market outcomes at the level of firms and workers (see, for instance, Berry et al., 2026; Jeffers, 2024; Baker et al., 2024; Cai et al., 2024; Bloom et al., 2026). A profile typically lists a worker's history of job positions, each with a title, a location, a start date, and an end date.[7] Revelio maps every job title to one of seven corporate seniority levels, from entry level (level 1) to senior executive (level 7). For the purpose of simplicity, I group the levels into three bands: entry (levels 1 and 2), middle (3 and 4), and director and above (5 to 7).[8] I only keep workers with at least one U.S.-based position at the companies in my sample. I define firms by their current ownership to prevent acquisitions from suddenly moving a firm's measured workforce from one year to the next. For instance, a company acquired in 2022 counts as part of its acquirer in 2016 as well.

I measure each firm's workforce on July 1 of each year. This gives each firm a single headcount per year and places the 2020 snapshot before the pledges of 95 of the 115 pledgers. A worker counts at firm f on July 1 of year t if the most recent position she started on or before that date is at f and her profile shows no end to it before that date. The panel runs from 2014 to 2025 and covers 1,504 firms, including the 1,449 index firms. From 2014 to 2024, it holds 23.5 million workers and 126.5 million worker-years.

There are two concerns about using LinkedIn data in this context. The first is the selection into LinkedIn. The second is differential coverage rates by race, especially if minorities face discrimination on the platform (Evsyukova et al., 2025). To address these concerns, I do two things. First, I compare the workers by company year with the official EEO-1 data. Second, I compare my final sample with a representative sample of private-sector employees using data from the Current Population Survey. At the median firm matched to an EEO-1 report, LinkedIn shows 74 percent of the U.S. headcount the firm

[7] Revelio links each employer to its ultimate parent. I make minor changes to this mapping, which I describe in Appendix B.2.

[8] In 2019, 77 percent of workers in the top band sat at level 5 (director), 21 percent at level 6, and 2 percent at level 7. The levels come from Revelio's model of job titles, which can place the same title at different levels in different firms. Appendix B.5 describes a correction for titles whose level falls when the worker is promoted. After I drop cells with fewer than five workers, each band has about 12,500 firm-year cells in the estimation panel, and the median cell at director and above holds about 300 workers.

reports to the government. By industry, coverage is highest in information and finance and lowest in utilities and transportation (Appendix B.3). However, since the outcome is a share, low coverage matters only if it differs by race or between pledgers and controls. By race the difference is small: in North Carolina, 16.7 percent of Black and 15.5 percent of white registered voters have a matching profile (Appendix B.3). Across groups of firms, the ratio of LinkedIn workers to reported employees has similar means at pledgers, general pledgers and controls, but its median is lower at controls (Appendix B.3). Compared to the private-sector employees of employers with 1,000 or more workers in the Current Population Survey (Appendix Table B.2), the panel is close on sex, region and the imputed Black share but over-represents college graduates, younger workers and professional and managerial jobs.

3.4 Measuring Race

Since LinkedIn profiles do not report race or ethnicity, I predict them using other pieces of information available in the profiles, as other studies do (e.g., Berry et al., 2026; Onorato et al., 2026). I classify people into one of four categories, white, Black, Hispanic of any race, and Asian or Pacific Islander, using Bayesian Improved First-name Surname Geocoding (BIFSG; Voicu, 2018). BIFSG adds first names to the surname-and-place method of Elliott et al. (2009) and Imai and Khanna (2016). BIFSG has been used to measure racial disparities in tax audits (Elzayn et al., 2025).[9] For group r, first name f, surname s, and county g,

$$P(r \mid f, s, g) \;\propto\; P(r \mid g)\, P(f \mid r)\, P(s \mid r),$$

which treats the terms as independent within a group. I take the name terms from the tables that Rosenman et al. (2023) estimate from the voter files of six states that record race. For locations, I use the Census Bureau's estimate of each group's share of the population in the county where a worker held most of her U.S. positions. I keep this county fixed over her career, so that a move does not change her predicted race (Appendix C.1). For graduates of colleges where most graduates are Black, I multiply the Black odds by the college's likelihood ratio from IPEDS graduates by race, with a weight fitted on North Carolina voters (education-augmented BIFSG; Appendix C). For the share outcomes, I average the classifier's probabilities over the workers in a cell, and in the hire-level designs

[9] Among North Carolina voters, BIFSG labels 71 percent of Black voters as Black, while Revelio's own race classifier labels them correctly only 55 percent of the time. Therefore, I use Revelio's race probabilities only as a robustness check (Appendix C.3).

I use the most likely group.

Given that the classifier is probabilistic, the dependent variable has non-classical measurement error, which attenuates the estimated effects on the shares. Following Mahajan (2006) and Hu (2008), the measured Black share of a cell, or its Black and Hispanic share, is

$$m = \pi_0 + (\pi_1 - \pi_0)\, s,$$

where s is the true share and π_1 and π_0 are the classifier's average probabilities for the group among its members and among non-members. If π_1 and π_0 are the same at pledgers and controls, before and after the pledges, a change in s translates into a change in m that is $\pi_1 - \pi_0$ times as large. Because the classifier is more likely to misclassify a Black worker as something else than the other way around, $\pi_1 - \pi_0$ is below one. I report the estimates below as measured. Dividing them by $\pi_1 - \pi_0$ scales them into true units: for the Black share, the factor is 1.30 on the EEO-1 reports of federal contractors and 1.62 on the North Carolina voter file, which records self-reported race (Appendix Table C.1). Scored by its most likely label instead, the classifier labels 76 percent of Black voters and 6 percent of the others as Black.

4 Empirical Design

I estimate the effect of a numerical race pledge by comparing the workforces of pledging firms with those of never-pledged S&P 1500 firms, before and after each pledge. Since firms announced their pledges at different dates, mostly between 2020 and 2022, I run this comparison as a staggered event study of the Black share at the company level and in each seniority band (Section 4.2). For outcomes at a worker-level, a triple difference adds a second comparison within each firm, between white men and Black workers hired in the same year (Section 4.4).[10]

4.1 Sample and Timing

In my main specification, the treated firms are the 115 firms that made a numerical race pledge between 2018 and 2022, and the controls are the 1,063 firms with no pledge signal in any source (see Table 2).[11] I define a firm's cohort G_f as the year of the first July 1 on

[10] The flow decomposition of Section 5.4 is described in Appendix E.2.

[11] I exclude four other groups of firms from the main comparison (Table 2), since they faced the same public pressure and may also have changed whom they hired: the general pledgers, which I use instead

or after its earliest numerical race pledge. A firm that announced its pledge in September 2020, for instance, belongs to the 2021 cohort. I keep a pledger in its cohort even if it later softened or withdrew its target. To separate the effect of a pledge from shocks that were common to all firms in 2020, I compare the change in a pledger's workforce after its pledge with the change at never-pledged firms over the same years. The four snapshots before each pledge let me see whether the two groups were already moving apart.

TABLE 2: SAMPLE CONSTRUCTION AND CLASSIFICATION OF S&P 1500 FIRMS

	Firms	Role in the analysis
A. Classification of the index		
Numerical race pledge, cohorts 2018 to 2022	136	Treated (all in a robustness check)
of which main estimation sample	115	Treated
Numerical race pledge, cohorts before 2018 or after 2022	7	Dropped
Other numerical pledges after 1 July 2022	17	Held out
Targets for the board only	11	Held out
Targets for women only	22	Held out
General pledgers (no numerical race target)	193	Alternative comparison group
No commitment	1,063	Controls
All firms in the index	1,449	
B. Pledge data: firms with a numerical pledge covering U.S. workers, any date		
Race and gender	108	
Race only	49	
Gender only	29	
All numerical pledgers (401 targets)	186	Description (Section 3.1)
C. Workforce panel, 2014 to 2024		
All firms in the panel	1,504	23.5 million workers, 126.5 million worker-years
Main estimation sample	1,177	14.3 million workers, 68.5 million worker-years

Notes: The table classifies the S&P 1500 firms by pledge status (Panel A), counts the numerical pledgers (Panel B) and gives the size of the workforce panel (Panel C). Panel B includes every firm with a numerical pledge covering U.S. workers, at any date; seven of the 193 general pledgers in Panel A set a numerical gender target and are counted under Gender only. The 1,504 panel firms of Panel C include the 1,449 index firms. Sources: pledge dataset; LinkedIn profiles via Revelio Labs (November 2025 release).

as an alternative comparison group, most of the firms that pledged after 1 July 2022 (Appendix A.3), the firms with targets for women only, and the firms whose only targets concern the board. Board members are not employees, so they do not show up in the positions recorded on LinkedIn. Because one control firm, Old Dominion Freight Line, has no firm-band-year cell with at least five workers, every share estimate uses 1,062 controls. The exhibits that count all the firms in the sample definition list 1,063 controls.

4.2 Event-Study Design

The main outcome of this section is the Black share of a firm's workforce both at the company level and in each of the three seniority bands defined in Section 3.3. For firm f, band b, and year t, $y_{f,b,t}$ is that share on July 1, in percent, so coefficients are in percentage points. In my main specification, I estimate race probabilistically (Section 3.4). Therefore, each worker enters the share with the classifier's probability of being Black rather than the most likely category. I perform robustness checks with the most likely category. I drop cells with fewer than five workers.

I estimate the following interaction weighted event study estimator of Sun and Abraham (2021):

$$y_{f,b,t} = \alpha_f^b + \delta_t^b + \sum_{g=2018}^{2022} \sum_{\substack{e=-4 \\ e\neq -1}}^{3} \beta_{g,e}^b \, \mathbf{1}\{G_f = g\} \, \mathbf{1}\{t - g = e\} + \varepsilon_{f,b,t}. \tag{1}$$

I use this estimator because, with pledges at different dates, a regression with a single post-pledge indicator would also use firms that had already pledged as controls for later pledgers. Such a regression can put negative weights on some of the effects if they differ across cohorts or grow over time (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020). In Equation (1), I instead compare each cohort only with the never-pledged firms. The estimates are similar with other staggered-adoption estimators (Appendix D.4; Callaway and Sant'Anna, 2021; Borusyak et al., 2024). To summarize the path, I first average the cohort coefficients at each event time. Each cohort's weight is its share of the treated firms observed at that event time. The headline number is the mean of these averages over event times 0 to 3, the first four snapshots after the pledge:

$$\tau_b = \frac{1}{4} \sum_{e=0}^{3} \sum_{g \in \mathcal{G}_e} \omega_{g,e} \, \beta_{g,e}^b, \qquad \omega_{g,e} = \frac{N_g}{\sum_{g' \in \mathcal{G}_e} N_{g'}}. \tag{2}$$

Here, $\mathcal{G}_e$ is the set of cohorts observed at event time e, and N_g is the number of treated firms in cohort g. Every cohort is observed in all four pre-pledge years and at event times 0 to 2; the panel ends in 2024, so event time 3 is observed only for the 2018 to 2021 cohorts (83 firms). Cells are unweighted, so the estimand is the average effect on a pledging firm, not the effect on the average worker at a pledging firm. I do the worker-weighted version, which weights each cell by its number of workers, in a robustness check (Appendix G; Solon et al., 2015).

Standard errors are clustered at the ultimate parent company level. I also report randomization inference, which reassigns pledge status and cohort years across firms within size and sector strata and compares the observed t-statistic with the resulting distribution.[12]

4.3 Identification and Threats

These estimates represent the average treatment effect on the treated (ATT) under three assumptions. First, parallel trends requires that, without their pledges, the measured Black share would have moved as it did at the controls. The closest empirical sign for this would be a violation of pre-treatment trends. Second, no anticipation requires that pledgers did not change whom they hired or promoted before event time 0, the first snapshot after their announcement. If they had, then the Black share would have already risen between event times -1 and 0. Third, the stable unit treatment value assumption (SUTVA) requires that a firm's share not depend on whether other firms pledged. In particular, if pledging firms poach minority managers from the controls, that would inflate the estimate. It is also feasible that control firms could quietly follow the pledgers in their attempts. That would shrink the coefficients. Section 5.4 empirically tests the poaching concern by looking at the company-to-company flows by treatment status, and poaching from the controls turns out to matter little for the estimate. For the economy as a whole, however, the movement of Black workers from other employers to the pledgers is what Section 7 measures: there, I follow these workers back to the employers they left and ask who filled the seats they left behind.

Pledging firms and never-pledged firms in the S&P 1500 are not equal in observables. Table 3 compares the race pledgers of the main estimation sample and the general pledgers with the controls; its workforce rows are for 2019 and its financial rows for fiscal 2019.[13] The numerical race pledgers are much larger: the average pledger has about six times the revenue, eight times the assets and six times the U.S. LinkedIn workforce of the average control. Pledgers were also more exposed to public pressure. Eighteen percent of them own one of the world's 100 most valuable brands, against almost none of the controls. More of the pledgers are consumer-facing firms and federal contractors, and a smaller share of their executives who give to political campaigns lean Republican. In the

[12] The test is exact under the sharp null that no pledge moved any firm's share; because the statistic is studentized, it remains approximately valid for a zero average effect.

[13] Fiscal 2019 is the fiscal year ending between June 2019 and May 2020, as first reported in an annual report.

LinkedIn data, the Black share at director level and above in 2019 is nearly the same at pledgers, general pledgers and controls (10.60, 10.99 and 10.26 percent). The Black and Hispanic share differs by less than a point (Table 3). On the EEO-1 reports that federal contractors file, pledgers were already ahead among officials and managers, with 5.6 percent Black against 4.1 percent at the controls.

TABLE 3: BASELINE FIRM CHARACTERISTICS BY DIVERSITY PLEDGE STATUS (2019)

				Race minus control	
	Race pledgers ($N = 115$)	General pledgers ($N = 193$)	Never-pledged controls ($N = 1{,}062$)	Difference	p-value
Panel A: Size and workforce					
Revenue ($ billions)	26.7	30.4	4.6	22.1	<0.001
Net income ($ billions)	3.19	3.02	0.30	2.89	<0.001
Total assets ($ billions)	95.0	90.7	11.5	83.5	0.001
LinkedIn workers, 2019 (thousands)	21.7	18.7	3.7	18.1	<0.001
EEO-1 U.S. employees, 2019 (thousands)	31.1	31.0	8.7	22.4	<0.001
Director and above, 2019 (percent)					
Black	10.60	10.99	10.26	0.34	0.343
Black and Hispanic	19.16	19.25	18.69	0.47	0.315
Female	36.77	35.15	32.40	4.36	<0.001
Panel B: Exposure to pressure (percent)					
Interbrand top-100 brand, 2020	18.3	8.8	0.2	18.1	<0.001
Consumer-facing sector	35.7	21.2	19.7	16.0	<0.001
Federal contractor, 2016 to 2020	77.4	62.7	52.5	24.8	<0.001
Right-leaning executive donors	33.7	43.5	47.3	−13.6	<0.001
Democratic vote in headquarters state	55.9	53.4	52.6	3.3	<0.001

Notes: The table reports means of baseline firm characteristics by pledge status. The last two columns give the difference between race pledgers and controls and the p-value of a Welch t-test. Financial information is from year 2019. EEO-1 employees: Official 2019 U.S. headcount reports by federal contractors. LinkedIn headcount and director-and-above shares: July 1, 2019 of my panel (Revelio levels 5 to 7; education-augmented BIFSG probabilities and first-name probabilities of being female, averaged over firms). Revenue is available for 115, 190 and 1,012 firms (race pledgers, general pledgers, controls), a 2019 EEO-1 report for 78, 105 and 455, executive donor data for 107, 158 and 729, and a headquarters state for 104, 187 and 1,032. In Panel B, a firm has an Interbrand top-100 brand if it owns a brand in Interbrand's 2020 Best Global Brands, and it is consumer-facing if its NAICS industry is retail, restaurants and hotels, airlines, media, telecommunications, retail banking or consumer goods. It is a federal contractor if it has an EEO-1 report for any year from 2016 to 2020 among those released under the Freedom of Information Act. Right-leaning executive donors are measured as the share, among the firm's executives who gave to political campaigns in 2012 to 2020, of those whose donations lean Republican (DIME CFscore above zero; firms with at least five such donors). Democratic vote is the Democratic share of the two-party presidential vote in 2016 in the headquarters state. Sources: pledge dataset; SEC XBRL company facts; EEO-1 reports; LinkedIn profiles via Revelio Labs (November 2025 release); Interbrand; DIME; 2016 presidential election returns.

Another concern is that large firms may have been hit harder by shocks in these years, such as the pandemic hiring freeze, the move to remote hiring or the later layoffs in tech-

nology and finance. To address this, I estimate a version with sector-by-year fixed effects, which absorb the part of these shocks that firms in the same sector share (Section 5.2). Finally, the race classifier makes errors, which only rescale the estimates if their rates are the same at pledgers and controls and stay the same after 2020. To check this, I use the North Carolina voters matched to the panel. Among these voters, the classifier's error rates are close at pledgers and controls and did not change differently after 2020 (Appendix C).

4.4 Triple Differences

The second question of this paper is whether the pledgers lowered the bar for the Black workers they hired, and whether the new positions were window dressing. The event study compares a demographic share across firms. Other outcomes are measured on hires, such as a new hire's university, previous employer, starting title or later promotion. For these outcomes, I add a second comparison group inside each firm, made up of the white men hired in the same year. The triple difference compares Black hires in the main specification, and Black and Hispanic hires in the secondary one, with these white-male hires, after the pledge relative to before, at pledgers relative to the controls. The superscript BH below stands for whichever group is compared.

The sample is every U.S. hire from outside the firm at the 115 pledgers and the 1,063 controls between 2016 and 2024. I count a position start as an outside hire if the worker's previous job was at another employer or she had none. Moves within the firm are left out, as in the decomposition of Section 5.4. Each hire gets its most likely ethnicity label (Section 3.4).

For an outcome y, firm f, band b, and hire year t, let $\bar{y}^{BH}_{f,b,t}$ and $\bar{y}^{WM}_{f,b,t}$ be the mean outcome among the compared group's hires and among white-male hires, and let $n^{BH}_{f,b,t}$ and $n^{WM}_{f,b,t}$ be their numbers. The regression mirrors Equation (1), with the gap between the two groups as the outcome:

$$\bar{y}^{BH}_{f,b,t} - \bar{y}^{WM}_{f,b,t} = \alpha^b_f + \delta^b_t + \sum_{g=2018}^{2022} \sum_{\substack{e=-4 \\ e\neq -1}}^{3} \theta^b_{g,e}\, \mathbf{1}\{G_f = g\}\, \mathbf{1}\{t - g = e\} + u_{f,b,t}, \tag{3}$$

estimated with weights $h_{f,b,t} = n^{BH}_{f,b,t}\, n^{WM}_{f,b,t} / (n^{BH}_{f,b,t} + n^{WM}_{f,b,t})$. With these weights, the regression on the gap is equivalent to the full triple difference with firm-by-year, firm-by-group and group-by-year fixed effects, and a gap built from a handful of hires gets little weight. I drop a cell if the firm hired no worker from one of the two groups in that band and year.

Since a pledge raises the number of Black hires, it changes both the weights and the set of cells. The post-pledge coefficients therefore lean toward the firms that responded most (Appendix D).

I estimate the equation as a stacked regression (Cengiz et al., 2019): each pledge cohort forms a stack with all never-pledged controls, and the fixed effects are specific to each stack. The window, reference year and clustering follow the event study, and θ_b averages the coefficients over event times 0 to 3. Since retention and promotion are measured 24 months after the hire, their window starts at -6 and their reference year is -3.

The identifying assumption of this triple differences in differences estimate is that, absent the pledge, the gap in observable variables between incoming Black hires and incoming white-male hires would have moved at pledgers as it did at the control firms. Because a fraction of Black hires would have been hired without the pledge, θ averages over all of them. Online Appendix OA.C.4 derives the bounds on the added hires and the size of the label shift.

5 Effects on Workforce Composition

In this section, I document how the composition of the pledging firms' workforces changed and through which flows the change in composition was achieved. I then test the effect against a placebo group and the general pledgers.

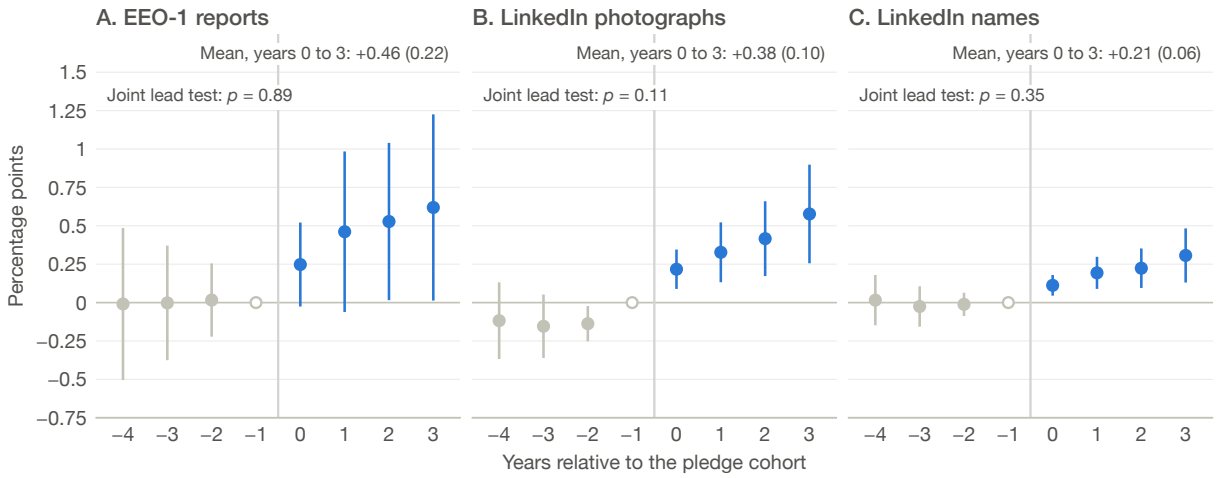
5.1 The Black Share

I document that the Black share of all employees at the pledging firms rose after a numerical race pledge regardless of whether I use official firm demographics, the photo-based measure of Onorato et al. (2026) or my measure predicting race from names, location and education. Figure 4 plots the coefficients of Equation (1) for three sources, each on every pledger and control it covers. The three sources are the firms' own EEO-1 reports, which count workers by self-reported race; race predicted from LinkedIn profile photographs and surnames by Onorato et al. (2026); and my name-based measure. Averaged over event years 0 to 3, the Black share rose by 0.47 percentage points in the EEO-1 reports (wild-bootstrap 95 percent interval 0.09 to 0.84), by 0.38 points in the photographs and by 0.21 points in the name-based measure. None of the joint tests rejects flat pre-trends at the 10 percent level (p = 0.89, 0.11 and 0.35). The differences in effect sizes across measures

are also consistent with the amount of measurement error one would expect from official data (little measurement error), photographs (more) or names, location and education (the most).

On the firm-years covered by both sources, the name-based estimate is 0.22, against 0.47 in the EEO-1 reports. Comparing the Black shares of the two sources across firms before 2020 gives a company-level correction factor of 1.8, which turns the 0.22 into 0.40, inside the EEO-1 interval. For the rest of the paper I use the name-based measure, since it is the only one of the three that covers almost every firm in every year and that I can split into the seniority bands of Section 3.3.

FIGURE 4: EFFECT OF NUMERICAL RACE PLEDGES ON THE FIRM-WIDE BLACK SHARE ACROSS RACE MEASURES



Notes: The figure plots event-study coefficients, in percentage points, for the Black share of all employees, measured in three sources. The estimator is the interaction-weighted event study of Sun and Abraham (2021), Equation (1), with firm and year fixed effects and 95 percent confidence intervals based on firm-clustered standard errors. Year −1 is the reference, and labels give the average over years 0 to 3 (standard error). Panel A uses EEO-1 reports of 102 pledgers and 687 controls, each report dated to the first July 1 after its count, with a separate fixed effect for each stretch of a firm's reports between breaks at mergers, acquisitions and divestitures (Appendix E.5). Its joint lead test also covers year −5, which is not shown. Panel B uses race predicted from profile photographs and surnames by Onorato et al. (2026), for 92 pledgers and 491 controls in their Fortune 1000 panel, dated by calendar year. Panel C uses the name-based measure (education-augmented BIFSG) for 115 pledgers and 1,062 controls. Sources: EEO-1 reports released by the OFCCP under FOIA, company reports, Onorato et al. (2026), Revelio Labs.

After a numerical race pledge the Black share rises in every band and keeps rising. Figure 5 plots the coefficients of Equation (1) for each band, and Table 4 reports their average over event years 0 to 3.[14] At director and above the average effect is 0.20 percentage

[14]The tables report standard errors clustered by firm and the figures 95 percent intervals. In the text I give significance levels, and intervals where a null or a bound is the result.

points, significant at the 1 percent level. The coefficient grows from 0.07 points in the first snapshot after the pledge to 0.38 three years later, 3.5 percent of the pledgers' 2019 share of 10.6 percent.[15] Panel A of the table gives the averages for the other bands. Multiplied by the pledgers' LinkedIn headcounts in 2023, the averages amount to about 1,200 more Black workers at director level and above and 5,800 across the firms. The Black and Hispanic share rises by about as much as the Black share, and the Hispanic share alone does not move in any band (Table 4, Panel A). This is congruent with the pledges, since only a fifth of them mention Hispanic workers directly, and always alongside Black workers.

Before the pledges the paths are flat outside the middle band. The joint test that the three pre-pledge coefficients are zero does not reject at director and above, at entry or for the whole firm (Table 4, Panel B). The exception is the middle band, whose coefficient three years before the pledge is -0.18, significant at the 5 percent level, with a joint p of 0.08.

The effect had not faded by the end of the window, when the backlash began.[16] In every band the coefficient at event year 3 is the largest of the path. The July 2025 snapshot adds event years 4 and 5 (shaded in Figure 5). Event year 4, which rests on 83 pledgers, is significant at the 1 percent level in every band: 0.34 points at director and above, 0.38 in the middle band, 0.43 at entry and 0.31 for the whole firm. Event year 5 rests on 20 pledgers and is imprecise (0.28 at director and above).[17] Aligned on each firm's first retreat, the firm-wide Black share at the first July 1 after it did not fall (Appendix Table E.1, Panel B). The 95 percent interval excludes falls larger than about two fifths of the pledge effect. The number of Black hires per 100 positions did not fall in the quarters after the retreat either (Panel A of that table; Appendix E.1). Half of the retreating firms are observed for only one or two quarters after their retreat.

As discussed in the measurement section, my race classifier is more likely to misclassify a Black worker as something else than the other way around. If its error rates are the same at pledgers and controls and did not change with the pledges, the true effect is the measured one times the correction factors of Appendix Table C.1. For the Black share, the correction factors range from 1.3 (EEO-1 reports) to 2.72 (proxy photographs), with 1.62

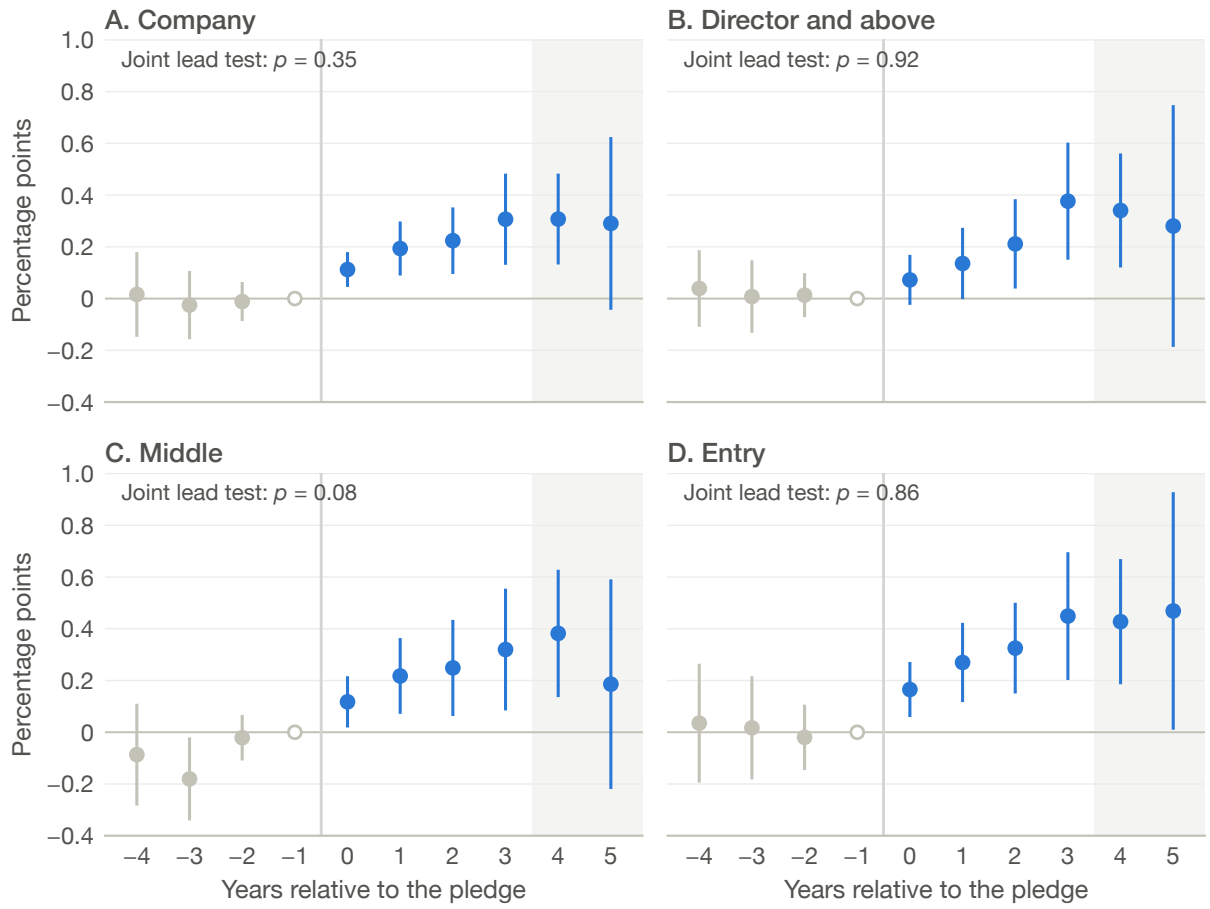
[15] Event year 3 is observed only for the 83 firms of the 2018 to 2021 cohorts, so part of the jump between years 2 and 3 reflects the change in which firms are observed. For those 83 firms the path at director and above rises steadily, from 0.11 to 0.19, 0.30 and 0.38.

[16] The firm-wide increase did not reverse at the first July snapshot after firms retreated, although half of the firms are observed for only one or two quarters after retreat, too briefly to rule out a slow reversal.

[17] Some job changes of 2025 had not yet reached the profiles when the data were extracted in October 2025. Reweighting the cells for incomplete recording moves the year-4 estimates by -0.05 to $+0.08$ points. In the run that adds 2025, years -4 to 2 are the same to the third decimal. Year 3, however, also includes the 2022 cohort, whose effect at director and above is near zero, and its coefficient there falls to 0.26.

from the voter file. Correcting by these factors would turn the coefficient at director level and above to between 0.26 and 0.54 percentage points, and the effect three years after the pledge to between 0.5 and 1.0 points.

FIGURE 5: EFFECT OF NUMERICAL RACE PLEDGES ON THE BLACK SHARE BY SENIORITY LEVEL



Notes: The figure plots event-study coefficients, in percentage points, for the Black share of the company and of each band. The estimator is the interaction-weighted event study of Sun and Abraham (2021), Equation (1), with firm and year fixed effects and 95 percent confidence intervals based on firm-clustered standard errors. Event year 0 is the first July 1 snapshot on or after the pledge, and year −1 is the reference. The sample is 115 numerical race pledgers and 1,062 never-pledged controls, in unweighted firm-band-year cells with at least five workers. Years −4 to 3 are the estimates of Table 4, and years 4 and 5 (shaded) add the July 2025 snapshot and rest on 83 and 20 pledgers. Source: Revelio Labs.

5.2 Placebo, General Pledgers and Robustness

The placebo in the second row of Table 5 shows no rise in a share that the pledgers' targets do not name: the 39 pledgers without a numerical gender target show no rise in the white-

TABLE 4: EFFECT OF NUMERICAL RACE PLEDGES ON BLACK AND HISPANIC WORKFORCE SHARES

	Name-based measure				Company, other sources	
	Director and above	Middle	Entry	Company	EEO-1 reports	Photographs
Panel A: ATT, mean of event years 0 to 3 (percentage points)						
Black	0.199***	0.226***	0.302***	0.209***	0.471**	0.384***
	(0.072)	(0.077)	(0.080)	(0.056)	(0.222)	(0.104)
Black and Hispanic	0.229**	0.248**	0.319***	0.225***	0.249	0.436***
	(0.098)	(0.102)	(0.114)	(0.081)	(0.298)	(0.127)
Hispanic	0.030	0.022	0.016	0.015	−0.223	0.052
	(0.071)	(0.077)	(0.079)	(0.055)	(0.285)	(0.071)
Panel B: Joint test of the leads (p-value)						
Black	0.92	0.08	0.86	0.35	0.89	0.11
Black and Hispanic	0.68	0.14	0.76	0.48	0.30	0.39
Hispanic	0.42	0.67	0.64	0.65	0.10	0.55
Black share in 2019 at the pledgers (percent)	10.6	11.7	13.8	12.7	10.5	9.2
Numerical race pledgers	115	115	115	115	102	92
Never-pledged controls	1,062	1,062	1,062	1,062	685	491

Notes: The table reports the effect of numerical race pledges on workforce shares. The estimator is the interaction-weighted event study of Sun and Abraham (2021), Equation (1), with firm and year fixed effects, event years −4 to 3 and reference year −1. Panel A gives the ATT of Equation (2), the mean of the coefficients in event years 0 to 3, with firm-clustered standard errors in parentheses. Panel B gives the p-value of the joint test that the leads (−4 to −2) are zero. The first four columns compare the pledgers of the main sample with the never-pledged controls on unweighted firm-band-year cells, with shares measured as education-augmented BIFSG posterior means; the 2019 row averages the pledgers' shares on July 1, 2019. The last two columns show the whole-firm estimates of Figure 4 for the EEO-1 reports (event years −6 to 3, leads −5 to −2, and a fixed effect for each stretch of a firm's reports between breaks in its series at mergers, acquisitions and divestitures; Appendix E.5) and for the photo panel of Onorato et al. (2026) (calendar years); their 2019 rows average the 84 and 91 pledgers observed in 2019. The EEO-1 Hispanic lead test gives $p = 0.097$. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sources: pledge dataset; LinkedIn profiles via Revelio Labs (November 2025 release); EEO-1 reports (OFCCP release under FOIA and company reports); Onorato et al. (2026).

women share.[18] If anything the white-women share fell, significantly only in the middle band and at the 10 percent level, and the placebo has no significant lead. The third row replaces the never-pledged controls with the 193 general pledgers, firms that made public diversity commitments under the same public pressure but set no numerical race target. Against them, the estimate at director and above is unchanged (0.22, significant at the 5 percent level), but below the top it falls by about half or more and loses significance. Relative to the never-pledged firms, the general pledgers' Black share thus rose somewhat below the top, through their own statements or a trend shared with the pledgers, but not

[18]Gender is predicted from first-name probabilities (Revelio Labs, 2025).

at the top. The part of the effect specific to a numerical target is largest at the level most targets named (Figure 3). The difference across bands is imprecise.

Changes to the panel or the fixed effects leave the estimates positive and of similar size, although some leads then fail. The fourth row keeps only the firms observed at every event year, which raises the estimates outside the middle band and leaves the middle band unchanged. The leads then fail in the middle band and for the whole firm ($p = 0.01$). The last row drops the 2024 snapshot, which profiles may not yet reflect, and the estimates change little. Sector-by-year fixed effects, which absorb shocks common to an industry, give 0.24 at director and above, significant at the 1 percent level.[19] With them the middle band's leads fail ($p = 0.05$). Only the estimate at director and above depends on the consensus rule that defines the main sample (Section 4.1 and Appendix A.3). With all 136 race pledgers of the 2018 to 2022 cohorts it falls to 0.14, still significant at the 5 percent level. The other bands, and the stricter set of 111 firms, barely move.[20] The 21 firms that the rule leaves out have a pre-trend of their own at director and above (joint lead $p = 0.01$). That fits the reason for leaving them out, since their pledges were the hardest to date. Randomization inference reassigns the 115 pledges and their cohort years at random within 33 sector-by-size strata, 200 times. It gives p-values of 0.020 at director and above, 0.010 at entry and 0.005 in the other two bands, the smallest value 200 draws allow. The permuted t-statistics have a standard deviation of 0.80 to 0.87, so the clustered standard errors, which treat firms as independent although most pledges fell in one wave, look if anything conservative.

5.3 Headcount and Other Groups' Shares

One hypothesis is that firms on an expansion path would find it easier to accommodate a growing minority share, since a high number of openings would give space to several people from various demographics. Figure 6 applies the same estimator to the logarithm of each band's LinkedIn headcount, and Appendix Table E.4 reports the averages. At the typical pledger, headcount over the four years after the pledge was about 5 percent lower relative to the controls than in the year before (-4.8 log points; first row, Company

[19] With sector-by-year fixed effects the estimates are 0.20 in the middle band, 0.28 at entry and 0.20 across the firm. Against the general pledgers, they are 0.27 at director and above, significant at the 5 percent level, and, below the top, larger than without these fixed effects, 0.17 to 0.21, significant at the 5 or 10 percent level.

[20] All 136 pledgers: 0.22 in the middle band, 0.28 at entry and 0.19 across the firm. The 111 firms: 0.21 at director and above, 0.23 in the middle band, 0.31 at entry and 0.21 across the firm.

TABLE 5: EFFECT OF NUMERICAL RACE PLEDGES ON THE BLACK SHARE: PLACEBO AND ROBUSTNESS CHECKS

					Firms	
	Director and above	Middle	Entry	Company	Treated	Control
Main estimate	0.199***	0.226***	0.302***	0.209***	115	1,062
	(0.072)	(0.077)	(0.080)	(0.056)		
Placebo: no gender target, white-women share	−0.219	−0.428*	−0.221	−0.239	39	1,062
	(0.176)	(0.238)	(0.169)	(0.169)		
General pledgers as controls	0.215**	0.071	0.140	0.108	115	193
	(0.097)	(0.090)	(0.091)	(0.066)		
Balanced cohorts	0.245***	0.234***	0.347***	0.236***	83	1,062
	(0.080)	(0.087)	(0.094)	(0.066)		
No 2024 snapshot	0.226**	0.202**	0.312***	0.221***	115	1,062
	(0.088)	(0.085)	(0.103)	(0.067)		

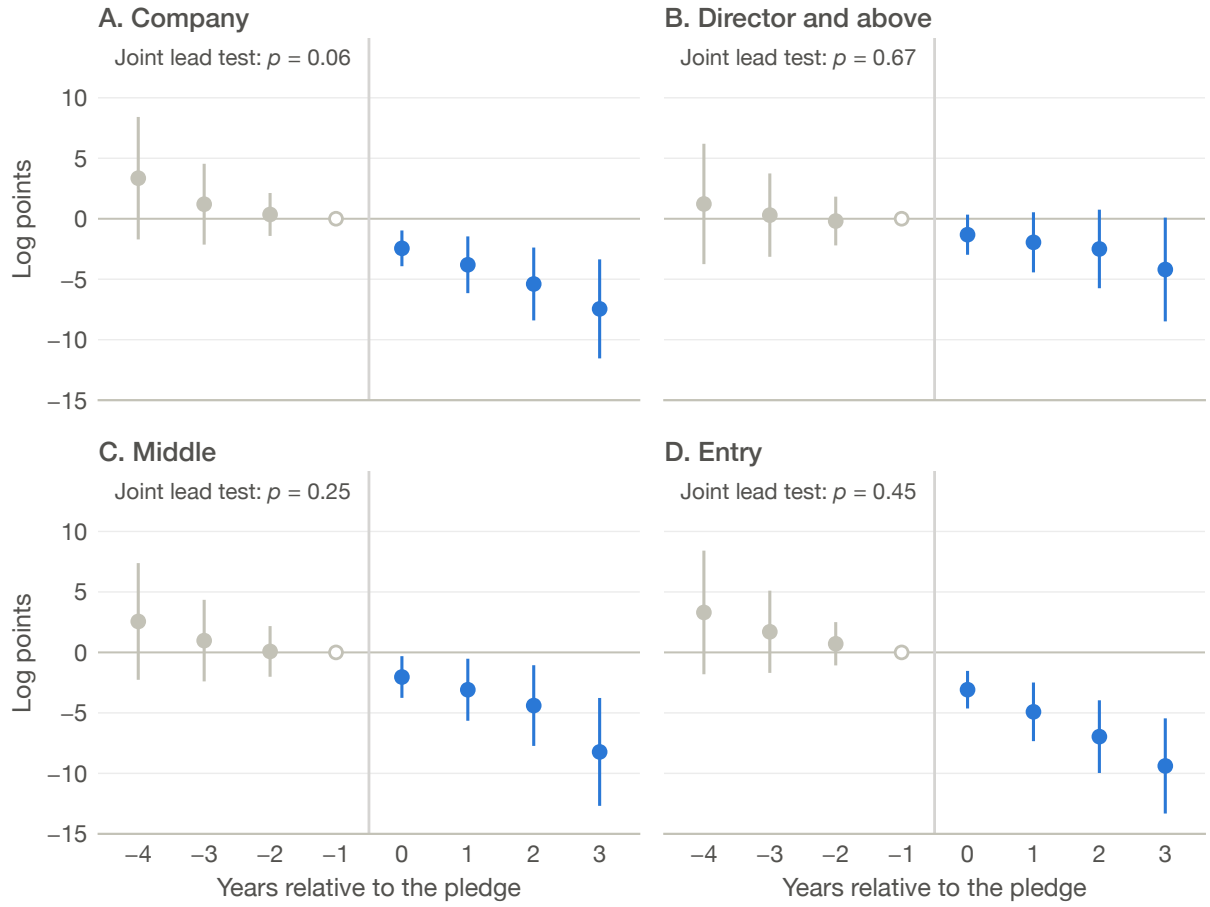
Notes: The table compares the main estimate of the effect of numerical race pledges with a placebo and robustness checks. The estimator is the interaction-weighted event study of Sun and Abraham (2021), Equation (1), with firm and year fixed effects, event years −4 to 3 and reference year −1. Estimates are ATTs of Equation (2), the mean of the coefficients in event years 0 to 3, in percentage points, on the LinkedIn firm-band-year cells of Section 4.2, with firm-clustered standard errors in parentheses. Unless the row says otherwise, the comparison is between the numerical race pledgers of the main sample and the never-pledged controls, and the outcome is the Black share. The placebo uses the 39 main-sample pledgers without a numerical gender target. The balanced-cohorts row keeps the 2018 to 2021 cohorts, each firm observed at every event time from −4 to 3. The last row ends the panel in 2023. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sources: pledge dataset; LinkedIn profiles via Revelio Labs (November 2025 release); education-augmented BIFSG race probabilities, first-name gender probabilities.

column).[21] Relative headcount had already fallen by about 3.4 percent between event years −4 and −1, and it fell by about 7.5 percent between years −1 and 3. Removing that earlier decline as a linear trend shrinks the average drop to 1.6 log points, which is not significant. The firms' own EEO-1 reports, filed by 98 pledgers and 612 controls, are too noisy to confirm the decline, but they rule out a relative expansion of more than about 6 percent (Appendix Table E.5). The top band's share of the workforce rose (third row of Appendix Table E.4) only because entry-level headcount fell fastest (first row, Entry column). That share was already rising before the pledge. At the typical pledger, then, the Black share rose without a relative expansion of positions, so some other group's share fell.

I next ask which groups' shares fell as the Black share rose. In raw terms, the white-male share fell by 0.4 to 0.5 points in every band, more than twice the rise in the Black

[21]The cells are unweighted, so these are statements about the average pledger. Summed over firms, the pledgers' LinkedIn headcount grew faster than the controls' from 2019 to 2023, only because of Amazon.

FIGURE 6: EFFECT OF NUMERICAL RACE PLEDGES ON FIRM HEADCOUNT BY SENIORITY LEVEL



Notes: The figure plots event-study coefficients for 100 times the log of LinkedIn headcount, approximately the percent difference, in the company and in each band. The estimator is the interaction-weighted event study of Sun and Abraham (2021), Equation (1), with firm and year fixed effects and 95 percent confidence intervals based on firm-clustered standard errors. Firm fixed effects absorb differences in LinkedIn coverage, so the coefficients compare growth and say nothing about levels. Timing is as in Figure 5, and the sample is 115 pledgers and 1,063 controls. Source: Revelio Labs.

share at director and above and across the firm (Panel A of Table 6). The five groups' shares (Black, Hispanic, Asian, white women and white men) sum to 100 percent in every cell, so their effects sum to zero. The Asian share rose, and the Hispanic share did not move. The white-women share moved little, significantly only in the middle band and at the 10 percent level.

Both movements had begun before the pledges and kept their pace afterwards. The white-male and Asian shares fail the lead test in most bands, and Figure 7 and the slope rows of Table 6 show why. Relative to the controls, the pledgers' white-male share was

already falling and their Asian share already rising, significantly so in every band except entry. Panel B removes each group's linear trend through its own pre-pledge coefficients, fitted by generalized least squares and extrapolated to event years 0 to 3. Against its own trend, no group's share other than the Black one moves significantly in any band. The Black share, in contrast, breaks from its trend. Its pre-pledge coefficients have almost no slope, and the trend-adjusted increase is 0.23 points at director and above, significant at the 5 percent level, and stays significant at entry and across the firm. Only in the middle band, with its early dip, does it lose significance. The sensitivity analysis of Rambachan and Roth (2023) gives the same answer at director and above, at entry and across the firm. The answer holds as long as the pre-pledge paths are close to straight.[22]

White men lost the most in raw terms, but their share was already falling at pledgers. One movement in the Asian share does coincide with the pledges. At director and above it stalls in the pledge year, 0.22 points below its pre-pledge line, significant at the 5 percent level. Almost all of the stall is among Asians educated abroad, at the time of the 2020 and 2021 restrictions on travel and visas (Appendix E.3).

5.4 Flows and Origins of the Hires

There are four different ways in which a firm could change its Black share at any particular step of its corporate ladder. It could hire Black workers from outside the firm, promote them into those positions from below, promote them out of those positions, or lose workers of a given group when they leave the firm. To disentangle these four channels, I run the event study of Section 5.1 as a stacked difference in differences, separately for each flow (Appendix E.2). The results are depicted in Figure 8. At director and above, hiring contributes 83 percent of the net yearly change in the Black share. Promotion into the band adds a fifth. Separations are not significant, which suggests that white incumbents did not leave faster after the pledge. In the middle band, hiring also drives the increase, promotion from the entry band adds a smaller amount, and separations offset about half of the total.

The fact that hiring contributes the majority of the effect raises the concern that untreated firms were poached by the pledgers, violating SUTVA and inflating the event study estimates. The fact that my worker panel includes the origin firm for every moving

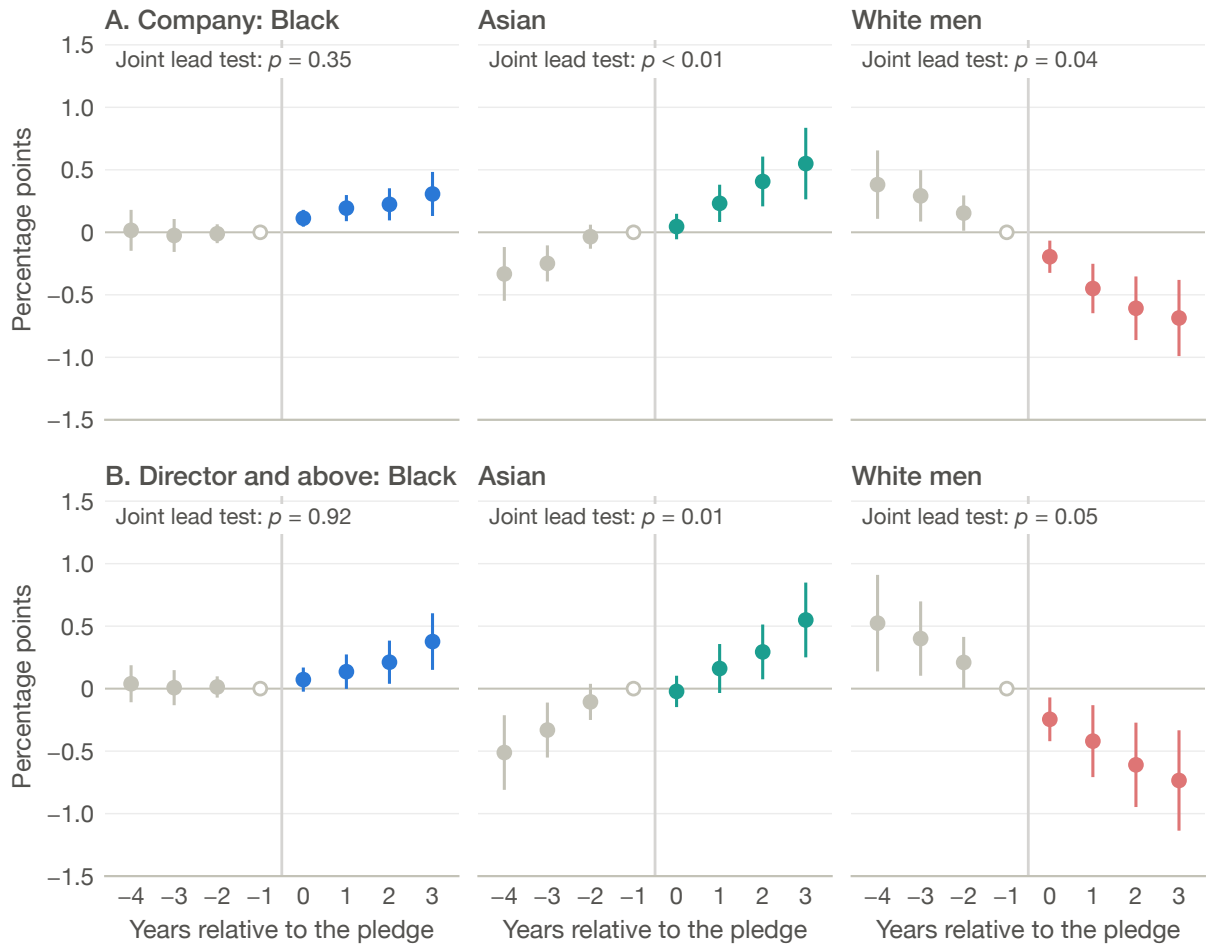
[22] Under a linear trend the 95 percent intervals are 0.05 to 0.43 at director and above, 0.08 to 0.55 at entry and 0.10 to 0.42 across the firm. They still exclude zero if the slope may change from one year to the next by 0.01 points at director and above and by 0.01 at entry and across the firm, and they reach zero at about 0.02 and 0.02.

TABLE 6: EFFECT OF NUMERICAL RACE PLEDGES ACROSS RACIAL AND GENDER GROUPS

	Director and above	Middle	Entry	Company
Panel A: Raw estimates				
Black	0.199***	0.226***	0.302***	0.209***
	(0.072)	(0.077)	(0.080)	(0.056)
	[0.921]	[0.082]	[0.864]	[0.351]
Hispanic	0.030	0.022	0.016	0.015
	(0.071)	(0.077)	(0.079)	(0.055)
	[0.416]	[0.669]	[0.637]	[0.655]
Asian	0.245***	0.420***	0.178	0.309***
	(0.094)	(0.139)	(0.114)	(0.085)
	[0.008]	[0.004]	[0.043]	[0.001]
White women	0.028	−0.262*	−0.060	−0.049
	(0.121)	(0.137)	(0.108)	(0.094)
	[0.736]	[0.098]	[0.594]	[0.117]
White men	−0.503***	−0.406**	−0.436***	−0.485***
	(0.134)	(0.158)	(0.114)	(0.103)
	[0.049]	[0.001]	[0.464]	[0.043]
Panel B: Adjusted for each group's pre-pledge trend				
Black	0.233**	0.109	0.330***	0.264***
	(0.096)	(0.114)	(0.121)	(0.083)
Hispanic	−0.073	−0.038	−0.107	−0.018
	(0.099)	(0.130)	(0.123)	(0.078)
Asian	−0.161	−0.036	0.044	0.034
	(0.152)	(0.172)	(0.148)	(0.097)
White women	0.065	−0.242	−0.074	−0.063
	(0.185)	(0.170)	(0.149)	(0.103)
White men	−0.056	0.212	−0.237	−0.182
	(0.198)	(0.206)	(0.183)	(0.131)
Pre-pledge slope, Asian (pp a year)	0.163***	0.182***	0.054	0.110***
	(0.050)	(0.055)	(0.046)	(0.035)
Pre-pledge slope, white men (pp a year)	−0.179***	−0.247***	−0.080	−0.121***
	(0.065)	(0.062)	(0.060)	(0.045)

Notes: The table reports the effect of a numerical race pledge on each group's share of a band and of the firm (Company column), in percentage points. Estimates are τ_b (Equation (2)), the mean of the event-year coefficients 0 to 3 of the interaction-weighted event study of Sun and Abraham (2021), Equation (1), with firm and year fixed effects, event years −4 to 3 and reference year −1. The sample is 115 numerical race pledgers and 1,062 never-pledged controls. Firm-clustered standard errors are in parentheses and p-values of the joint lead test in brackets. Shares average education-augmented BIFSG probabilities, and the white probability is split by the first name's probability of being female. Panel B subtracts a line fitted by generalized least squares through the pre-pledge coefficients, anchored at zero in year −1; the last two rows report its slope. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sources: LinkedIn profiles via Revelio Labs (November 2025 release); pledge dataset.

FIGURE 7: EFFECT OF NUMERICAL RACE PLEDGES ON WORKFORCE SHARES BY DEMOGRAPHIC GROUP



Notes: The figure plots event-study coefficients for the Black, Asian and white-male shares of the whole firm (Panel A) and of the director-and-above band (Panel B), in percentage points. The estimator is the interaction-weighted event study of Sun and Abraham (2021), Equation (1), with firm and year fixed effects; bars are 95 percent confidence intervals based on firm-clustered standard errors. The white probability is split between white women and white men according to the first-name probability of being female. Sample and timing follow Figure 5. Sources: LinkedIn profiles via Revelio Labs; pledge dataset.

worker allows me to categorize Black inflows into pledging firms in different bins and see how those particular flows changed after the pledge (Table 7). I find that poaching accounts for little of the increase. At director level and above, the increase came in similar parts from other numerical race pledgers, general pledgers, private firms with at least 1,000 U.S. workers, and smaller private firms. By contrast, never-pledged index firms sent 9 percent of the entrants and supplied 5 percent of the increase. Even at the upper 95 percent limit of the Black hires from controls, the bias is at most 0.033 points, a sixth of the estimate, and at the point estimate it is zero.

FIGURE 8: DECOMPOSITION OF THE POST-PLEDGE INCREASE IN THE BLACK SHARE: HIRING, PROMOTIONS AND SEPARATIONS



Notes: The figure decomposes the event-study estimate of Table 4 into four flows, each labeled with its share of the net yearly change in the Black share at pledgers relative to controls. The flows come from a stacked difference in differences that compares the change after the pledge (event years −1 to 3) with the change before (−4 to −1), cohort by cohort. Workers are pooled across firms and demotions are excluded. Appendix Table E.2 gives the flows with 95 percent bootstrap intervals. Sources: LinkedIn profiles via Revelio Labs; pledge dataset.

I also find that the pledges did not widen the set of companies from which a company hires. For the 95 pledgers of the 2021 and 2022 cohorts, the gap between Black and white-male hires in the share coming from employers new to the firm did not change, and the quarterly Black hire rate shows no step at the announcement either.[23] Total and white-male hiring fell around the pledges, while the Black hire rate rose by an insignificant 0.05 per 100 positions a year (Table 7, total row).[24] Black hires thus ran 0.11 per 100 positions a year above what the fall in total hiring would imply at the earlier Black share, a gap significant at the 1 percent level. Therefore, the pledges tilted a shrinking flow of hires toward Black candidates.

6 Black Hires Compared with White-Male Hires

Since the majority of the compositional shift came through hiring, I now turn to the characterization of those workers. From the LinkedIn profiles, I build the hire-level outcomes defined in Table 8. The estimates are the triple differences of Section 4.4, which compare Black and white-male hires in the same firm and hire year, before and after the firm's pledge, at the pledgers relative to the controls. The reported effect is the average over event years 0 to 3.

6.1 Placement in DEI, Human-Resources and Staff Roles

A firm under pressure to show progress could place Black hires in visible roles that carry little authority, such as diversity officer or a human-resources post, or in staff functions away from the line jobs that lead to the top. Table 9 tests this for hires into director and above. None of these placements rises. DEI and human-resources titles fell relative to white men. The share in staff rather than line functions, the path-to-power score of the function, the chance of a role new to the firm and the chance of managing people did not move.

Relative to white men, the Black hires' chance of holding a DEI title fell by 0.42 points. The decline is relative: after 2020 DEI titles became more common among Black hires at the pledgers and at the controls, but much more so at the controls, where they had

[23] The step is $+0.002$ per 100 positions a quarter, on a level of about 0.32. The 95 percent interval rules out a step larger than about 0.03 in either direction, under a tenth of the level.

[24] Dated by start month, total hiring fell by 0.21 hires per 100 positions a quarter and white-male hiring by 0.07, both significant at the 5 percent level.

TABLE 7: EXTERNAL HIRING CONTRIBUTIONS TO THE BLACK SHARE AT DIRECTOR AND ABOVE BY ORIGIN EMPLOYER

Origin (percent of entrants)	Black hires per 100 positions	White-male hires per 100 positions	Contribution to the Black share (pp)	Share of the increase (percent)
Other numerical race pledgers (11.1)	0.0026	−0.077	0.0165***	21.0
	(0.0131)†	(0.050)	(0.0048)	(7.5)
General pledgers (10.1)	0.0038	−0.043	0.0108**	13.7
	(0.0108)	(0.039)	(0.0043)†	(4.8)
Never-pledged S&P 1500 firms (9.3)	0.0024	−0.005	0.0038	4.8
	(0.0068)	(0.025)	(0.0039)	(4.8)
Private, 1,000+ U.S. workers (21.0)	0.0128	−0.056	0.0200**	25.4
	(0.0160)	(0.051)	(0.0079)	(7.4)
Smaller private, self-employed (21.1)	0.0047	−0.060	0.0187***	23.7
	(0.0133)	(0.060)	(0.0072)	(7.3)
Government (1.7)	0.0022	−0.002	0.0017	2.2
	(0.0031)	(0.006)	(0.0025)	(3.4)
Education, nonprofits, health (4.6)	0.0087**	0.015	0.0022	2.7
	(0.0044)	(0.011)†	(0.0027)	(3.4)
No prior job or a 2+ year gap (12.0)	0.0028	−0.002	0.0006	0.8
	(0.0132)†	(0.028)†	(0.0077)†	(10.6)
Firm abroad, other index, unknown (9.1)	0.0090**	0.012	0.0045	5.7
	(0.0043)	(0.017)†	(0.0032)	(4.3)
Total	0.0490	−0.217	0.0787***	100
	(0.0639)	(0.246)	(0.0237)	

Notes: The table reports the change in external hiring into director and above, and its contribution to the Black share, by the hire's previous employer (the origin). Each cell is the change at pledgers minus the change at controls, event years 0 to 3 against −3 to −1, with cohorts stacked and workers pooled across firms, so the origin rows add up to the total. Black hires are weighted by the education-augmented BIFSG probability of being Black and white-male hires by the probability of being white and male. Hires are per 100 positions a year (before the pledge, 1.34 Black and 3.97 white-male hires), and contributions are in percentage points a year. Bootstrap standard errors (1,000 firm-cluster draws) are in parentheses. A dagger (†) after a standard error marks an estimate whose joint lead test gives $p < 0.10$. The sample is 115 numerical race pledgers and 1,063 never-pledged controls. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sources: LinkedIn profiles via Revelio Labs (November 2025 release); pledge dataset.

been rare. DEI titles are rare in any case: after the pledges, 0.9 percent of the Black hires into director and above at the pledgers held one. Relative to white men, the Black hires' chance of a human-resources title fell by 0.87 points.

6.2 Qualifications, Pay and Career Outcomes

I find that the pledges did not change how the Black hires into director and above compare with the white men hired alongside them. Table 10 reports the triple difference for six outcomes, which compares Black and white-male hires in the same firm and hire year,

TABLE 8: DEFINITIONS OF HIRE-LEVEL VARIABLES AND OUTCOMES

Outcome	Definition	Sample
Previous employer's rank	Where the hire's previous employer ranks among U.S. employers, in percentiles. The ranking uses the moves workers chose to make in 2014 to 2019 (Sorkin, 2018).	Hires who came from another employer
Staff function	Equals 1 if the job is in administration, marketing, sustainability or reception (32 of Revelio's 150 functions), and 0 if it is in a line function such as engineering, finance, operations, sales or science.	Placement sample, job recorded under the parent company
Path-to-power score	How often jobs in the function start at Revelio level 6 or 7, standardized across the 150 functions. It does not vary across firms.	Same as above
New role	Equals 1 if nobody at the firm started a job in the same role (one of Revelio's 1,500 roles) in the three years before the hire.	Same as above
Manages people	Equals 1 if the title says manager, director, head of or supervisor, names a team, group, squad or shift lead, or is at director level or above outside the individual-contributor track. Product, program, project, account, category, relationship, portfolio and vendor managers get a 0.	Placement sample
Bottom fifth of pay	Equals 1 if the predicted salary is below the 20th percentile among the firm's 2016 to 2019 outside hires into director and above who are labeled Black, Hispanic or white male (Table H.2).	External hires into director and above
Bottom fifth of previous employers	Equals 1 if the previous employer's rank is below the 20th percentile among the same hires, keeping only those who came from another employer (Table H.2).	Hires who came from another employer
Lower-tail cuts for the bounds	Two cutoffs: the 10th percentile of predicted salary among the white men the firm hired into director and above in 2016 to 2019, moves within the firm included, and the 25th percentile of the previous employer's rank among the white men it hired from other employers (Table H.3).	White men hired in 2016 to 2019

Notes: The table defines each hire-level variable and outcome and its sample. The previous employer is the employer of the latest position that started before the hire. Ranks come from the revealed-preference method of Sorkin (2018), estimated on the 32,787 employers with at least 100 moves in or out. The path-to-power score uses positions started in 2015 to 2019 at panel firms that did not pledge. The manager rule uses the title rules of Appendix B.5. The placement sample leaves out titles that mark board members, advisers, contractors, consultants to the company, former or retired employees, and volunteers. Predicted salary comes from Revelio. Source: LinkedIn profiles via Revelio Labs (November 2025 release).

TABLE 9: JOB PLACEMENT AND FUNCTIONAL ROLES OF EXTERNAL BLACK HIRES (DIRECTOR AND ABOVE)

	Black hires at pledgers after the pledge (1)	Triple difference θ (2)	Joint lead p-value (3)
DEI title (pp)	0.87	-0.42^{***} (0.15)	0.46
HR title (pp)	1.54	-0.87^{***} (0.27)	0.70
Staff rather than line function (pp)	30.17	0.63 (1.11)	0.13
Path-to-power score of the function (standard deviations)	0.467	-0.038 (0.030)	0.24
Role new to the firm, rolling definition (pp)	1.13	0.04 (0.44)	0.56
Manages people (pp)	52.28	0.28 (1.04)	0.77

Notes: The table reports how numerical race pledges changed the placement of external Black hires into director and above, relative to white-male hires. The sample is U.S. hires from outside the firm (previous job elsewhere, or none), 2016 to 2024, at the 115 numerical race pledgers and 1,063 never-pledged controls. The estimator is a stacked triple difference, Equation (3), with stack-by-firm and stack-by-year fixed effects, event years -4 to 3 and reference year -1. Column 1 gives the percent share among Black hires at pledgers in event years 0 to 3 (the mean for the path-to-power score). Column 2 gives θ, the mean of the event-year coefficients 0 to 3 of the Black minus white-male gap within firm and July 1 hire year, pledgers relative to controls, with firm-clustered standard errors. Column 3 reports the joint test of the three leads. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sources: LinkedIn profiles via Revelio Labs (November 2025 release); education-augmented BIFSG most likely race label; pledge dataset.

before and after each firm's pledge, at the pledgers relative to the controls. Specifically, the gaps in predicted pay, schooling and retention show no change distinguishable from zero. The gap in the rank of the previous employer, in contrast, narrowed by 1.8 percentile points, significant at the 1 percent level and with flat leads (Figure 9, panel C), indicating that pledging firms managed to hire Black workers from better firms. The picture is the same when I count every position start, moves within the firm included, and under the alternative designs of Appendix Table H.1.

These estimates reflect changes in gaps that already existed. Before and after 2020, Black hires into director and above entered lower-paid positions than the white men hired alongside them and had less experience, at the pledgers and at the controls alike. For example, after 2020 Black hires at the pledgers entered positions predicted to pay $13,200

less and had 2.8 fewer years since their bachelor's degree.[25]

The pledges also slightly changed how far up the ladder the Black hires had already climbed. Among hires with an earlier job on record, the share of Black hires taking their first director-level job fell by 4.0 percentage points relative to white men, significant at the 1 percent level.[26] The hires were not bid up: relative to white men, they did not climb more levels on the way in, and the predicted pay of their new position did not rise more relative to their previous one (Online Appendix Table OA.12).[27]

The two remaining coefficients cannot be read as effects of the pledges. The experience gap narrowed relative to the controls, but it had already been narrowing before the pledges and fails the lead test (Figure 9, panel B). More than half of the insignificant drop of 0.66 points in promotion within 24 months comes from event year 3, which rests on three pledgers (panel F); over event years 0 to 2 it is an insignificant -0.32.

These comparisons average over all Black hires, most of whom would have been hired without the pledges. On average, the pledges moved the predicted pay of Black hires at the top, relative to white men, by between $-$\$2,260 and \$3,040 (95 percent). Relative to white men, I find no change distinguishable from zero in their retention and promotion over 36 months or in the share entering positions in the bottom fifth of the firm's pay. The bottom-fifth test would detect a shift of a tenth of the base share with 80 percent power (Appendix Table H.2).[28] This rules out a lowered bar large enough to show in the average Black hire, but not a moderate one among the few hires the pledges added, about 7 percent of the Black hires into director and above after the pledges. At the 95 percent limits, these could still have entered positions predicted to pay about \$45,000 less than the other Black hires, or been 13 points more likely to fall in the bottom fifth of pay. Yet they came from previous employers at least as well ranked as the white men hired alongside

[25] The first four columns of Table 10 give the raw gaps, pooled over firms, before and after 2020. The gaps in experience, retention and promotion are about the same at the pledgers and at the controls, and the pay gap, about \$3,000 smaller at the pledgers, changed by the same amount at both after 2020. The schooling gap is smaller at the pledgers throughout, and the previous-employer gap was wider there before 2020 and narrowed afterwards.

[26] Among hires with an earlier job on record, 64 percent of Black hires into director and above had held a director-level job before, against 73 percent of white men. At the pledgers that share rose from 60 to 67 percent after the pledges, and at the controls from 62 to 66 percent. Among hires whose profiles cover at least three earlier years, the decline is 3.8 points.

[27] For all hires, the 95 percent intervals rule out average increases larger than 0.02 levels and 2.3 log points. Revelio predicts pay from the title, place and employer, so these estimates cannot show a premium within a title.

[28] The 36-month estimates rest on event years 0 to 2 of the 2020 to 2022 cohorts, the last year on 16 pledgers alone, and are no tighter than the 24-month ones; the retention estimate nearly fails its lead test ($p = 0.05$) because of one early cell of a single cohort. For the previous employer's rank, the same test passes the joint lead test ($p = 0.06$) but fails a linear pre-trend test ($p = 0.009$).

them (Appendix Tables H.3 and H.2).

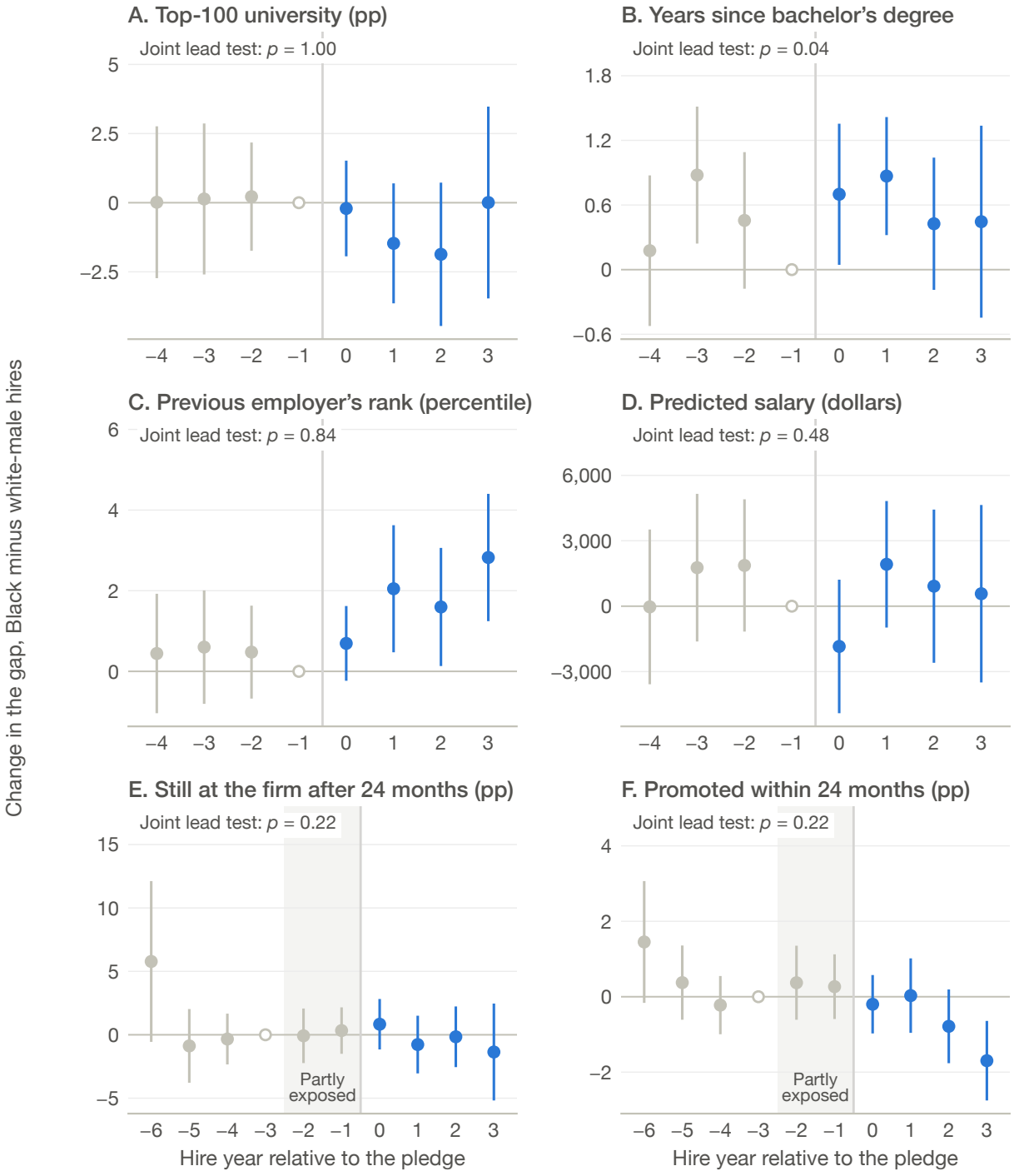
Below the top, three gaps changed within firms (Online Appendix Table OA.15 and Appendix Table H.1, which use all position starts). In the middle band, the predicted pay of Black hires fell relative to white men (5 percent level, with flat leads), as did their chance of a top-100 degree (1 percent level, although it fails its lead test). At entry, Black hires shifted into staff functions (5 percent level). The advantages of Black entry-level hires over white men in retention and promotion that appear across firms shrink by more than half within firms and vanish beyond the pre-pledge trend. The entry-level pay advantage across firms reflects which firms hired more Black workers, since within firms it is close to zero.

TABLE 10: QUALIFICATIONS, PREDICTED PAY AND CAREER OUTCOMES OF EXTERNAL BLACK HIRES (DIRECTOR AND ABOVE)

	Raw gap, Black minus white men				Triple	
	Pledgers		Controls		difference	Lead
	Before	After	Before	After	θ	p-value
Top-100 university (pp)	−4.25	−4.07	−6.86	−5.89	−0.88	1.00
					(1.09)	
Years since bachelor's degree	−3.34	−2.85	−3.55	−3.00	0.61**	0.04†
					(0.28)	
Previous employer's rank (percentile, movers only)	−4.19	−3.14	−2.69	−2.46	1.79***	0.84
					(0.57)	
Predicted salary (dollars)	−14,217	−13,204	−17,362	−16,282	391	0.48
					(1,352)	
Still at the firm after 24 months (pp)	−2.42	−2.54	−2.49	−2.39	−0.36	0.22
					(1.00)	
Promoted within 24 months (pp)	−0.08	−0.15	−0.32	−0.38	−0.66	0.22
					(0.41)	
External Black hires	19,601	31,589	26,468	34,085		
External white-male hires	89,907	104,275	136,564	153,548		

Notes: The table reports the gap between external Black and white-male hires into director and above in six outcomes. The sample is U.S. hires from outside the firm (previous job elsewhere, or none), 2016 to 2024, at the 115 numerical race pledgers and 1,063 never-pledged controls. Raw gaps are Black minus white-male means for 2016 to 2019 and 2020 to 2024 (24-month outcomes to 2022). The estimator is a stacked triple difference, Equation (3), with stack-by-firm and stack-by-year fixed effects, event years −4 to 3 and reference year −1 (for 24-month outcomes, event years −6 to 3 and reference year −3). θ is the mean of the event-year coefficients 0 to 3 of that gap within firm and July 1 hire year, pledgers relative to controls, with firm-clustered standard errors. Lead p-values come from the joint test of the three leads, and † marks a lead test that rejects at 5 percent. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sources: LinkedIn profiles via Revelio Labs (November 2025 release); education-augmented BIFSG most likely race label; pledge dataset.

FIGURE 9: CHARACTERISTICS AND CAREER OUTCOMES OF EXTERNAL BLACK VS. WHITE-MALE HIRES (DIRECTOR AND ABOVE)



Notes: The figure plots coefficients of the stacked triple difference, Equation (3), with stack-by-firm and stack-by-year fixed effects, for six outcomes of external hires into director and above. Each coefficient is the change in the gap between Black and white-male hires in the same firm and hire year, at pledgers relative to controls; bars are 95 percent confidence intervals based on firm-clustered standard errors. Hires are dated by the first July 1 on or after their start. Event years run from −4 to 3 with reference year −1; for the two 24-month outcomes they run from −6 to 3 with reference year −3, and the shaded years are partly exposed to the pledge. Sources: LinkedIn profiles via Revelio Labs; pledge dataset.

7 Net New Black Directors

Section 5.4 showed that the pledgers raised their Black share at director level and above mostly by hiring from outside the firm, and that these hires came from many kinds of employers, with only a small part from the controls. A Black worker hired away from a director job at another firm leaves that seat behind, so a gain at the pledgers does not need to be a gain for the U.S. labor market as a whole. In this section, I count how many of the Black directors the pledges added were new to the economy.

Table 11 splits the increase at director level and above by the job that each new Black director held just before. Of every 100 Black directors added, 21 were promoted inside the firm and 79 were hired from outside. Since 34 of these outside hires were not directors in their previous job, about 55 in 100 were new to the director level. The other 45 came straight from a director-level job at another firm (95 percent interval 28 to 83), and few of them moved up a level on the way in. Such moves are not new, and they are as common at the controls. At the pledgers, 39 percent of the Black outside hires into director and above came straight from a director-level job before the pledges and 42 percent after, against 40 and 43 percent at the controls. Counted by heads, about four in ten of the added outside hires also came from a director-level job. If I instead count anyone who had ever held a director-level job, outside hires who had been directors make up 70 of every 100 added, and half of the inside promotions went to people who had been directors at another firm (Online Appendix OA.D.2).

Whether the pledges created Black directors or moved them therefore depends on the seats the movers left behind. Because the LinkedIn data cover workers at every U.S. employer, these seats can be followed (Appendix F). Across 9,348 moves of Black directors from a director-level job elsewhere to a pledger after its pledge, the seat left behind was filled again in the same role within a year 78 percent of the time. A Black worker filled it in the same role within a year only 7.4 percent of the time (95 percent interval 6.8 to 8.0). Among the refilled seats, the Black share was no higher than the origin's usual share among its director hires (9.5 against 10.2 percent), and it was about the same when Black directors had left for pledgers before the pledges or for the controls. Firms with strong ties to the pledgers before 2020 show no detectable fall in their Black director share.

In Table 11, the inside promotions and the outside hires who were not directors just before count as new Black directors, while a hire who moved from a director job elsewhere counts only if a Black worker took the seat she left. Only a hire whose job just before was a director job leaves a director seat behind, so the count classifies hires by that job. For

every 100 Black directors the pledges added, the net number of Black directors in the U.S. labor market therefore rose by about 58 (95 percent interval 23 to 75). That is about 700 of the 1,200 added on average, and 1,330 of the 2,300 three years after the pledge. Figure 10 shows this accounting. It starts from the 100 Black directors the pledges added, takes away the 45 director seats that the movers left at other firms, and adds back the 3 of those seats that a Black worker filled. Most of the uncertainty comes from the share of hires who were already directors, since the refill rate is measured precisely. These shares come from the pooled accounting of Section 5.4, in which larger pledgers weigh more. Without Amazon, 40 of every 100 added came straight from a director-level job and about 63 were new to the economy. Counting each pledger equally gives 34 and about 69, although the share of sitting directors then has an interval that includes zero.

TABLE 11: MARKET-LEVEL DECOMPOSITION OF ADDED BLACK POSITIONS AT DIRECTOR AND ABOVE

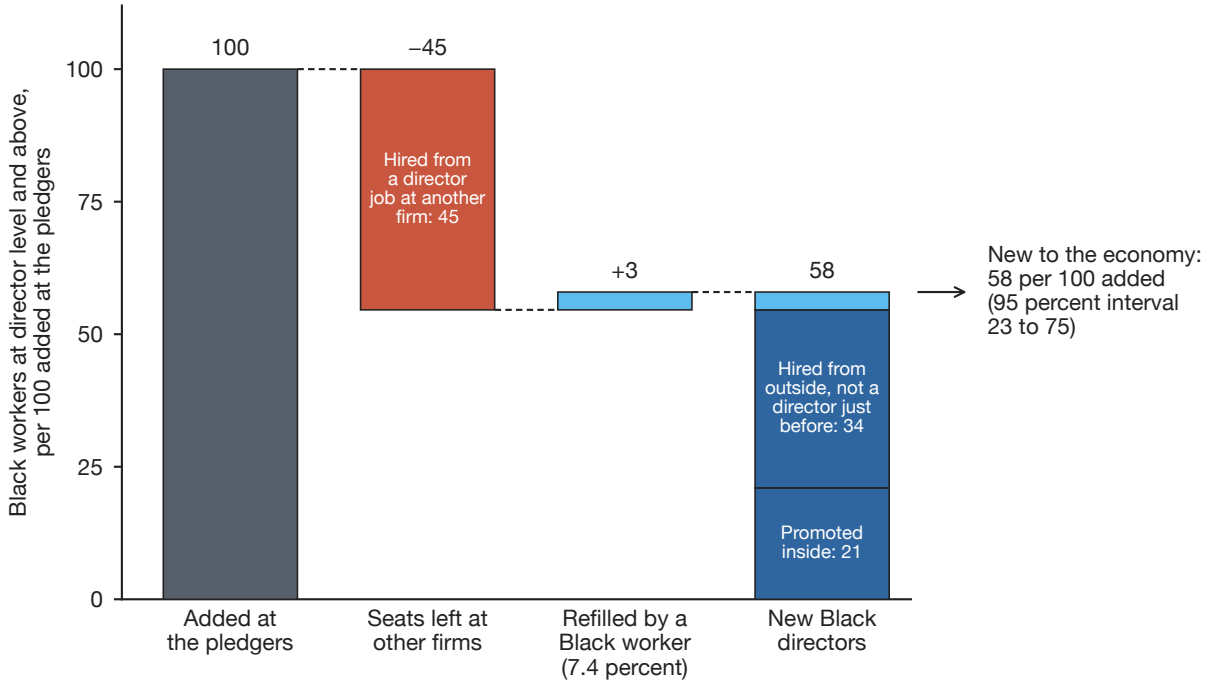
	Per 100 added	95% interval	Black workers	
			Average	Year 3
A. Routes into the added seats				
Promoted inside the firm	21	[−4, 43]	250	480
Outside hire, not a director just before	34	[−4, 53]	400	770
Outside hire, already a director	45	[28, 83]	540	1,040
of which moving up a level	6	[−1, 13]	70	140
Total	100		1,200	2,300
B. Seats left by the movers				
Moved seats refilled by a Black worker (7.4%)	3		40	80
New Black directors	58	[23, 75]	700	1,330

Notes: The table splits the Black workers the pledges added at director level and above by the job each held just before (Panel A) and counts the new Black directors (Panel B). Panel A estimates are stacked differences in differences of pledgers relative to controls, pooled across firms (Section 5.4; Appendix Table E.2). Shares are per 100 entries through inside promotions and outside hires, with 95 percent intervals from 1,000 firm-bootstrap draws, and leave out extra separations (about 5 percent of the net increase). Counts use the 1,200 added on average over event years 0 to 3 and the 2,300 in year 3. Panel B adds the moved seats that a Black worker refilled within 12 months in the same role (Appendix F.1). Sources: LinkedIn profiles via Revelio Labs (November 2025 release); pledge dataset.

Across all U.S. employers on LinkedIn, the Black share of director-level jobs rose by 0.22 points from 2019 to 2023, from 11.41 to 11.63 percent, against 0.63 points at the pledgers and 0.21 at the never-pledged index firms (Figure 11). The pledgers held 5 percent of director-level jobs and accounted for about 15 percent of the economy-wide rise, and their share of all Black directors grew from 4.75 percent in 2019 to 5.13 percent in 2024.[29]

[29]Non-pledging employers in the industries and metropolitan areas where the pledgers held more director jobs did not move in a consistent direction (Appendix F.2).

FIGURE 10: AGGREGATE INCIDENCE OF DIVERSITY PLEDGES: REALLOCATION VS. NET NEW BLACK DIRECTORS

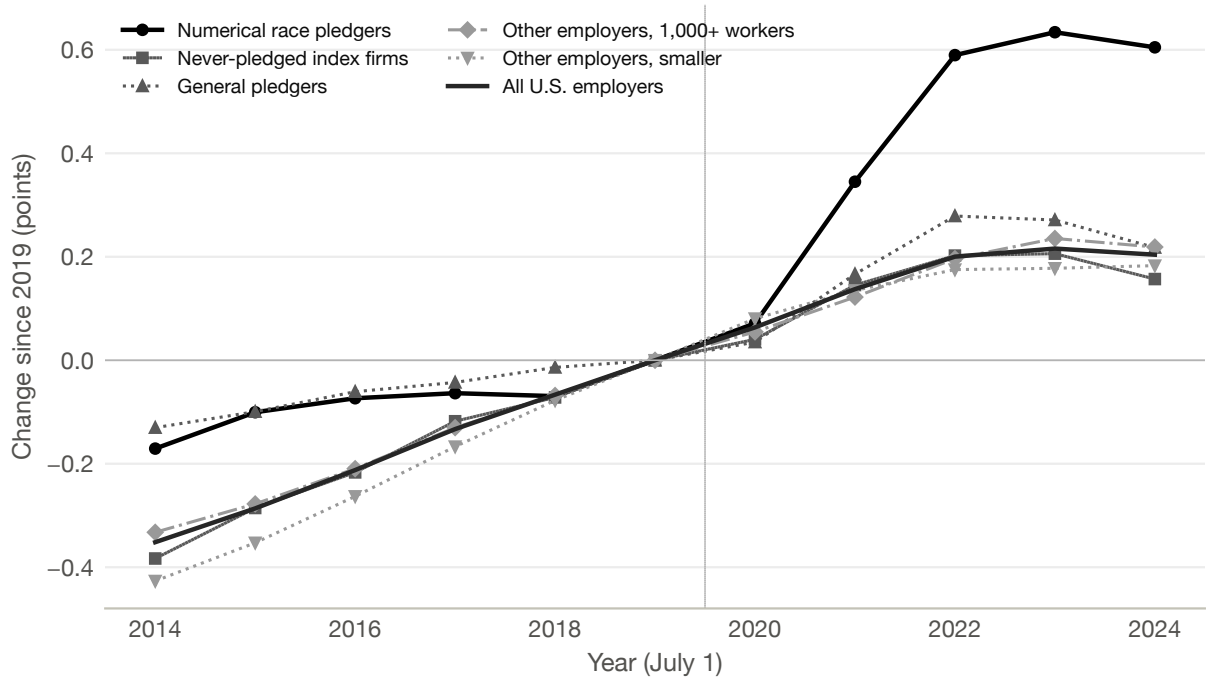


Notes: The figure classifies every 100 Black workers the pledges added at director level and above by the job each held just before and shows how many were new Black directors. Dark blue shows new Black directors and red shows hires from a director job at another firm. Light blue shows the 7.4 percent of those seats that a Black worker refilled within 12 months in the same role. Added workers are estimated as stacked differences in differences of pledgers relative to controls, pooled across firms (Section 5.4), averaged over event years 0 to 3. The 95 percent interval comes from 1,000 firm-bootstrap draws. Sources: Revelio Labs, pledge dataset.

8 Conclusion

Between 2018 and 2022, more than a hundred of the largest publicly traded U.S. companies promised to raise the share of minorities in their workforce. Using both official workforce data and LinkedIn profiles, I find that pledging firms indeed raised the share of Black people in their workforce. This increase came mostly through outside hiring and these new hires were not differentially placed in diversity or staff roles. However, following these hires back to the firms they left shows that the pledgers' own numbers overstate the change in the labor market as a whole. Of every 100 Black workers added at director level and above, 45 came straight from a director-level job at another firm. Those firms refilled the vacated seats with a Black worker only 7.4 percent of the time, so the net number of Black directors in the U.S. labor market rose by about 58 for every 100 the pledgers added.

FIGURE 11: EVOLUTION OF THE BLACK SHARE AT DIRECTOR AND ABOVE ACROSS U.S. EMPLOYERS ON LINKEDIN (2014 TO 2024)



Notes: The figure plots the change since July 1, 2019, in the share of Black workers at director level and above, by group of employers, in percentage points. Shares are weighted by workers and use the education-augmented BIFSG posterior (Section 3.4). Pledgers are the 115 numerical race pledgers. General pledgers made public diversity commitments without a numerical race target. Other employers (those outside the S&P 1500) are split at 1,000 U.S. workers on LinkedIn, using each employer's largest count from 2014 to 2024. Pseudo-employers such as "Retired" are excluded. Sources: Revelio Labs, pledge dataset.

These findings speak to two classic theories of affirmative action. First, I find no sign of a lowered bar like it's suggested in Coate and Loury (1993). After the pledges, the gaps between Black hires into director level and above and the white men hired alongside them did not widen. Second, this reallocation of sitting directors is the type of sorting that Welch (1976) would have predicted. In this episode, it accounts for about 42 of every 100 Black workers the pledgers added at director level and above.

My findings imply that firm-level estimates of corporate diversity initiatives overstate their aggregate effect. For the firm's purposes, a minority hired away from another firm counts the same as one promoted from within. In this episode, counting every added Black director as new would overstate the increase in the labor market by about 70 percent. Since an internal pipeline takes years to build and a sitting director can be hired at once, firms end up hiring them from similar positions in similar firms. Targets and disclosures defined over promotions, and over hires from below the director level, would

instead credit only the margins that add Black directors one for one, which in this episode produced 55 of the 58.

By their 2025 filings, nearly every pledger had softened or withdrawn its target. At the first snapshot after these retreats, the Black share at the pledgers had not fallen, although half of them are observed for only one or two quarters after retreating.

References

Arrow, Kenneth J. 1973. "The Theory of Discrimination." In *Discrimination in Labor Markets*, edited by Orley Ashenfelter, and Albert Rees, 3–33, Princeton, NJ: Princeton University Press.

Baker, Andrew C, David F Larcker, Charles G McClure, Durgesh Saraph, and Edward M Watts. 2024. "Diversity Washing." *Journal of Accounting Research* 62 (5): 1661–1709.

Balakrishnan, Karthik, Rafael Copat, Daniela De la Parra, and K. Ramesh. 2023. "Racial Diversity Exposure and Firm Responses Following the Murder of George Floyd." *Journal of Accounting Research* 61 (3): 737–804. 10.1111/1475-679X.12484.

Bayer, Patrick, and Kerwin Kofi Charles. 2018. "Divergent Paths: A New Perspective on Earnings Differences Between Black and White Men Since 1940." *Quarterly Journal of Economics* 133 (3): 1459–1501.

Becker, Gary S. 1957. *The Economics of Discrimination*. Chicago: University of Chicago Press.

Berry, Alexander, Molly Maloney, and David Neumark. 2026. "The Missing Link? Using LinkedIn Data to Measure Race, Ethnic, and Gender Differences in Employment Outcomes at Individual Companies." In *Race, Ethnicity, and Economic Statistics for the 21st Century*, edited by Randall Akee, Lawrence F. Katz, and Mark A. Loewenstein, University of Chicago Press, NBER Working Paper 32294, 2024.

Bertrand, Marianne, Sandra E. Black, Sissel Jensen, and Adriana Lleras-Muney. 2019. "Breaking the Glass Ceiling? The Effect of Board Quotas on Female Labour Market Outcomes in Norway." *Review of Economic Studies* 86 (1): 191–239.

Bertrand, Marianne, Rema Hanna, and Sendhil Mullainathan. 2010. "Affirmative Action in Education: Evidence from Engineering College Admissions in India." *Journal of Public Economics* 94 (1–2): 16–29.

Bertrand, Marianne, and Sendhil Mullainathan. 2004. "Are Emily and Greg More Employable Than Lakisha and Jamal? A Field Experiment on Labor Market Discrimination." *American Economic Review* 94 (4): 991–1013.

Bezrukova, Katerina, Chester S. Spell, Jamie L. Perry, and Karen A. Jehn. 2016. "A Meta-Analytical Integration of Over 40 Years of Research on Diversity Training Evaluation." *Psychological Bulletin* 142 (11): 1227–1274. 10.1037/bul0000067.

Bhagwat, Yashoda, Nooshin L Warren, Joshua T Beck, and George F Watson IV. 2020. "Corporate Sociopolitical Activism and Firm Value." *Journal of Marketing* 84 (5): 1–21.

Bleemer, Zachary. 2022. "Affirmative Action, Mismatch, and Economic Mobility after California's Proposition 209." *The Quarterly Journal of Economics* 137 (1): 115–160.

Bloom, Nicholas, Gideon Moore, Lisa K. Simon, and Caelan Wilkie-Rogers. 2026. "Time Travel on Professional Profiles." NBER Working Paper35546, National Bureau of Economic Research.

Bock, Laszlo. 2014. "Getting to Work on Diversity at Google." Blog post, 28 May 2014. https://blog.google/topics/diversity/getting-to-work-on-diversity-at-google/.

Bolton, Patrick, and Marcin Kacperczyk. 2026. "Firm Commitments." *Management Science* 72 (3): 2134–2167. 10.1287/mnsc.2024.07298.

Borusyak, Kirill, Xavier Jaravel, and Jann Spiess. 2024. "Revisiting Event-Study Designs: Robust and Efficient Estimation." *Review of Economic Studies* 91 (6): 3253–3285.

Bourveau, Thomas, Rachel W. Flam, and Anthony Le. 2026. "Behind the Curtain of Workforce Diversity: Evidence from EEO-1 Reports." *Management Science* 72 (2): 1699–1726. 10.1287/mnsc.2024.07432.

Cai, Wei, Yue Chen, Shiva Rajgopal, and Li Azinovic-Yang. 2024. "Diversity Targets." *Review of Accounting Studies* 29 (3): 2157–2208. 10.1007/s11142-024-09831-x.

Callahan, Chrissy. 2020. "'Pull Up or Shut Up' Campaign Urges Beauty Brands to Reveal Staff Diversity." TODAY, June 8, 2020. https://www.today.com/today/amp/tdna183593.

Callaway, Brantly, and Pedro H C Sant'Anna. 2021. "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics* 225 (2): 200–230.

Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer. 2019. "The Effect of Minimum Wages on Low-Wage Jobs." *Quarterly Journal of Economics* 134 (3): 1405–1454.

de Chaisemartin, Clément, and Xavier D'Haultfœuille. 2020. "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects." *American Economic Review* 110 (9): 2964–2996.

Chang, Edward H., Katherine L. Milkman, Dena M. Gromet, Robert W. Rebele, Cade Massey, Angela L. Duckworth, and Adam M. Grant. 2019. "The Mixed Effects of Online Diversity Training." *Proceedings of the National Academy of Sciences* 116 (16): 7778–7783. 10.1073/pnas.1816076116.

Chen, AJ Yuan, Patricia M. Dechow, and Samuel T. Tan. 2026. "Corporate Response to the Black Lives Matter Movement: Determinants of Speaking Out in Support of Social Causes." *Review of Accounting Studies* 31 (2): 1245–1300. 10.1007/s11142-026-09951-6.

Chetty, Raj, Nathaniel Hendren, Maggie R. Jones, and Sonya R. Porter. 2020. "Race and Economic Opportunity in the United States: An Intergenerational Perspective." *Quarterly Journal of Economics* 135 (2): 711–783.

Coate, Stephen, and Glenn C Loury. 1993. "Will Affirmative-Action Policies Eliminate Negative Stereotypes?" *American Economic Review* 83 (5): 1220–1240.

Crépon, Bruno, Esther Duflo, Marc Gurgand, Roland Rathelot, and Philippe Zamora. 2013. "Do Labor Market Policies have Displacement Effects? Evidence from a Clustered Randomized Experiment." *Quarterly Journal of Economics* 128 (2): 531–580. 10.1093/qje/qjt001.

Derby, Elena, Connor Dowd, and Jacob Mortenson. 2025. "Statistical Bias in Racial and Ethnic Disparity Estimates Using BIFSG." *National Tax Journal* 78 (4): 919–938.

Derenoncourt, Ellora, Chi Hyun Kim, Moritz Kuhn, and Moritz Schularick. 2024. "Wealth of Two Nations: The U.S. Racial Wealth Gap, 1860–2020." *Quarterly Journal of Economics* 139 (2): 693–750.

Edmans, Alex, Caroline Flammer, and Simon Glossner. 2023. "Diversity, Equity, and Inclusion." *ECGI Working Paper Series in Finance*.

Elliott, Marc N, Peter A Morrison, Allen Fremont, Daniel F McCaffrey, Philip Pantoja, and Nicole Lurie. 2009. "Using the Census Bureau's Surname List to Improve Estimates of Race/Ethnicity and Associated Disparities." *Health Services and Outcomes Research Methodology* 9 (2): 69–83.

Elzayn, Hadi, Evelyn Smith, Thomas Hertz, Cameron Guage, Arun Ramesh, Robin Fisher, Daniel E. Ho, and Jacob Goldin. 2025. "Measuring and Mitigating Racial Disparities in Tax Audits." *Quarterly Journal of Economics* 140 (1): 113–163.

Evsyukova, Yulia, Felix Rusche, and Wladislaw Mill. 2025. "LinkedOut? A Field Experiment on Discrimination in Job Network Formation." *Quarterly Journal of Economics* 140 (1): 283–334.

Ferracci, Marc, Grégory Jolivet, and Gerard J. van den Berg. 2014. "Evidence of Treatment Spillovers Within Markets." *Review of Economics and Statistics* 96 (5): 812–823. 10.1162/REST_a_00428.

Flammer, Caroline. 2015. "Does Corporate Social Responsibility Lead to Superior Financial Performance? A Regression Discontinuity Approach." *Management Science* 61 (11): 2549–2568.

Fryer, Roland G., and Steven D. Levitt. 2004. "The Causes and Consequences of Distinctively Black Names." *Quarterly Journal of Economics* 119 (3): 767–805.

Gautier, Pieter, Paul Muller, Bas van der Klaauw, Michael Rosholm, and Michael Svarer. 2018. "Estimating Equilibrium Effects of Job Search Assistance." *Journal of Labor Economics* 36 (4): 1073–1125. 10.1086/697513.

Giuliano, Laura, David I. Levine, and Jonathan Leonard. 2009. "Manager Race and the Race of New Hires." *Journal of Labor Economics* 27 (4): 589–631.

Goodman-Bacon, Andrew. 2021. "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics* 225 (2): 254–277.

Gormley, Todd A., Vishal K. Gupta, David A. Matsa, Sandra C. Mortal, and Lukai Yang. 2023. "The Big Three and Board Gender Diversity: The Effectiveness of Shareholder Voice." *Journal of Financial Economics* 149 (2): 323–348. 10.1016/j.jfineco.2023.04.001.

Hinrichs, Peter. 2012. "The Effects of Affirmative Action Bans on College Enrollment, Educational Attainment, and the Demographic Composition of Universities." *Review of Economics and Statistics* 94 (3): 712–722.

Holzer, Harry J., and David Neumark. 1999. "Are Affirmative Action Hires Less Qualified? Evidence from Employer-Employee Data on New Hires." *Journal of Labor Economics* 17 (3): 534–569.

Holzer, Harry J., and David Neumark. 2000a. "Assessing Affirmative Action." *Journal of Economic Literature* 38 (3): 483–568.

Holzer, Harry J, and David Neumark. 2000b. "What Does Affirmative Action Do?" *Industrial and Labor Relations Review* 53 (2): 240–271.

Hsieh, Chang-Tai, Erik Hurst, Charles I. Jones, and Peter J. Klenow. 2019. "The Allocation of Talent and U.S. Economic Growth." *Econometrica* 87 (5): 1439–1474.

Hu, Yingyao. 2008. "Identification and Estimation of Nonlinear Models with Misclassification Error Using Instrumental Variables: A General Solution." *Journal of Econometrics* 144 (1): 27–61.

Imai, Kosuke, and Kabir Khanna. 2016. "Improving Ecological Inference by Predicting Individual Ethnicity from Voter Registration Records." *Political Analysis* 24 (2): 263–272.

Imai, Kosuke, Santiago Olivella, and Evan T. R. Rosenman. 2022. "Addressing Census Data Problems in Race Imputation via Fully Bayesian Improved Surname Geocoding and Name Supplements." *Science Advances* 8 (49): eadc9824.

Imbens, Guido W., and Jeffrey M. Wooldridge. 2009. "Recent Developments in the Econometrics of Program Evaluation." *Journal of Economic Literature* 47 (1): 5–86. 10.1257/jel.47.1.5.

Jeffers, Jessica S. 2024. "The Impact of Restricting Labor Mobility on Corporate Investment and Entrepreneurship." *Review of Financial Studies* 37 (1): 1–44.

Kalev, Alexandra, Frank Dobbin, and Erin Kelly. 2006. "Best Practices or Best Guesses? Assessing the Efficacy of Corporate Affirmative Action and Diversity Policies." *American Sociological Review* 71 (4): 589–617.

Kalinoski, Zachary T., Debra Steele-Johnson, Elizabeth J. Peyton, Keith A. Leas, Julie A. Steinke, and Nathan A. Bowling. 2013. "A Meta-Analytic Evaluation of Diversity Training Outcomes." *Journal of Organizational Behavior* 34 (8): 1076–1104. 10.1002/job.1839.

Kline, Patrick, Evan K Rose, and Christopher R Walters. 2022. "Systemic Discrimination Among Large U.S. Employers." *The Quarterly Journal of Economics* 137 (4): 1963–2036.

Kurtulus, Fidan Ana. 2016. "The Impact of Affirmative Action on the Employment of Minorities and Women: A Longitudinal Analysis Using Three Decades of EEO-1 Filings." *Journal of Policy Analysis and Management* 35 (1): 34–66.

Lalive, Rafael, Camille Landais, and Josef Zweimüller. 2015. "Market Externalities of Large Unemployment Insurance Extension Programs." *American Economic Review* 105 (12): 3564–3596. 10.1257/aer.20131273.

Lang, Kevin, and Jee-Yeon K. Lehmann. 2012. "Racial Discrimination in the Labor Market: Theory and Empirics." *Journal of Economic Literature* 50 (4): 959–1006.

Law, Kelvin K. F., and Jingdan Tan. 2026. "Diversity Tokenism." *Journal of Accounting Research* 64 (1): 317–363. 10.1111/1475-679X.70019.

Leonard, Jonathan S. 1990. "The Impact of Affirmative Action Regulation and Equal Employment Law on Black Employment." *Journal of Economic Perspectives* 4 (4): 47–63.

Lundberg, Shelly J, and Richard Startz. 1983. "Private Discrimination and Social Intervention in Competitive Labor Markets." *American Economic Review* 73 (3): 340–347.

Mahajan, Aprajit. 2006. "Identification and Estimation of Regression Models with Misclassification." *Econometrica* 74 (3): 631–665.

Mercan, Yusuf, and Benjamin Schoefer. 2020. "Jobs and Matches: Quits, Replacement Hiring, and Vacancy Chains." *American Economic Review: Insights* 2 (1): 101–124. 10.1257/aeri.20190023.

Miller, Conrad. 2017. "The Persistent Effect of Temporary Affirmative Action." *American Economic Journal: Applied Economics* 9 (3): 152–190.

Muralidharan, Karthik, Paul Niehaus, and Sandip Sukhtankar. 2023. "General Equilibrium Effects of (Improving) Public Employment Programs: Experimental Evidence From India." *Econometrica* 91 (4): 1261–1295. 10.3982/ECTA18181.

Nichols, Shaun. 2014. "Tim Cook on Pale, Male Apple: 'As CEO, I'm NOT Satisfied'." The Register, 13 August 2014. https://www.theregister.com/2014/08/12/apple_diversity_report/.

Office of the New York City Comptroller. 2020. "Comptroller Stringer and Three New York City Retirement Systems Call on 67 S&P 100 Companies Who Issued Supportive Statements on Racial Equality to Publicly Disclose the Composition of Their

Workforce." Press release, July 1, 2020. https://comptroller.nyc.gov/newsroom/comptroller-stringer-and-three-new-york-city-retirement-systems-call-on-67-sp-100-com

Onorato, Donato, Kamelia Stavreva, Adina Sterling, and Emmanuel Yimfor. 2026. "From Callbacks to Careers: Do Audit Studies Predict Workforce Diversity?", research paper, Columbia Business School. 10.2139/ssrn.6022958.

Pajuste, Anete, Maksims Dzabarovs, and Romans Madesovs. 2024. "Boardroom Racial Diversity: Evidence from the Black Lives Matter Protests." *Corporate Governance: An International Review* 32 (2): 170–189. 10.1111/corg.12497.

Phelps, Edmund S. 1972. "The Statistical Theory of Racism and Sexism." *American Economic Review* 62 (4): 659–661.

Raghunandan, Aneesh, and Shiva Rajgopal. 2024. "Do Socially Responsible Firms Walk the Talk?" *Journal of Law and Economics* 67 (4): 767–810. 10.1086/728855.

Rambachan, Ashesh, and Jonathan Roth. 2023. "A More Credible Approach to Parallel Trends." *Review of Economic Studies* 90 (5): 2555–2591.

Revelio Labs. 2025. "Revelio Labs Workforce Intelligence Platform." Individual-level position, education, and demographics data. November 2025 release. https://www.reveliolabs.com.

Rosenman, Evan T. R., Santiago Olivella, and Kosuke Imai. 2023. "Race and Ethnicity Data for First, Middle, and Surnames." *Scientific Data* 10 299.

Solon, Gary, Steven J. Haider, and Jeffrey M. Wooldridge. 2015. "What Are We Weighting For?" *Journal of Human Resources* 50 (2): 301–316.

Sorkin, Isaac. 2018. "Ranking Firms Using Revealed Preference." *Quarterly Journal of Economics* 133 (3): 1331–1393.

Stainback, Kevin, and Donald Tomaskovic-Devey. 2009. "Intersections of Power and Privilege: Long-Term Trends in Managerial Representation." *American Sociological Review* 74 (5): 800–820. 10.1177/000312240907400506.

Sun, Liyang, and Sarah Abraham. 2021. "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects." *Journal of Econometrics* 225 (2): 175–199.

U.S. Department of Labor. 2024. "Affirmative Action Programs, 41 C.F.R. Part 60-2." Code of Federal Regulations, Title 41, Part 60-2 (2024 edition); placement goals in § 60-2.16.

U.S. Equal Employment Opportunity Commission. 2018. "2018 Job Patterns for Minorities and Women in Private Industry (EEO-1)." National table. Data source: Employer Information Reports (EEO-1 Single and Establishment Reports), 2018. https://www.eeoc.gov/statistics/employment/jobpatterns/eeo1/2018/national/table, accessed 25 October 2026.

Voicu, Ioan. 2018. "Using First Name Information to Improve Race and Ethnicity Classification." *Statistics and Public Policy* 5 (1): 1–13.

Weise, Karen. 2020. "Amazon Employees Want More Done to Address Racial Inequalities at Work." The New York Times, June 24, 2020.

Welch, Finis. 1976. "Employment Quotas for Minorities." *Journal of Political Economy* 84 (4, Part 2): S105–S141.

White, Harrison C. 1970. *Chains of Opportunity: System Models of Mobility in Organizations*. Cambridge, MA: Harvard University Press.

Williams, Maxine. 2014. "Building a More Diverse Facebook." Newsroom post, 25 June 2014. https://about.fb.com/news/2014/06/building-a-more-diverse-facebook/.

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A The Pledge Dataset

A.1 Sources and Candidate Firms

The universe is the S&P 1500 constituent snapshot, with each firm matched to its ultimate parent in the workforce data. Out of that sample, I decided to drop the following three firms: Sodexo, which is not an index member, Sylvamo, which has no pre-period workforce, and a mismatched LinkedIn record under Amazon's ticker. This leaves 1,449 firms.

A firm becomes a candidate if any of five sources shows a diversity commitment. The first source is the signatory lists of nine industry coalitions formed during the sample period:

- CEO Action for Diversity & Inclusion
- Paradigm for Parity
- White House Tech Inclusion Pledge
- OneTen
- Black Economic Alliance
- Catalyst CEO Champions for Change
- Just Capital Racial Equity Tracker
- As You Sow Racial Equity Audit campaign
- Bloomberg Gender-Equality Index

Signing one of the coalitions does not suffice to make the firm a pledger. The second is a keyword search of every firm's proxy statements (DEF 14A): 9,104 filings of 1,399 firms from 2015 to 2021, and 8,462 proxy filings of the 965 control firms with an SEC identifier from 2022 to 2025. The remaining three are the firm's annual and current reports with their exhibits, its sustainability, ESG and diversity reports, and its press releases and newsroom statements, found by web search.

For each candidate, a language model reads the collected text and returns one record per workforce target under a fixed extraction schema. Claude Haiku 4.5 (Anthropic) does the bulk extraction and Claude Sonnet 4.6 reviews every record that could make a firm a pledger; both are called through structured tool use. A record holds the verbatim quote, its source document, the groups named, the number and its kind, the baseline and its

year, the deadline, the positions covered, the first announcement date stated in the text, whether executive pay depends on the target, and whether it comes with a commitment of money. Every record that could make a firm a pledger was then checked by hand against the primary text (Appendix A.4).

A dictionary of 99 terms, frozen on 9 September 2026, selects the passages that reach the extraction. It has 40 group terms (Black, African American, Hispanic, Latino, people of color, BIPOC, underrepresented minorities, ethnically diverse, and the like), 45 commitment terms (commit, pledge, aspire, target, goal, will, and the like) and 14 quantifiers (percentages, counts, dollar amounts, multipliers and year deadlines). A window of text is a hit when a group term and a commitment term fall within 300 characters of each other, and a strong hit when a quantifier accompanies them. Of 474 pledge quotes verified by hand, 98.3 percent are dictionary hits and 83.5 percent strong hits. A simpler screen, a pair of regular expressions that looks for a race or diversity term within 300 characters of a target, goal, percentage, deadline or dollar term, ignores word boundaries and flags many false hits ("relating" for Latin, "BlackRock" for Black). Of 300 windows it flagged, 42 percent are dictionary hits and 14 percent strong hits. I deduplicate the windows and queue them for extraction.

A.2 Classification Rules

A numerical race pledge is a commitment that meets the following five conditions.[30]

(i) The target is the firm's own and covers its U.S. workforce or part of it (the whole firm, management, officers, or hiring and promotion flows), explicitly or as a firm-wide or global goal. Goals of a coalition, a parent or a subsidiary, board-only targets, and targets for suppliers, customers, communities, philanthropy, lending or marketing do not count.

(ii) The target names a racial or ethnic group of U.S. workers that includes Black or Hispanic workers: a specific group (Black, African American, Hispanic, Latino) or an umbrella group (underrepresented minorities, people of color, BIPOC, racial or ethnic minorities, women of color). Targets for Asian workers only and for nationalities do not count. A joint target for women and minorities counts only if race carries a number of its own.

[30] Each condition must be established in a primary document of the firm: a proxy statement, an annual or current report, a sustainability or diversity report, or a company press release.

(iii) The target gives a number: a share, a percentage increase, a multiple such as doubling, or a count of people holding positions. Parity with no numeric level does not count, and neither does a count of hires or program places with no share attached.

(iv) The target has a deadline: a calendar or fiscal year, or a horizon measured from a dated document (e.g., "within five years").

(v) The target is a public goal for a future year. Three things that resemble one do not count: a metric in a single year's bonus scorecard that the firm never published as a multi-year goal, a report that a past goal was met, and a commitment to a process (training, surveys, diverse candidate slates) rather than to an outcome.

Rules (a) and (b) settle two kinds of hard cases, and rules (c) to (i) show how the conditions apply to the closest cases:

(a) A goal the firm calls aspirational or an ambition counts if it meets the five conditions. After 2020 the word served mostly as a legal hedge, so excluding such goals would remove many of the most specific targets.

(b) A target published only in a sustainability report or a press release counts, provided its date can be verified. A robustness check keeps only the pledges that appear in an SEC filing.

(c) Condition (iii): the parity goals of SAIC, Exxon and Carnival set no numerical level, so they are general.

(d) Condition (ii): Lumentum and CF Industries set only a joint number for "women and minorities", which is not a race target.

(e) Condition (ii): Philip Morris International's target of 20 percent Asian representation in senior roles covers no group that includes Black or Hispanic workers, so it is general.

(f) Condition (iv): the targets of Eli Lilly, General Mills and Morgan Stanley have no deadline.

(g) Condition (iii): a share of hires or promotions counts, but JPMorgan's commitment to hire 4,000 Black students is a count with no share attached.

(h) Condition (i): Astec, Inspire Medical, Acadia, Euronet, Harmonic, Azenta, Quanex and Dynatrace set race targets only for the board, so they are general. The 11 firms whose only targets concern women on the board (Thermo Fisher, Central Pacific Financial, Balchem, Repligen, Edgewell, ACI Worldwide, CoreCivic, HealthEquity,

Ryman Hospitality, Interface and BJ's Restaurants, each in a proxy statement filed between 2022 and 2024) are not pledgers, and I hold them out of every estimate (Section 4). For Live Nation, Brinker and Ventas I leave out the board targets and count the others. Ball's only race target covers its South American packaging business, so it is general.

(i) Condition (v): all of American Airlines' numbers are retrospective.

The same conditions and rules, with women in place of the racial or ethnic group in condition (ii), define a numerical gender pledge. The 29 numerical pledgers without a race target include seven firms whose race commitment is a general statement but whose gender target meets the five conditions (TransUnion, TD SYNNEX, Yum! Brands, Eastman Chemical, Autodesk, Jacobs and Dow). They do not include three firms whose gender targets cover only workers outside the United States (DXC Technology, Aflac and Northern Trust). Of the race pledgers in the main estimation sample, 94 name the United States explicitly, 14 state no geography but use U.S. race categories, and 7 set global or multi-region targets that include the United States.

A.3 Dating and the Estimation Samples

I date a pledger by the day of the first public announcement of its earliest qualifying race target. A pledger with no race target is dated by its first gender target; for most of these only the year is known, and I set the date to 1 January of that year. The source can be a press release, a report, a proxy statement or a news article reporting the firm's announcement. A restatement does not reset the date, while an earlier qualifying target moves it back. Reports are dated by their release date, which can differ from the year on the cover. The cohort is the year of the first July 1 on or after the date. A firm that announced on or before 1 July 2022 is treated, however its pledge was found. A firm whose first announcement came later is held out as having pledged after the window. The exceptions are Domino's, WEX and IDEXX: their first announcements fall in the second half of 2022, and I code them as pledgers with the 2023 cohort. Neither group enters the estimates. For robustness checks, I keep a second date, that of the first SEC filing that mentions the pledge.

Two further rules make a firm's years before the pledge comparable to its years after it. I exclude firms created inside their event window, such as a spin-off whose earlier years in the panel belong to another workforce. The one such firm in the data is Sylvamo, spun

off from International Paper in 2021. Mergers and divestitures inside the window are documented and handled with the same rules for treated and control firms (Appendix G).

The final re-audit added 17 firms to the window or re-dated them within it, and these firms went through a blind audit. Its readers were not told about the pattern in the data. They confirmed 13 and corrected 4: HP and International Paper to the 2020 cohort, Progressive to 2021 and BNY Mellon to 2022. The main estimation sample of Section 4.1 keeps the 115 firms with the same classification and cohort in the first complete verification and in the final re-audit. Three of them (HP, International Paper and BNY Mellon) had their cohort changed in an intermediate round and restored by the blind audit. A stricter sample of 111 firms also requires that no audit disputed the date.

The consensus rule compares two rounds of the classification. The first complete verification is the classification at the point when every pledger had been read against its primary text and the hardest cases had been read again by three readers (Appendix A.4). It had 129 race pledgers with cohorts 2018 to 2022 and 1,076 controls. The final re-audit went over that classification again. Three independent readers and a tiebreak re-read 76 of the 129 pledgers. Of these, 72 held, 13 of them with a new date (11 within the window; Iron Mountain and Airbnb were dated to 2016 and leave it), and 4 became general pledgers (Leidos, Marathon Petroleum, 3M and United Airlines). A screen of all 1,076 controls found ten with a target that gives a number for a racial or ethnic group. Six had pledged within the window and two had pledged after it; the targets of Philip Morris International and Ball are general under rules (e) and (h) of Appendix A.2. The 17 firms that these two steps added to the window or re-dated within it are the ones that went through the blind audit above. One or two readers, with a judge for disputed cases, then checked the dates of the other pledgers with cohorts from 2017 to 2023 and of the race pledgers of the 2023 cohort that were at the time held out as having pledged after the window. This step skipped 49 firms whose announcement was dated to the day and fell more than 45 days from 1 July. It moved five of the held-out race pledgers into the window, and three others to the general pledgers, since none of the three had a qualifying race target. A partial second screen, of 352 controls, added XPO and Masco to the window and Omnicom to the firms held out as having pledged after the window. A separate review of the held-out firms made general pledgers of the four whose only race targets concerned the board. After these steps, the final re-audit has 136 pledgers in the window and 1,063 controls.

Table A.1 reconciles this classification with the estimation samples and comparison groups of Sections 5 and 6. It does not list the hire-level subsamples or the robustness samples of Appendix G. Of the 157 race pledgers, 14 are held out as having pledged after

the window and seven have cohorts outside 2018 to 2022: Starbucks and Intel (2015), Iron Mountain and Airbnb (2017), and Domino's, WEX and IDEXX (2023). Of the 136 pledgers in the window, the final re-audit added 13 (nine former controls and four firms previously dated after the window; eight of the 13 have the 2022 cohort) and re-dated eight: Caesars Entertainment, Etsy, Marriott, Owens Corning, Progressive, Prudential, RTX and Union Pacific. The other 115 form the main sample. Panel B starts from the first complete verification, where 14 of the 129 pledgers in the window are not in the main sample: the four new general pledgers, Iron Mountain and Airbnb, and the eight re-dated firms. The strict sample leaves out four of the 115. Three are the firms that the blind audit restored. The fourth is Meta, whose stored date the last dating step judged too late, although the earlier announcement it found falls in the same cohort. The triple differences on hires and the retreat designs of Appendix E.1 use the same 115 pledgers and, where they need controls, the same 1,063.

The other samples follow from the classification of the index in Table 2. The 193 general pledgers are 186 firms whose commitments were all general statements and seven firms with a general race statement and a numerical gender target (TransUnion, TD SYNNEX, Yum! Brands, Eastman Chemical, Autodesk, Jacobs and Dow). Section 3.1 counts these seven among the 29 numerical pledgers without a race target. The 22 firms classified as having targets for women only are held out of the race comparisons. Three of them, DXC Technology, Northern Trust and Aflac, set gender targets only for workers outside the United States, so Section 3.1 does not count them as numerical gender pledgers (Appendix A.2). The placebo of Table 5 uses the 39 firms of the main sample that set no numerical gender target. The 17 firms held out as having pledged after the window are the 14 with a race target and three with a gender target only (Newmont, Chart Industries and Avery Dennison).

A.4 Verification and Validation

Each pledger's classification, source and date were read against the primary text, and the quote was checked to belong to the firm. The firms whose status or date was hardest to establish, together with a random sample of the rest, were then read again by three independent readers. One checked the source, one the five conditions and the date, and one tried to refute the classification; a fourth resolved disagreements. One round of this check covered 51 firms: the three readers agreed on 36, and the fourth decided the other 15. The round included 12 randomly drawn firms that the first two passes had classified

TABLE A.1: RECONCILIATION OF THE ANALYSIS SAMPLES

	Firms	Used in
A. From the race pledgers to the main sample		
Numerical race pledgers, any date	157	Section 3.1
Pledged after 1 July 2022, except Domino's, WEX and IDEXX	−14	Held out
Cohorts 2015 and 2017	−4	Dropped
Cohort 2023	−3	Dropped
Cohorts 2018 to 2022	136	Robustness, Section 5.2
Classification or cohort changed by the final re-audit	−21	Pre-trend, Section 5.2
Added	13	
Re-dated	8	
Consensus sample (main sample)	115	Sections 5 and 6
Date disputed in an audit	−4	
Strict consensus sample	111	Robustness, Section 5.2
B. The consensus rule		
Cohorts 2018 to 2022 in the first complete verification	129	
Reclassified as general pledgers	−4	
Re-dated outside 2018 to 2022	−2	
Re-dated within 2018 to 2022	−8	
Same classification and cohort in the final re-audit	115	
C. Subsets of the main sample		
Cohorts 2018 to 2021	83	Event years 3 and 4; Table 5
Cohorts 2018 to 2020	20	Event year 5, Figure 5
Cohorts 2021 and 2022	95	New employers, Section 5.4
No numerical gender target	39	Placebo, Table 5
D. Other groups of the index		
Never-pledged controls	1,063	Every comparison
General pledgers	193	Comparison group, Table 5
General statements only	186	
General race statement, numerical gender target	7	
Targets for women only	22	Held out
Numerical pledge after 1 July 2022, except Domino's, WEX and IDEXX	17	Held out
With a race target	14	
With a gender target only	3	
Targets for the board only	11	Held out

Notes: Panel A follows the race pledgers of Section 3.1 to the main sample, and Panel B compares the pledgers with cohorts 2018 to 2022 in the first complete verification with that sample (Appendix A.3). Panel C splits the main sample and Panel D gives the other groups of Table 2. Rows with a minus sign are subtracted to give the next row that is not indented; other indented rows detail the row above them. Source: pledge dataset.

as pledgers. All 12 held, and 3 were re-dated.

Two exercises measure the quality of the stored fields. In a blinded double-coding, 100 randomly sampled pledges were coded again from their verbatim quotes alone, without access to the stored fields. Agreement is 89 percent on whether a pledge targets race ($\kappa = 0.78$) and 93 percent on gender ($\kappa = 0.86$). The numerical target matches exactly in 93 percent of cases (correlation 0.88), the deadline agrees within a year in 98 percent, and the classification as numerical agrees in 89 percent. The weakest field is the link to executive pay ($\kappa = 0.27$), which I use only to compare the pledgers that tied a target to

pay with the others (Online Appendix OA.G).

The second, a re-sourcing exercise, traced every stored quote back to its primary text. Of the 345 pledges checked, 48 appeared verbatim in SEC documents already on disk. The other 297 were traced to company pages, press releases, archived snapshots and a web search. In the end, 215 quotes (62 percent) match the source word for word, 87 differ by an elision or a few words, 25 are confirmed on group, number and deadline, and 18 (5 percent) could not be located. Eight rows had a substantive discrepancy with the source. Four assigned a percentage or a scope to the wrong level and were corrected, and two attributed another company's quote to the firm. A wider check of every race pledger confirmed these two attribution errors. It also found thirteen more firms whose race target rested on a commitment outside the workforce or on ambiguous text. I classify the fifteen firms concerned as general.

A last check compares the data with an independent catalog. The appendix of Cai et al. (2024) lists diversity targets found in firms' sustainability reports. It contains two pledgers that the proxy-based passes had missed, because their only target appeared in a standalone report. I added both and classified them under the rules above.

A.5 Screening the Controls

A control must show no commitment in any source. It appears on none of the nine coalition rosters, its proxy statements from 2015 to 2025 contain no qualifying language, and its annual reports, current reports and sustainability reports contain no target with a number and a deadline; every hit in these reports was read by hand. Every control flagged by any screen was also checked on the web. A board-diversity commitment alone does not disqualify a control.

I drew a random sample of 30 of the 362 controls whose documents use race language without a stated target. None had a numerical pledge, 13 had a general statement, and 17 had nothing that qualifies as a commitment. Their general statements escaped the screens, so these firms stayed among the controls. Misses of this kind make the controls resemble the pledgers and attenuate the estimates. Reading the 2022 to 2025 proxy statements turned up firms that pledged after the estimation window. These 17 firms are held out of every estimate. The same reading found the 11 firms whose only targets concern women on the board. They are not pledgers (rule (h) in Appendix A.2) and are held out as well.

A.6 Specificity and Ambition of the Targets

Targets also differ in how precisely they name positions. I classify the scope each target states as a named title (director, vice president, managing director, officer and similar), a generic scope ("leadership," "senior management") or the whole workforce. Just over a third of the 401 targets name a title, two fifths a generic scope and a quarter the whole workforce (Panel A of Table A.5). Race targets name a title more often than gender targets do (39.5 against 29.9 percent), and they cover the whole workforce more often (29.2 against 20.4 percent).

TABLE A.2: PLEDGING FIRMS BY INDUSTRY

		Numerical pledgers						
Sector (GICS)	Index firms	All	Race	Gender only	Main sample	General pledgers	Controls	Race pledgers (%)
Consumer Discretionary	185	32	31	1	22	13	140	17
Industrials	246	38	29	9	17	26	182	12
Financials	237	22	21	1	16	27	185	9
Information Technology	171	24	21	3	16	14	133	12
Health Care	168	19	17	2	13	16	130	10
Consumer Staples	71	14	14	0	12	13	43	20
Materials	74	19	12	7	10	8	48	16
Communication Services	42	8	8	0	7	7	27	19
Real Estate	94	3	2	1	1	11	79	2
Energy	63	3	1	2	0	10	50	2
Utilities	56	3	1	2	1	7	46	2
Not classified	42	1	0	1	0	41	0	0
Total	1,449	186	157	29	115	193	1,063	11

Notes: Number of S&P 1500 firms in each sector. Numerical pledgers are the firms with a numerical pledge covering U.S. workers made between 2015 and 2024, including those that pledged after 1 July 2022; race pledgers set a race or ethnicity target (with or without a gender target). Main sample: the race pledgers of the main estimates. General pledgers and controls as in Table 2; the seven firms whose race commitment is a general statement but whose gender target is numerical appear both under Gender only and among the general pledgers, and the 11 firms whose only targets concern the board and three firms whose gender targets cover only workers outside the United States count only among the index firms. The last column gives race pledgers as a percentage of the sector's index firms. Sectors are the Global Industry Classification Standard sectors recorded in Compustat; 42 firms, 41 of them general pledgers, have no sector code in the extract.

The targets in Table A.3 asked for a median of 1.0 point a year, as did the first race targets of 73 of the 115 main-sample pledgers that I can express as a change in points, 32 with a starting share from the firm's reports or EEO-1 filing. On LinkedIn, the targeted share had risen by a median 0.33 to 0.39 points a year in the five pre-pledge years (its Black part scaled by 1.62, or by 1.8 company-wide to 2.72 at director level and above), and 84 to 86 percent asked for a faster rise, as did 14 of the 17 company-wide targets against the

TABLE A.3: ANNUAL INCREASE REQUIRED BY THE RACE TARGETS

	Targets	Firms	25th pct.	Median	75th pct.	Above 1.5 (percent)
All targets	72	42	0.67	1.00	1.56	26
By the lowest level covered						
From entry level	31	20	0.51	0.75	1.11	13
From levels 2 to 4	18	16	0.69	1.34	1.95	39
Director and above	23	17	0.95	1.13	1.75	35
By the group named						
Umbrella term	49	34	0.75	1.00	1.67	31
Named group	23	13	0.45	0.67	1.23	17

Notes: Race targets stated as a share that also state the starting share, its year and the deadline. The required annual increase is the target share minus the starting share, divided by the years between them, in percentage points. Umbrella terms are underrepresented minorities, people of color and similar; named groups are Black, Hispanic, or both.

TABLE A.4: MINORITY GROUPS MENTIONED IN THE RACE AND ETHNICITY PLEDGES

	Race only	Race and gender	All race pledgers	Main estimation sample
A. Groups named in any numerical race target (a firm can use several)				
Black or African American	28 (57)	31 (29)	59 (38)	48 (42)
Hispanic or Latino	14 (29)	19 (18)	33 (21)	26 (23)
Asian	2 (4)	4 (4)	6 (4)	6 (5)
Native American or Indigenous	2 (4)	4 (4)	6 (4)	5 (4)
Underrepresented minorities	9 (18)	29 (27)	38 (24)	29 (25)
People of color or BIPOC	16 (33)	30 (28)	46 (29)	29 (25)
Racial or ethnic minorities	2 (4)	22 (20)	24 (15)	14 (12)
Racially or ethnically diverse	9 (18)	23 (21)	32 (20)	22 (19)
B. Each firm once, in the first row that fits				
Black and Hispanic	14 (29)	19 (18)	33 (21)	26 (23)
Black, not Hispanic	14 (29)	12 (11)	26 (17)	22 (19)
Underrepresented minorities only	2 (4)	19 (18)	21 (13)	16 (14)
Umbrella that counts Asian workers	18 (37)	58 (54)	76 (48)	51 (44)
Historically Black colleges only	1 (2)	0 (0)	1 (1)	0 (0)
Firms	49	108	157	115

Notes: Number of firms (column percent in parentheses) whose numerical race targets use each wording, coded from the verbatim quote. Panel A counts every wording a firm uses; Panel B places each firm once, in the first row that fits. The first three columns follow Figure 2 and include the firms whose race pledges came after 1 July 2022; the last is the main estimation sample.

TABLE A.5: SENIORITY SCOPE AND FEATURES OF RACE AND GENDER TARGETS

A. Seniority scope of the target (percent of the row)

	Named title	Generic scope	Whole workforce	Targets
Race targets	39.5	31.3	29.2	233
Gender targets	29.9	49.7	20.4	157
Joint targets	27.3	45.5	27.3	11
All targets	35.4	38.9	25.7	401

B. Race and gender targets of the same 103 firms (percent of targets)

	Race targets	Gender targets
Names a title	38.0	34.1
Covers the whole workforce	29.3	21.1
States a baseline share	47.3	43.9
Tied to executive pay	22.0	24.4
Targets	150	123

Notes: Panel A: the 401 numerical targets of the 186 numerical pledgers, by whether the stated scope names a title (director, vice president, officer, partner, manager), a generic band ("leadership," "senior management") or the whole workforce; rows sum to 100 up to rounding. Panel B: the 103 firms with both a race and a gender target, and the share of their 150 race and 123 gender targets with each feature.

EEO-1 trend. With firms' own figures where they match the target and LinkedIn or EEO-1 otherwise, 26 to 29 percent of the 58 targets due by 2025 had been met by the deadline or, for the 40 still running, by the last year observed, and the median had covered about half of its promise; on LinkedIn and EEO-1 alone, 9 to 14 percent had. Summed over the 73, the event study's break relative to never-pledged firms amounts to a tenth to an eighth of the promised pace.

The 103 firms with separate race and gender targets write the two alike in most respects. Similar shares name a title (38.0 and 34.1 percent), state a baseline share (47.3 and 43.9 percent) and are tied to executive pay (22.0 and 24.4 percent). The race targets cover the whole workforce more often, 29.3 percent against 21.1 percent of the gender targets (Panel B of Table A.5).

Numerical and general pledgers also differ in what they put on record. Almost every numerical pledge promised some annual public reporting. Yet by text matching of their filings, only about half of the numerical pledgers disclosed the share they were starting from and reported progress in later years; among general pledgers, 4 and 16 percent did. Numerical pledgers are also ahead on pay links (30 against 3 percent of firms), published EEO-1 reports (23 against 2) and targets limited to the senior levels (40 against 12).

Table A.2 counts firms by sector, and Table A.4 groups the wording of the race targets. Of the 157 race pledgers, 59 name Black workers explicitly, 33 of them together with Hispanic workers. No firm names Hispanic workers without Black workers.

A.7 Firm-Level Pledge Tables

Table A.6 lists the 115 race pledgers of the main estimation sample. For each firm it gives the cohort (the year of the first July 1 after the firm's first numerical race pledge), the month of that pledge, a short summary of its race targets that meet the five conditions, and the document that states them. Joint targets for women and minorities and parity goals without a level are left out, as the rules above require. When a firm later restated its target, that document can postdate the first announcement. A file with the quote, source, date, group, number, deadline and levels of every target, including the gender targets and the firms outside the main sample, is available from the author.

TABLE A.6: THE 115 RACE PLEDGERS OF THE MAIN ESTIMATION SAMPLE

Firm	Cohort	Date	Race target(s)	Source
Accenture Plc	2021	Sep 2020	By 2025: raise African American and Black share of the organization from 9% to 12% and of managing directors from 2.8% to 4.4%; raise U.S. Hispanic American and Latinx share from 9.5% to 13%.	Press release, Sep 10, 2020 (quotes CEO announcement Sep 1, 2020); DEF 14A 2021
Adobe, Inc.	2021	Sep 2020	Double underrepresented minorities in leadership positions by 2025; double Black representation as percentage of U.S. employees by 2025.	Proxy statement, 2021
Air Products & Chemicals, Inc.	2021	Oct 2020	At least 20% minority representation from a 17% base in U.S. professional and managerial population by 2025 (October 2020); raised to 30% by 2025 (November 2021).	Press release, November 2021; proxy statement, 2021
Alaska Air Group, Inc.	2021	Apr 2021	Increase racial diversity at all levels, including leadership, to at least 30% by 2025.	Press release, April 2021
Alphabet, Inc.	2020	Jun 2020	Improve leadership representation of underrepresented groups by 30% by 2025; more than double the number of Black+ Googlers at all other levels by 2025.	2021 DEF 14A proxy statement (CEO announcement Jun 17, 2020)
Altria Group, Inc.	2020	Jun 2020	Increase VPs and Directors who are Asian, Black, Hispanic or two or more races to at least 30% within five to ten years.	Corporate responsibility report, June 2020

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Table A.6, continued

Firm	Cohort	Date	Race target(s)	Source
Amazon.com, Inc.	2021	Nov 2020	Double number of Black directors and vice presidents again in 2021, after a goal to do so in 2020 was achieved; increase hiring of Black mid-level employees by 30% in 2021.	Proxy statement, Apr 2021
Arrow Electronics, Inc.	2022	Mar 2022	Grow U.S. leaders of underrepresented race and ethnicity (supervisors and above) by 0.5 percentage points in 2022 and by 2 percentage points by end of 2025.	Environmental, Social, and Governance Report, Mar 2022
AvalonBay Communities, Inc.	2021	Dec 2020	Increase under-represented minorities in leadership (officers and director-level associates and above) from 15% to 20% by 2025 and to 25% by 2030.	Press release, December 2020
Axalta Coating Systems Ltd.	2022	Apr 2022	Fill 30% of management positions in U.S. and Canada with underrepresented racial/ethnic groups by 2030.	Proxy statement, Apr 2022
Axcelis Technologies, Inc.	2022	Mar 2022	U.S. employees 10% URM by 2025 (8% in 2020); URMs in Engineering and Product Support 8% (4%); URMs in Leadership (Bands L2-L4) 5% (2.7%); all by 2025.	ESG report, October 2023
Baxter International, Inc.	2022	Jul 2021	Increase representation of ethnic minorities in leadership roles (director and above) in United States to 25% by 2030.	Corporate Responsibility Report, 2022
Becton, Dickinson & Co.	2022	Oct 2021	Improve ethnic diversity, and separately gender diversity, by 1% year over year at management and executive levels through 2030.	Press release, Oct 18, 2021
Best Buy Co., Inc.	2021	Dec 2020	Fill one of three new non-hourly corporate positions with BIPOC (Black, Latinx, Indigenous) employees by 2025.	Press release, Dec 9, 2020
Biogen, Inc.	2021	Apr 2021	Increase diversity in U.S. manager positions and above by 30% by end of 2021 (share held by Black, African American and Latinx employees, plus Asian employees where underrepresented).	Proxy statement, April 2021
BlackRock, Inc.	2020	Jun 2020	Double representation of Black senior leaders (directors and above), from 3%, by 2024; increase overall U.S. Black workforce representation by 30%, from 5%, by 2024.	Company memo by Larry Fink and Rob Kapito, Jun 22, 2020
Brinker International, Inc.	2022	Sep 2021	Increase minority representation in restaurant operations leadership from 35% (2021) to 40% by fiscal 2025.	Sustainability report, Sep 2021

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Table A.6, continued

Firm	Cohort	Date	Race target(s)	Source
Bristol Myers Squibb Co.	2021	Aug 2020	Double U.S. Black/African American executives (VP+) from 3.0% to 6.0% and Hispanic/Latino executives from 3.7% to 7.4%, from June 2020 levels, by year-end 2022.	2020 ESG Report
Brown-Forman Corp.	2020	Jun 2020	Achieve 25% people of color in United States by 2030.	Form 10-K, June 2020
Cardinal Health, Inc.	2022	Sep 2021	Increase Black and African American representation at manager level and above in the U.S. to 11% (from 5%) by 2030; all other ethnically diverse groups to 23% (from 17%) by 2030.	Proxy statement, September 2021
Cisco Systems, Inc.	2021	Sep 2020	Achieve 25% increase in Black/African American employees entry-level through manager by 2023; 75% increase in Black/AA from Director through VP+ by 2023.	Company document 'Social Justice Beliefs and Actions', released Sep 2020
Citigroup, Inc.	2019	Aug 2018	Black AVP to MD share in North America: 8.1% to 11.5% by 2025. U.S. Hispanic and Latino AVP to MD share: 13.7% to 16% by 2025. 2018 goal: U.S. Black, 6% to 8% within three years.	Reuters/SRN News, Sep 2022; Bloomberg and internal memo, Aug 2018
Conagra Brands, Inc.	2021	Feb 2021	Double representation of people of color in management and middle-management roles within five years.	Press release, February 2021
Constellation Brands, Inc.	2022	Apr 2022	Achieve 30% ethnic diversity among U.S. salaried population by fiscal 2026.	Form 10-K, Apr 2022
Danaher Corp.	2021	Nov 2020	Raise people of color among U.S. associates to 35% (from 31% in 2019) by 2025; later raised goal to 38% by 2025.	Press releases, November 2020 and October 2021
Dave & Buster's Entertainment, Inc.	2022	Mar 2022	Improve representation of BIPOC team members in corporate and field leadership by 33% by end of FY2025.	Proxy statement, May 2022
Dell Technologies, Inc.	2020	Nov 2019	Black/African American and Hispanic/Latino U.S. workforce from 13% to 25% by 2030; U.S. people leaders (managers/executives) from 10% to 15% by 2030.	FY20 Progress Made Real Report (published Jul 21, 2020); announced Nov 12, 2019
Delta Air Lines, Inc.	2021	Jan 2021	Double representation of Black officers and director-level employees by 2025 compared to 2020 baseline.	ESG Report, 2022; CEO memo, Aug 2020
Dick's Sporting Goods, Inc.	2022	Aug 2021	Grow BIPOC representation in leadership roles by 30% by 2025.	Sustainability report (2020 Purpose Playbook), August 2021

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Table A.6, continued

Firm	Cohort	Date	Race target(s)	Source
DoorDash, Inc.	2022	Oct 2021	Raise underrepresented people of color from 32% to 40% of the U.S. corporate team by 2025; from 7% to 10% in technical roles; from 12% to 20% in director level and above.	Company newsroom, Oct 4, 2021
Eaton Corp. Plc	2022	Oct 2021	Increase salaried U.S. minorities to 34% by 2030.	Press release, Oct 28, 2022 (goals first set 2021)
Ecolab, Inc.	2021	2020	Increase management-level ethnic/racial diversity to 25% by 2030.	Form 10-K, fiscal 2020 (filed February 2021)
Emerson Electric Co.	2021	2021	Double representation of U.S. minorities at leadership level by 2030.	Proxy statement, Dec 2021
Expedia Group, Inc.	2021	Feb 2021	25% of U.S. external hires from racially and ethnically under-represented groups by end of 2021.	Inclusion and Diversity Report, 2020
FirstEnergy Corp.	2021	Jan 2021	Increase underrepresented racial/ethnic employees by 30% by 2025 from 2019 baseline: overall 10% to 13%, supervisor-and-above leadership 7.5% to 9.7%.	Press release, Jan 2021; restated in 2021 proxy
FMC Corp.	2021	Feb 2021	Reach 14% Black/African American representation in the U.S. workforce by 2027.	Form 10-K, fiscal 2020 (filed February 2021)
Fortive Corp.	2021	Jun 2021	BIPOC representation: 37% of U.S. workforce by 2025 (from 34% in 2020); 36% in professional and management roles by 2025 (from 26%).	Sustainability report, June 2021
FTI Consulting, Inc.	2021	Apr 2021	Reach 120 underrepresented minority Senior Managing Directors by 2025 (more than double the 2020 level).	Press release / 2020 Corporate Sustainability Report, Nov 2021
Gap, Inc.	2020	Jun 2020	Double Black and Latinx employees at all levels in U.S. HQ offices by 2025; increase Black Store Leader representation by 50% by 2025.	Company website, 'Equality & Belonging' page (corroborated by press release, Jun 2021)
Gilead Sciences, Inc.	2021	2020	Double the percentage of Black employees at Gilead by 2025; diverse hires at least 15% of all hiring by 2025.	Proxy statement, 2021
Global Payments, Inc.	2022	Aug 2021	Increase people of color from 17% to 29% of U.S. leadership roles and from 31% to 39% of the U.S. workforce by 2025.	Global responsibility report, Aug 2021

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Table A.6, continued

Firm	Cohort	Date	Race target(s)	Source
Granite Construction, Inc.	2021	Mar 2021	Increase persons of color from 14.7% to 20% workforce-wide by 2025; increase persons of color in leadership from 17% to 20% by 2025.	Form 10-Ks, fiscal 2020 and 2021
Hasbro, Inc.	2018	Apr 2018	Reach 25% minority representation in the U.S. workforce by 2025; raise U.S. annual hiring rate of minorities to 40% by 2025.	Proxy statements, DEF 14A April 2018 and DEFA14A May 2022
Hewlett Packard Enterprise Co.	2022	Feb 2022	Double U.S. Black and Hispanic executive headcounts by 2027, from 2020 levels.	Proxy statement, Feb 2022; Form 10-K, FY2022
Hilton Worldwide Holdings, Inc.	2021	Apr 2021	25% ethnic diversity at corporate leadership levels in the U.S. by 2027.	Newsroom Q&A, April 2021
HP, Inc.	2020	Jun 2020	Double the number of Black and African American executives by 2025.	Form 10-K, December 2020
Huntington Bancshares, Inc.	2022	Oct 2021	15% ethnically diverse in middle and executive management by end of 2021 (from 13.5% at end of 2020).	American Banker, October 2021; proxy statement, 2021
Hyatt Hotels Corp.	2022	Feb 2022	Double representation of Black leaders in key leadership roles by 2025 (Hyatt outward materials: U.S. vice president-plus and general manager levels).	Form 10-K, Feb 2022
Ingersoll Rand, Inc.	2021	Feb 2021	Increase underrepresented populations in the U.S. workforce from 25% to 30% by 2025.	Form 10-K, fiscal 2020 (filed February 2021)
Ingredion, Inc.	2022	May 2022	Black, Indigenous and People of Color (BIPOC) at 38% of management level in the U.S. by 2030.	Press release (GlobeNewswire), May 2022
Intercontinental Exchange, Inc.	2022	Mar 2022	Double representation of under-represented minorities in U.S. officer population from 6% to 12% by 2027.	Proxy statement, Mar 2022
International Flavors & Fragrances, Inc.	2022	Dec 2021	Reach 40% people of color in U.S. management roles (executive, upper, middle) by 2030, up from 27.2% in 2021.	Press release, Dec 2021; ESG+ Report, Jun 2022
International Paper Co.	2020	Feb 2020	30% minority representation in U.S. salaried positions by 2030.	Form 10-K, February 2021
Intuit, Inc.	2021	Feb 2021	Increase Black, Latino, and other underrepresented ethnic minorities from 12% to 16% of total workforce by 2023.	Company blog, 2021 (confirmed in Fast Company interview, Apr 2021, and DEI Report 2023)

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Table A.6, continued

Firm	Cohort	Date	Race target(s)	Source
Janus Henderson Group Plc	2022	Aug 2021	Increase racially and ethnically diverse employees in senior management positions from 11% to 16% by 2023.	DEF 14A proxy statement, Mar 24, 2022
JetBlue Airways Corp.	2021	Feb 2021	Double race and ethnic minority representation at officer and director level from 12.5% to 25% by 2025.	Press release, Feb 2021
Johnson & Johnson	2021	Nov 2020	Achieve 50% growth of African American talent at manager and above levels in United States by 2025.	Proxy statement, Mar 2021
Johnson Controls International Plc	2021	Jan 2021	Double the representation of minority leaders in the United States within five years (by 2026).	Press release, January 2021
Keurig Dr Pepper, Inc.	2021	Jun 2021	Increase by 25% the representation of people of color in Director and above roles (U.S.) by 2025 (from 17% in 2020).	Press release, June 2021
KeyCorp	2022	Feb 2022	Increase representation of people of color in senior leadership by 25% over five years (by 2026) and by 50% over ten years (by 2031), from 2021.	2022 ESG Report / investor relations press release, May 2023
Keysight Technologies, Inc.	2021	May 2021	45% of U.S. new hires underrepresented minorities by fiscal year 2021; increase to 47.4% of U.S. new hires by fiscal year 2022.	CSR press release, May 17, 2021 and Form 10-K (FY2021)
Lincoln National Corp.	2021	Sep 2020	Grow minority representation at Officer level (Assistant Vice President and above) by 50% by 2023, with focus on Black officers.	Press release, Sep 2020
Live Nation Entertainment, Inc.	2021	Jul 2020	Double Black leadership (Director and above) by 2025; 30% racially/ethnically diverse U.S. leadership (Director and above) by 2025.	Proxy statement, 2021
lululemon athletica, Inc.	2021	2021	40% racially diverse representation for stores and 30% racially diverse representation of directors and assistant store managers and above, by 2023.	Demographic survey findings report, 2020 (undated)
Macy's, Inc.	2020	Sep 2019	Achieve 30% ethnic diversity at senior director level and above by 2025.	Press release via BusinessWire, Sep 10, 2019
Mastercard, Inc.	2020	Jun 2020	Grow Black leadership at Vice President level and above globally by 50% by 2025.	Proxy statement, 2021
McDonald's Corp.	2021	Feb 2021	Increase representation of historically underrepresented groups in Senior Director and above roles in the U.S. from 29% to 35% by 2025.	Proxy statement, 2021
McKesson Corp.	2021	Jun 2021	Increase representation of people of color in leadership ranks (VP and above) by 20% relative to FY2021 (21% baseline) by 2025.	Proxy statement, 2021

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Table A.6, continued

Firm	Cohort	Date	Race target(s)	Source
Medtronic Plc	2019	Nov 2018	Maintain or exceed 20% ethnically diverse representation in U.S. management positions in fiscal 2020.	Inclusion, Diversity, and Equity Report, 2019
Merck & Co., Inc.	2022	Feb 2022	Raise U.S. Black/African American representation in senior management (VP or SVP) from 3% in 2020 to 10% by 2024; U.S. Hispanics/Latinos from 5% in 2020 to 10% by 2024.	Form 10-K (FY2021), filed Feb 25, 2022
Meta Platforms, Inc.	2020	Jun 2020	50% of workforce from underrepresented populations by 2024; increase people of color in U.S. leadership, including Black leadership, by 30% from 2020 to 2025; double Black and Hispanic employees within 5 years (by 2025).	Form 10-K, 2021 (filed 2021)
Microsoft Corp.	2020	Jun 2020	Double the number of Black and African American people managers, senior individual contributors, and senior leaders in the U.S. by 2025; also double Hispanic and Latinx in the same roles by 2025.	DEF 14A 2020-2021 (announcement Jun 2020)
Molson Coors Beverage Co.	2021	Jul 2020	Increase representation of people of color in U.S. operation by 25% by 2023 across salaried employees and leadership positions.	Earnings call, Jul 2020; restated in 2021 proxy
Mondelez International, Inc.	2021	Sep 2020	Double Black representation in U.S. management team by 2024.	Form 10-K, fiscal 2020 (filed February 2021)
NIKE, Inc.	2021	Mar 2021	35% racial and ethnic minorities in U.S. corporate workforce by 2025; 30% racial and ethnic minorities at Director level and above in the United States by 2025.	Proxy statement, March 2021 (quoting FY20 Impact Report)
NXP Semiconductors NV	2022	Feb 2022	Achieve 50% underrepresented minorities in U.S. workforce by 2025.	Form 10-K, FY2021, and Sustainability Report, Apr 2022
Palo Alto Networks, Inc.	2022	Oct 2021	Achieve 20% underrepresented minorities in U.S. workforce by 2025.	DEF 14A proxy statement, Oct 29, 2021
Paramount Skydance Corp.	2022	Oct 2021	A 40% U.S. hire and promotion rate for ethnically diverse vice presidents by end of 2021.	ViacomCBS CEO letter, Oct 1, 2021
PepsiCo, Inc.	2020	2020	Increase U.S. Black and Hispanic managerial population to 10% by 2025.	Proxy statements, 2024 and 2021
Pfizer Inc.	2020	Oct 2019	Raise U.S. minority representation at vice president level and above to 25%, from 19% in 2019, by 2025.	Press release, Oct 17, 2019

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Table A.6, continued

Firm	Cohort	Date	Race target(s)	Source
Pinterest, Inc.	2021	May 2021	Increase underrepresented races and ethnicities to 20% in technical roles and leadership by 2025.	Newsroom report, May 2021
PPG Industries, Inc.	2022	Mar 2022	Increase Black, Latinx and Asian employees by more than 45% in non-frontline, U.S.-based workforce by 2025, from a January 1, 2021 baseline.	Press release, 2022
PVH Corp.	2021	Jun 2021	Double Black and Hispanic/Latinx representation at Director and VP levels (U.S.) by 2026; increase total BIPOC at SVP+ by 50% by 2026.	Bloomberg sponsored article, Feb 2022; company website I&D page
QUALCOMM, Inc.	2021	Feb 2021	Increase underrepresented minorities in leadership (principal-and-above technical; director-and-above non-technical) by 15% by 2025; increase overall workforce by 20% by 2025.	Form 10-K, Nov 2021
Ralph Lauren Corp.	2021	Aug 2020	People of color at least 20% of the Global Leadership Team by 2023 (stated Aug 2020); at least 10% Black, African or African American added in May 2021.	Earnings call, Aug 4, 2020; Form 10-K, May 2021
Salesforce, Inc.	2020	Jul 2020	Double U.S. representation of Black employees in leadership (VP+) by end of 2023; increase U.S. Black representation 50% by end of 2023; increase underrepresented minorities (Black, Latinx, Indigenous, multiracial) employees and leaders 50% by 2023.	Press release, Jul 1, 2020; third target from update, Feb 11, 2021
Sealed Air Corp.	2021	May 2021	Increase representation of racial and ethnic minorities in the U.S. workforce to above 35% by 2025 (34% in 2021).	Form 10-K, 2022
Sensata Technologies Holding Plc	2022	Sep 2021	Reach 25% racial/ethnic diverse representation in manager and above roles in the U.S. by 2026.	Press release, 2021
Shake Shack, Inc.	2022	Sep 2021	Raise people of color to 50% (from 34.7%) of Shack leadership (general managers through regional vice presidents of operations) by 2025; to 30% (from 19.6%) of home office leadership (director through senior vice president) by 2025.	Proxy statement, April 2022; sustainability report, April 2021
Signet Jewelers Ltd.	2021	Jun 2021	Increase diversity in Director+ leadership by 25% by 2030; increase Black, Indigenous, and People of Color (BIPOC) in Director+ support center roles by 20% by 2030.	FY2022 Corporate Citizenship and Sustainability Report, Jun 2022 (goals announced Jun 30, 2021)
Southwest Airlines Co.	2021	Sep 2020	Double the percentage of racial diversity in the Senior Management Committee by 2025.	Press release, September 2020

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Table A.6, continued

Firm	Cohort	Date	Race target(s)	Source
State Street Corp.	2021	Jul 2020	Triple Black and Latinx leadership (senior vice presidents and above) by 2023; double percentage of Black and Latinx populations (all levels) by 2023.	Company web statement, July 2020
Sunrun, Inc.	2022	Apr 2022	Increase BIPOC leaders in manager roles by 25% by 2025 (from about 35% in 2021).	Impact report press release, Apr 22, 2022
Target Corp.	2021	Sep 2020	Increase representation of Black team members across the company by 20% over three years (by 2023).	Press release, Sep 10, 2020
TEGNA, Inc.	2021	Mar 2021	Increase BIPOC share of content teams (news, digital, marketing) to about 36% by 2025; increase BIPOC representation in content leadership roles by 50% by 2025.	Form 8-K, 2021
The Bank of New York Mellon Corp.	2022	Jul 2021	U.S., by year-end 2023: raise Black representation by 15%, to 12%, and Latinx by 15%, to almost 8%; raise Black senior leaders (Levels M/S) by 30%, to over 4%, and Latinx senior leaders by 30%, to over 5.5%.	2020 Enterprise ESG Report (published Q2 2021)
The Boeing Co.	2021	Aug 2020	Increase Black representation in the U.S. by 20% by 2025.	Proxy statements, 2023 and 2024
The Chemours Co.	2019	2018	20% of all U.S. positions filled with ethnically diverse employees by 2030 (2020 report); raised to 30% by 2030 (2023 proxy).	Corporate responsibility report, 2020; proxy statement, March 2023
The Coca-Cola Co.	2021	Mar 2021	Mirror U.S. census data at all job grade levels for U.S. roles by 2030: 13% Black, 18% Hispanic, 6% Asian.	Proxy statement, Mar 2021
The Goldman Sachs Group, Inc.	2021	Aug 2020	Reach 7% Black professionals in Americas and UK vice president population by 2025; reach 9% Hispanic/Latinx professionals in Americas vice president population by 2025; double U.S. campus analyst hires recruited from HBCUs by 2025.	Press release, 2020
The Kraft Heinz Co.	2022	Oct 2021	Reach 30% people of color (from 24%) in the U.S. salaried employee population by 2025.	ESG report and news release, October 2021
The New York Times Co.	2021	2021	Double share of Black/African American and Latino/Hispanic colleagues in leadership by 2025.	Diversity and Inclusion Report, 2021
The Sherwin-Williams Co.	2021	Jun 2021	Increase underrepresented racial/ethnic groups in U.S. management roles to 30% by 2025.	Sustainability report press release, June 2021

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Table A.6, continued

Firm	Cohort	Date	Race target(s)	Source
Tractor Supply Co.	2022	Sep 2021	Increase People of Color at manager level and above by 50% by 2026, from a 2021 baseline.	Press release, September 2021
Trane Technologies Plc	2021	Apr 2021	Increase racial and ethnic diversity of salaried U.S. workforce from 17% to 26% (a 50% increase) by 2030.	2020 ESG Report (published Apr 30, 2021)
Truist Financial Corp.	2021	Sep 2020	Racially and ethnically diverse employees in senior leadership from about 12% to at least 15% in three years (by 2023); hiring from HBCUs from 3% to 5% by 2023.	Press release, September 2020; press release, July 2021
Uber Technologies, Inc.	2021	Jul 2020	Double Black in Director titles and above (5 most senior levels at Uber) by 2025.	Proxy statement, 2021
Valmont Industries, Inc.	2021	May 2021	Increase representation of people of color by 50% by 2025 and double it by 2030, workforce-wide, from a 2018 baseline.	Form 8-K, May 2021
VF Corp.	2021	Feb 2021	BIPOC representation of 25% within Director and above population by 2030 (from 16% in 2021).	Press release, February 2021
Visa, Inc.	2021	Jul 2020	Increase underrepresented employee population by 50% at Vice President and above level in the U.S. by 2023 (three years from July 2020); increase overall U.S. underrepresented employee population by 50% by 2025 (five years).	Proxy statement, 2020
Walker & Dunlop, Inc.	2021	Sep 2020	Minorities in management (assistant VP/associate director and above) from 11% (2020) to 20% by 2025; underrepresented groups in management from 14% (2021) to 25% by 2025; minorities among top earners to 15% by 2025 (6% baseline).	Proxy statement, Mar 2021; ESG reports and EEO disclosure, 2021
Wells Fargo & Co.	2020	Jun 2020	Double Black leadership (senior managers and executive ranks) by 2025, baseline 5.8% in 2020.	CEO memo, June 2020; press coverage, September 2020
Whirlpool Corp.	2021	Jul 2020	Increase by 50% the number of Black employees across all levels and increase by 50% Black director-level and above employees by 2025.	Sustainability Report press release, Mar 2021
Workday, Inc.	2021	Sep 2020	Increase Black and Latinx employees in U.S. by 30% by 2023 (baseline 7.56% in 2021); double Black and Latinx leaders in U.S. by 2023.	Company blog post, Sep 2020
Zoetis, Inc.	2021	Feb 2021	Increase people of color from 21% to 25% in U.S. workforce by 2025; increase Black colleagues from 4% to 5%; increase Latinx colleagues from 5% to 6%.	Form 10-K, Feb 2021

B The Workforce Panel

B.1 Construction

The panel uses the November 2025 release of Revelio Labs' LinkedIn data. I keep positions in the United States that have a start date. For each worker and each July 1 from 2014 to 2024, the worker counts at the employer of the most recent position that started on or before that date and has not ended by then; if two positions started on the same day, the more senior one counts. Many people do not record end dates, so a later start at another employer ends a spell even when the profile never closed it. Positions still open when the data were extracted, on 1 October 2025, are right-censored, and I do not impute an end for them. Counting only the most recent position removes about 8 percent of the workers whose position at a firm is still open on the snapshot date, because their profiles show a later job elsewhere.

Revelio assigns each employer to its ultimate parent under current ownership, so a company acquired in 2022 counts as part of its acquirer in 2016 as well. For each parent I keep its largest affiliates up to 95 percent of its U.S. workers. This drops 4 percent of workers and removes small, often mismatched units. I drop positions whose title marks the holder as retired, former, a board member or a volunteer; they are 2.3 percent of the worker-years in the top band. I keep frontline positions.

The same release also gives a snapshot on 1 July 2025, which I build with the rules above and those of Appendices B.2 and B.5. Its 298,305 workers without BIFSG probabilities, all of them new to the panel in 2025, are scored with BIFSG without the education term (Section 3.4). Run for 2024, the same code reproduces 99.78 percent of the panel's rows by worker, firm and level, and the cells for 2014 to 2024 do not change. I use the snapshot only in the checks on persistence. It adds the shaded event years 4 and 5 of Figure 5, where event year 4 covers the cohorts 2018 to 2021 (83 pledgers) and event year 5 the cohorts 2018 to 2020 (20 pledgers), and it gives the Black share at the first July 1 after a pledger's retreat in Panel B of Table E.1. The data were extracted on 1 October 2025, three months after this snapshot, so it likely records recent moves less completely than earlier snapshots do. The share of workers whose position started in the preceding twelve months falls from 19.3 percent in 2024 to 15.6 percent in 2025 at pledgers, and from 18.7 to 14.5 percent at controls, although part of this fall continues a decline in entries under way since 2022.

The hire-level designs use a panel of position starts instead of snapshots. It holds

every U.S. position that started between 2016 and 2024 at the 115 pledgers and the 1,063 controls, 18.0 million starts in all, each assigned to its firm with the reassignments of Appendix B.2. I date each start by the first July 1 on or after it, the clock of the cohorts, so a position that started in August 2024 counts in 2025. Its band is Revelio's level with the correction of Appendix B.5. A start is an external hire when the worker's previous position (the latest one at any employer and in any country that started before it) was at another employer, with or without a Revelio identifier, or when there is no previous position. When the previous position is at the same firm, the start is an internal move. I look up previous positions only for starts at Revelio levels 3 to 7, so entry-level starts cannot be classified, and the entry-level results of Online Appendix OA.F.1 use all starts. The triple differences compare hires by their most likely label. At director level and above, 702,265 of the 1,570,897 starts labeled Black, Hispanic or white male are external hires: 111,743 Black, 106,228 Hispanic and 484,294 white-male hires. In the middle band the count is 1,446,447 of 2,498,307 (290,978, 271,158 and 884,311). The flow decomposition of Appendix E.2 finds the workers who join a firm by comparing consecutive snapshots. Of those who join at director level and above and match a start by the same worker at the same firm, dated to the same July 1, 94 percent are external hires by this rule.

B.2 Company Mapping

Because every position is assigned to a parent, a wrong parent moves whole workforces between firms. I screened the estimation firms for three signs of a wrong parent: a large unit, at least a tenth of the parent's U.S. workers or 3,000 workers, whose name shares no word with the parent's; a jump in the firm's headcount of more than 50 percent up or a third down from one year to the next; and a panel headcount far outside the firm's EEO-1 total. Independent readers checked each flagged unit against public records of ownership, and 47 units were reassigned. Eleven are The Huntington National Bank and ten units of its lineage, which Revelio had placed under Associated Banc-Corp, a control. The other 36 are mostly tenants' staff that Revelio had placed under their real-estate landlords, such as casino operators under the real-estate trusts that own their buildings. One further unit, the Four Points by Sheraton hotels, was moved from Blackstone to Marriott. When I rebuild the panel without the reassignments, the affected workers' histories reproduce more than 99.6 percent of their worker-years in Revelio's mapping, so the rebuild changes nothing beyond the reassigned units. Moving the Four Points hotels changes the main estimates by less than 0.002 percentage points.

The static mapping also breaks histories when ownership changes. When a firm sells or spins off a unit, workers whose profiles name the parent before the sale leave its history; when it acquires a firm, the acquired workers enter its earlier years. These breaks are unrelated to pledges. Appendix G reports estimates that drop or trim the firms involved, on both sides of the comparison.

B.3 Coverage

Not every worker has a profile. For the 461 panel firms whose names match a consolidated EEO-1 report for 2020 and whose ratio of LinkedIn to reported workers lies between 0 and 4 (Online Appendix OA.B.1), the median firm's LinkedIn headcount on 31 December 2020, inside the report's payroll period, is 74 percent of the U.S. headcount it reported to the government, with an interquartile range of 50 to 106 percent. Coverage is highest in information and finance and lowest in utilities and transportation, where it is about half. Because the outcomes are shares, low coverage matters only if it differs by race or between pledgers and controls. In North Carolina, 16.7 percent of Black and 15.5 percent of white registered voters have a matching profile. Table B.1 compares the panel's headcount on 1 July 2019 with the 2019 reports of the firms in the main comparison. The means are close, 86 percent at race pledgers and at general pledgers and 83 percent at controls, but the medians are 79, 77 and 66 percent. Part of the means comes from six firms whose LinkedIn headcount is more than three times the reported one, five of them controls. Without these six the means are 83, 86 and 73 percent.

TABLE B.1: LINKEDIN COVERAGE OF REPORTED U.S. EMPLOYMENT BY GROUP, 2019

			Percentile		Firms with
	Mean	Median	25th	75th	a 2019 report
Race pledgers (115)	86	79	51	112	78
General pledgers (193)	86	77	56	109	105
Never-pledged controls (1,062)	83	66	43	97	455
All three groups	84	69	45	100	638

Notes: Ratio, in percent, of a firm's LinkedIn U.S. workers on 1 July 2019 to the U.S. employees in its 2019 EEO-1 report, for the firms of each group with such a report (last column). Race pledgers are the main estimation sample, general pledgers the firms with general statements and no numerical race target, controls the never-pledged firms; the last row pools the three groups. EEO-1 reports are those of federal contractors released under the Freedom of Information Act, matched as in Online Appendix OA.B.1. Workers and reports are the inputs of Table 3.

Profiles are updated with a delay, so the last years of the panel are recorded less com-

pletely. Headcounts in the panel fall slightly in 2023 and 2024, by about 5 percent at pledgers and 1 percent at controls from 2022 to 2024, which may reflect late updating as well as layoffs. The estimates hold when I drop the 2024 snapshot (Table 5).

B.4 Comparison with the Current Population Survey

Table B.2 compares the panel with the March 2019 Current Population Survey (CPS). The main benchmark is column 5: private wage and salary workers at firms with 1,000 or more employees. The panel is close to these workers on sex, region and the Black share (13.7 percent, against 14.4 percent in the CPS), but holds fewer Hispanic and more Asian workers. I impute race in the panel (Section 3.4), so the race rows compare an estimate with self-reported race. LinkedIn users are a selected group: the panel over-represents college graduates, younger workers and professional and managerial jobs. The industry mix reflects LinkedIn's uneven coverage across industries and the S&P 1500 frame, which includes no nonprofit hospitals or universities. Pledgers and controls are balanced when each firm counts once, as in the estimates (column 10: no normalized difference exceeds 0.16); pooled over workers, large firms widen the Asian, West, South and education and health gaps (column 9, at most 0.27).

B.5 Seniority Levels

Revelio assigns every title one of seven seniority levels. I group them into three bands: entry (levels 1 and 2), middle (3 and 4) and director and above (5 to 7). In 2019, 77 percent of the workers in the top band are at level 5, 21 percent at level 6 and 2 percent at level 7. The levels come from a model of job titles, so the same title can sit at different levels in different firms: for 91 of the 150 most common titles in the top band, fewer than 80 percent of firms place the title at its most common level. I correct one systematic error. When a worker's title at the same firm clearly goes up (manager to senior manager, associate director to director) but Revelio's level falls, I place the new job one level above the old one for as long as the worker holds that title. This recodes 43,506 promotions and moves no estimate by more than 0.013 percentage points. As an alternative I built bands from my own crosswalk of titles to levels, calibrated on officers named in SEC filings and on wages in H-1B applications. It tracks the EEO-1 categories about as well as Revelio's levels do and gives a smaller and noisier top band. Each band has about 12,500 firm-year cells in the estimation panel; the median cell in the top band holds about 300 workers, and fewer than 0.3 percent of cells fall below the floor of five workers.

TABLE B.2: THE WORKFORCE PANEL AND CPS BENCHMARKS, 2019

	Panel, 1 July 2019			CPS ASEC 2019			Normalized difference			
				Employed	Private,	(5), BA	(1) vs		(2) vs (3)	
	All	Pledgers	Controls	in March	1,000+	or MBSA	(5)	(6)	Workers	Firms
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Demographics										
Female	43.7	42.6	44.5	47.5	48.2	50.3	−0.06	−0.09	−0.03	0.01
Age under 30	37.4	37.9	37.1	23.7	28.5	20.5	0.13	0.27	0.01	0.03
Age 30 to 44	42.6	43.7	41.8	32.1	31.8	36.9	0.16	0.08	0.03	0.01
Age 45 or older	20.0	18.4	21.2	44.2	39.7	42.5	−0.31	−0.35	−0.05	−0.04
Mean age (years)	35.2	34.8	35.5	42.4	40.6	42.5	−0.30	−0.42	−0.05	−0.05
Race and ethnicity										
Black	13.7	13.3	14.0	11.6	14.4	10.7	−0.01	0.07	−0.01	0.01
Hispanic	11.8	11.8	11.7	17.9	15.8	9.8	−0.08	0.05	0.00	0.00
Asian, Pacific Islander	13.0	17.3	10.2	6.8	7.9	11.4	0.12	0.03	0.15	0.07
White, non-Hispanic	61.9	57.8	64.5	63.7	61.9	68.2	0.00	−0.09	−0.10	−0.05
Education										
BA+, coded degree	80.0	82.2	78.5	38.8	39.2	76.6	0.65	0.06	0.07	0.03
BA+, lower bound	53.1	58.3	49.7	38.8	39.2	76.6	0.20	−0.36	0.12	0.08
Occupation										
MBSA occupations	59.5	61.4	58.4	41.1	40.3	78.8	0.28	−0.30	0.04	0.00
Frontline occupations	40.5	38.6	41.6	58.9	59.7	21.2	−0.28	0.30	−0.04	0.00
Industry										
Goods-producing	30.5	31.5	29.9	18.8	16.5	14.2	0.24	0.28	0.02	0.11
Trade, transport, util.	22.3	26.7	19.5	18.3	28.0	15.8	−0.09	0.12	0.12	0.04
Info., finance, prof.	37.7	37.3	37.9	21.0	22.7	33.9	0.23	0.06	−0.01	−0.15
Education and health	4.2	0.0	6.9	23.5	20.3	29.4	−0.36	−0.51	−0.27	−0.16
Other services, public	5.3	4.5	5.8	18.5	12.5	6.7	−0.18	−0.04	−0.04	0.11
Census region										
Northeast	17.6	19.4	16.4	17.5	17.5	19.8	0.00	−0.04	0.05	0.08
Midwest	19.3	16.8	20.8	21.6	23.0	21.8	−0.06	−0.04	−0.07	−0.01
South	35.2	29.2	39.1	37.2	36.8	35.1	−0.02	0.00	−0.15	−0.09
West	27.9	34.6	23.7	23.8	22.7	23.3	0.08	0.07	0.17	0.03
Obs. (thousands)	6,418	2,501	3,917	84.4	27.6	13.7				
Firms	1,177	115	1,062							

Notes: Percent unless stated. Panel: worker-years in estimation cells; sex is imputed from first names, race from names and locations (Black also from colleges), and age from education dates (42 percent of rows). Coded degrees (2025 profiles, 66 percent of rows) count unless the first bachelor's started or ended after 2019; the lower bound counts the rest as none. CPS: weighted persons 16 or older; job variables refer to the longest job of 2018. MBSA: SOC groups 11 to 29. Column 10 weights firms equally. Normalized difference: $(\bar{x}_1 - \bar{x}_2)/\sqrt{s_1^2 + s_2^2}$ (Imbens and Wooldridge, 2009).

B.6 Index Membership over Time

The universe is the index as it stands today. Firms that were in the S&P 1500 during the pledge wave but have since been acquired, gone private or dropped out of the index are therefore missing, and firms that joined later are included. To measure how much this matters, I reconstruct the index on 30 June of each year from 2020 to 2024 from the holdings reported to the SEC by an exchange-traded fund that replicates the S&P 1500. Of the 1,454 companies in the index on 30 June 2020, 1,075 are in the universe. Most of the others left the index after 2020, and a few are current members that the name matching had missed. About 385 firms of the universe joined the index after June 2020. Appendix G reports the main estimates on the firms that were in the index in June 2020.

C Measuring Race

C.1 The BIFSG Classifier

On the statistical properties of BIFSG and its refinements, see Derby et al. (2025) and Imai et al. (2022). The name terms use the first-name and surname tables of Rosenman et al. (2023), which I downloaded in August 2026 as distributed with the `wru` package for R: 1,044,047 first names and 1,484,515 surnames. For each group, the tables give the probability of a name, the terms $P(f \mid r)$ and $P(s \mid r)$ of Section 3.4. When a surname is missing from the surname table, I look it up in the Census Bureau's list of the 162,253 surnames that occur at least 100 times in the 2010 Census, which adds almost no matches among the panel's workers. Since the tables report no counts, I do not smooth the probabilities of rare names. I also leave out the authors' middle-name table, because LinkedIn profiles record no middle name. Ninety-one percent of all workers are scored based on all three of first name, surname and location. Since some first names or surnames are not in the name tables, 6 percent of workers are scored on first name and place, 2 percent on surname and place, and 1 percent on place alone.

The place term comes from the Census Bureau's Vintage 2023 population estimates by county, race and Hispanic origin, for 1 July 2023 and all ages. The Census asks about Hispanic origin and about race in two separate questions, so Hispanic Americans can be of any race. I treat Hispanic as an absorbing category: it includes white Hispanics, Black Hispanics and Hispanics of any other race. A county's prior is its shares of non-Hispanic white, non-Hispanic Black, non-Hispanic Asian and Hispanic residents, with everyone else in a residual group, and state and national priors add up the counties. The classifier computes the posterior over these five groups and then rescales the paper's four groups to sum to one.

Each worker gets one geography for her whole career. I resolve the city of each of her U.S. positions to a county with the Census Bureau's 2020 list of places by county, searching only within the position's state; a place that spans several counties gets the average of their priors, weighted by population. Her county is the one that occurs most often among her positions. Each position counts once, whatever its length, and a tie goes to the county of the position she started most recently. If none of her positions resolves to a county, she gets the most frequent metropolitan area among her positions, then the most frequent state, then the national mix. Revelio's metropolitan labels match the Office of Management and Budget's 2009 delineation, and a metropolitan area's prior averages its

counties by population. When the positions do not give a county, I also parse the location in the profile and keep the finer of the two answers. Of the 25,426,806 workers scored, 73.2 percent get a county from their positions and 11.3 percent from their profiles. If the mapping is not clear (15 percent of the workers in the panel), I use either the metropolitan area (1 percent), their state (6 percent) or the national mix (8 percent). Puerto Rico and the other territories have no county prior, so positions there fall through to the national mix.

The prior is fixed over time. I use the 2023 estimates for every year of the panel, and the county comes from all of a worker's positions, whenever she held them, so she keeps a single score throughout. Among the North Carolina voters matched to the panel (Appendix C.3), this rule places 73.5 percent in the county where they are registered to vote. Scoring them with their registered county instead lowers the factor for the Black share from 1.62 to 1.60 (from 1.43 to 1.41 with most likely labels).

The education-augmented BIFSG of the main text adds one term. If a worker's college has a Black share of at least one half among its graduates, I multiply the Black-versus-rest odds of her posterior by that college's likelihood ratio: the odds that its graduates are Black over the national odds, from the IPEDS counts of graduates by race. The rule covers 138 colleges, 87 of them historically Black, with the campuses of a college system pooled. The term enters with a weight of about 0.66, fitted on the North Carolina voters (Appendix C.3). A vision model labels the proxy photographs of 4,450 directors and officers, and 343 of them appear Black. Among these Black directors and officers, the share whose most likely label is Black rises from 41 to 46 percent, while the false-positive rate stays at 4 percent.

C.2 Measurement Model and Correction Factors

Consider a cell of N workers, a share s of whom belong to the group, and let p_i be the score the classifier gives worker i: her probability of belonging to the group for the share outcomes, or an indicator that the group is her most likely label for the hire-level designs. The measured share is the average score,

$$m = \frac{1}{N}\sum_{i=1}^{N} p_i = s\,\pi_1 + (1-s)\,\pi_0 = \pi_0 + (\pi_1 - \pi_0)\,s,$$

where π_1 is the average score of the members of the cell and π_0 that of the non-members. With most likely labels, π_1 is the classifier's sensitivity and π_0 its false-positive rate. For the Black and Hispanic share, a member belongs to either group, and the score is the sum

of the two probabilities or an indicator that the most likely label is one of the two.

The event studies compare changes in m at pledgers and controls. Within a band, write the measured share of firm f in year t as $m_{ft} = \pi_{0,ft} + (\pi_1 - \pi_0)\, s_{ft}$. If $\pi_{0,ft}$ is the sum of a firm term and a year term, the firm and year fixed effects of Section 4.2 absorb it. If in addition $\pi_1 - \pi_0$ is the same in every cell, each coefficient of the event study on m equals $\pi_1 - \pi_0$ times the corresponding coefficient on s, and the true effect is the measured one times $1/(\pi_1 - \pi_0)$, the correction factor. The classifier misses some members ($\pi_1 < 1$) and gives some weight to non-members ($\pi_0 > 0$), so $\pi_1 - \pi_0$ is below one and the factor exceeds one.

The voter file records each voter's race (Appendix C.3), and a vision model labels the apparent race of each photographed director or officer (Appendix C.5). For these two sources, π_1 and π_0 are the average scores of the members and of the non-members, and the factor is $1/(\pi_1 - \pi_0)$. The EEO-1 reports record no individual's race, only each firm's official share s_{ft} (Appendix C.4). Across firms the model gives $m_{ft} = \pi_0 + (\pi_1 - \pi_0)\, s_{ft}$, so a regression of the measured share on the official one has intercept π_0 and slope $\pi_1 - \pi_0$, and the factor is the inverse of the slope. That slope also picks up coverage and the mapping of Revelio's levels into EEO-1 job categories, and, since it compares firms, where each firm's workers live.

The correction rests on two assumptions: the classifier errs at the same rates at pledgers and controls, and its rates did not change with the pledges. Since each worker carries one score (Appendix C.1), the rates in a cell change only when the names and places of its workers change. Appendix C.7 tests both assumptions among the North Carolina voters. The sources also cover different people. The voters are a broad population, the EEO-1 categories only approximate the top band, and the photographed directors and officers sit higher than most workers in that band. Table C.1 reports the factors from each source for education-augmented BIFSG, BIFSG and Revelio's classifier, scored by probabilities and by most likely labels.

C.3 Voter Files

North Carolina's voter file records the race and Hispanic ethnicity that each voter reports. I match the panel's workers whose LinkedIn location is in North Carolina to the 8,275,932 voters with a recorded race, on first name, last name and city, and keep the 239,207 workers who match exactly one voter. The sample is the 230,940 of them recorded as white, Black, Asian or Hispanic (Hispanic ethnicity takes precedence over race), of whom 53,323

TABLE C.1: CLASSIFIER CORRECTION FACTORS BY VALIDATION SOURCE, INCLUDING PROXY PHOTOGRAPHS

		Education-augmented BIFSG		BIFSG		Revelio	
Source	Population	Black	Black and Hispanic	Black	Black and Hispanic	Black	Black and Hispanic
Panel A: Posterior probabilities, used for the share outcomes							
North Carolina voter file	230,940 voters in the panel	1.62	1.65	1.75	1.76	2.48	2.41
EEO-1 reports	Managers at 190 firms	1.30	2.23	1.37	2.31	1.59	2.68
Proxy photographs	1,351 executives and directors	2.72	2.94	3.00	3.20	4.86	4.77
Panel B: Most likely label, used for the hire-level designs							
North Carolina voter file	230,940 voters in the panel	1.43	1.45	1.54	1.54	2.10	2.05
EEO-1 reports	Managers at 190 firms	1.44	2.33	1.53	2.43	1.59	2.55
Proxy photographs	1,351 executives and directors	2.48	2.67	2.80	2.97	4.75	4.40

Notes: Each entry is the factor that scales a measured change in the share to the true change. Voter file and photographs: $1/(\pi_1 - \pi_0)$, with π_1 and π_0 the average scores of group members and non-members (probabilities in Panel A, most likely labels in Panel B). EEO-1: the inverse slope of the measured share on the official share across 739 firm-years (Revelio levels 6 and 7 against officials and managers, 2016 to 2020), which also absorbs coverage and the level-category mismatch. BIFSG as in Section 3.4; education-augmented BIFSG raises the Black probability of graduates of majority-Black colleges and rescales the other groups in proportion; Revelio is the vendor's classifier, which uses names and location.

are Black and 8,738 Hispanic. The voter-file factors in Table C.1 use these workers because the paper's education-augmented BIFSG probabilities, which place each worker in the county of her positions, exist only for workers in the panel.

With most likely labels, education-augmented BIFSG labels 76.0 percent of the Black voters Black (BIFSG: 70.8) and 6.1 percent of the others, so its factor for the Black share is $1/(0.760 - 0.061) = 1.43$. Revelio's classifier labels 55.3 and 7.7 percent, a factor of 2.10. With probabilities, education-augmented BIFSG gives the Black voters an average probability of 73.3 percent of being Black and the others 11.7 percent, a factor of 1.62 (bootstrap 95 percent interval 1.62 to 1.63). North Carolina is one of the six states whose voter files Rosenman et al. (2023) use to estimate the name tables, so here the names are tested on a population like the one they were estimated from.

Revelio's probabilities, which also draw on first names, surnames and location, exist for every LinkedIn user, so I can also check its classifier on all 1,182,744 one-to-one

matches of LinkedIn users in North Carolina. With most likely labels, and with every voter not recorded as Black counted as a non-member, its factor for the Black share is 2.07. On the 1,146,785 of these voters recorded as white, Black, Hispanic or Asian, its factors are 2.04 for the Black share and 2.01 for the Black and Hispanic share, close to the 2.10 and 2.05 among panel workers. On the same voters, a model with surnames and each voter's registered county alone gives 2.06 and 2.03; it labels 59.9 percent of the Black voters Black and 11.3 percent of the others (rates on all matched users, not only on the 230,940 panel workers). A BIFSG variant that uses each voter's registered county gives 1.57 for both. For this sample I report only factors from most likely labels.

Florida's voter file also records the race each voter reports. When I built the classifier, I scored a random sample of 500,000 Florida voters recorded as white, Black, Hispanic or Asian, with first, middle and last names and the registered county. Two features of this check favor the classifier relative to the panel: Florida is another of the six states behind the name tables, and the check uses middle names, which LinkedIn profiles do not record. In that sample BIFSG labeled 72 percent of the Black voters Black and 2 percent of the others, a factor of 1.42, against 2.00 for surnames and county alone. Its sensitivity is 60 to 64 percent among Black voters born in the 1940s to 1960s and 78 to 82 percent among those born in the 1980s to 2000s.

New York does not record race, so a commercial voter file for New York, from TargetSmart, carries a race field that the vendor models. Matching it to LinkedIn users in New York gives 1,423,733 pairs. The modeled field and Revelio's classifier agree for 86.9 percent of the pairs (Cohen's κ 0.67) and for 55 percent of the people Revelio labels Black. Neither field records what the person reports, so this comparison measures agreement between two classifiers, and it does not enter the factors.

C.4 EEO-1 Reports

The consolidated EEO-1 reports that federal contractors filed for 2016 to 2020 count each firm's U.S. employees by race, sex and job category. I match them to firms by company name, after removing punctuation and legal suffixes such as Inc., Corp. and Holdings. The comparison uses an earlier build of the panel that covers 395 S&P 1500 firms, of which 190 match a report exactly, for 739 firm-years. In each firm-year, the measured share is the average score the classifier gives the firm's LinkedIn workers at Revelio levels 6 and 7. The official share is the share of the firm's officials and managers (EEO-1 categories 1.1 and 1.2) who are Black, or Black or Hispanic.

Across the 739 firm-years, a regression of the measured Black share on the official one has a slope of 0.77 with education-augmented BIFSG's probabilities, so the factor is 1.30 (95 percent interval 1.03 to 1.65, bootstrap over firms). For the Black and Hispanic share the slope is 0.45 and the factor 2.23 (1.80 to 2.92), because the measured Hispanic share follows the official one less closely (slope 0.36). With most likely labels the factors are 1.44 and 2.33, and with Revelio's probabilities 1.59 and 2.68 (Table C.1).

Because it compares firms, the slope picks up the classifier's use of place, in addition to coverage and the mapping of levels into job categories. BIFSG gives every worker a higher score in a place with many Black residents, so across places the measured share follows the true share more closely than it does within a place. I gauge the size of this effect with the North Carolina voters matched to the panel. Across the 361 employers with at least 50 matched workers in 2019, a regression of each employer's measured Black share on its voter-file Black share gives a factor of 1.41 for education-augmented BIFSG's probabilities and 1.22 for its most likely labels. Within employers (all 1,309 employers of these workers), the gap between Black and other workers gives 1.68 and 1.48, close to the 1.66 and 1.45 for the same workers pooled. Across the state's counties the factor is about one. The event study measures a change within a firm, and a comparison across firms understates the correction for such a change. For the Black share, the EEO-1 factor is the lowest of the three sources.

C.5 Proxy Photographs

Proxy statements filed on EDGAR carry photographs of executive officers and directors. I use a vision model to label the apparent race of each person in them. The model never sees the person's name, and no image is kept. After the match to the panel, the labels cover 1,351 people, most of whom sit on corporate boards. Education-augmented BIFSG calls 44 percent of the people who appear Black in their photographs Black, well below its 76 percent among voters. It gives them an average probability of 45.0 percent of being Black, and everyone else 8.1 percent. These averages imply factors of 2.72 for the Black share and 2.94 for the Black and Hispanic share, or 2.48 and 2.67 with most likely labels (Table C.1). The factors are imprecise, because only 108 of the 1,351 appear Black: the bootstrap 95 percent interval for the Black-share factor runs from 2.3 to 3.3. Revelio's classifier calls 27 percent of them Black, with factors of 4.86 and 4.77 (4.75 and 4.40 with most likely labels). A study that takes this vendor field as given, as Cai et al. (2024) do, therefore understates a race effect at the top by a factor of almost five. The classifier misses Black workers most

often at the top, so the estimates for the top band are the most understated. The labels are a model's reading of a photograph, and the main text uses these factors only as the upper end of the correction range (Section 5.1).

The photographs also give the precision of the most likely label, which the hire-level designs use. Of the 100 photographed people whose most likely education-augmented BIFSG label is Black, 48 appear Black in their photographs, a precision of 48 percent (bootstrap 95 percent interval 38 to 58 percent). Of the other 52, 46 appear white. Errors in the match to LinkedIn lower this number. Among the 1,009 people matched on their exact names, 43 of the 76 labeled Black appear Black, or 57 percent. Online Appendix OA.C.4 applies the 48 percent to the hires labeled Black at director level and above. Precision also depends on how many people in a population are Black, and the photographs cannot measure that for these hires. If I carry over to the hires the share that appears Black within each range of BIFSG probability, precision is 49 percent. Carrying over the label's sensitivity and false-positive rate instead gives 52 percent. If the Black share of these hires were 5.9 percent, precision would be 31 percent. That is the share among officials and managers in the 2016 to 2020 EEO-1 reports of the 451 sample firms that match a report.

C.6 Historically Black Universities

The Integrated Postsecondary Education Data System (IPEDS) records the race of each school's students. In the fall enrollment surveys of 2010 to 2023, Black students made up 93 to 99 percent of the undergraduates at six historically Black universities. These shares count only the four groups Black, Hispanic, Asian or Pacific Islander and white, and so leave out nonresident aliens, students of two or more races and students of unknown race (Table C.2). Among the panel's workers who completed a bachelor's degree at one of these schools in 2014 to 2024, Revelio's classifier labels 48 to 60 percent Black and BIFSG 64 to 74 percent. The share labeled Black rises with the year of graduation. With the six schools pooled, BIFSG labels 57 percent of those who graduated in 1999 or earlier Black, 62 percent of the 2000 to 2013 graduates and 68 percent of the 2014 to 2024 graduates (Revelio: 44, 46 and 51 percent). The Florida voters show the same gradient by year of birth (Appendix C.3). Distinctively Black first names are less common among older Black Americans (Fryer and Levitt, 2004). Since first names carry much of the classifier's information, part of the gap at the top likely comes from naming cohorts (Appendix C.5).

TABLE C.2: BLACK SHARE OF GRADUATES OF SIX HISTORICALLY BLACK UNIVERSITIES

			Revelio		BIFSG	
School	IPEDS	Graduates	Probability	Label	Probability	Label
Howard University	93.1	5,798	46.9	49.2	62.6	65.9
Spelman College	98.3	1,917	49.1	51.6	68.1	72.2
Morehouse College	99.4	2,054	46.0	48.5	61.9	64.1
Hampton University	96.0	2,445	46.9	50.2	63.2	66.6
Tuskegee University	98.0	1,505	48.0	50.9	65.8	69.1
Jackson State University	95.2	2,517	56.0	59.6	69.7	74.2

Notes: IPEDS is the Black share of the undergraduates in four groups (Black, Hispanic, Asian or Pacific Islander, white), excluding nonresident aliens and students of two or more races or of unknown race, in each school's fall enrollment surveys of 2010 to 2023. Graduates are the workers scored by the classifiers whose LinkedIn profiles list a U.S. bachelor's degree from the school completed in 2014 to 2024. Probability is their average probability of being Black; label is the share whose most likely group is Black. Entries in percent except graduates.

C.7 Error Rates at Pledgers and Controls

The correction factors rescale the estimates only if the classifier's error rates are the same at pledgers and controls and did not change with the pledges. I check this on the North Carolina voters matched to the panel. At the level of worker-years, I compare the gap between the education-augmented BIFSG scores of Black and other voters at the 115 pledgers and at the 1,004 controls that employ matched voters (113 and 806 at director level and above), in July 2014 to 2019 and July 2021 to 2024. Before 2020 the gaps imply factors of 1.64 at pledgers and 1.72 at controls. After 2020 the gap moved by 0.002 more at pledgers than at controls, and by 0.011 more at director level and above (firm-cluster bootstrap SEs 0.009 and 0.021). If I read this as a difference in differences of the measured Black share net of the true change scaled by the classifier's pre-2020 gap between Black and other voters, the classifier's own contribution is -0.04 percentage points (SE 0.27), and -0.15 at director level and above (SE 0.36). Both are centered on zero, but too imprecise to rule out a bias the size of the effect. A change in the names or places of the workers that pledgers hire could still weaken the assumption, since the classifier reads both.

D Estimation and Inference

D.1 The Event-Study Estimator

I estimate Equation (1) separately for each band and for the whole company, on cells of firm, band and July 1 year from 2014 to 2024, with the bands of Appendix B.5. The outcome is 100 times the mean education-augmented BIFSG probability of being Black among a cell's workers. I drop cells with fewer than five workers, at pledgers and controls alike. A pledger enters only from event year -4 to event year 3; the never-pledged controls enter in every year. The pledgers' windows are balanced. Each of the 115 pledgers has a cell in every band at every event year from -4 to 2, and so do the 83 pledgers of the 2018 to 2021 cohorts at event year 3. This gives 888 pledger cells in each band. The 1,062 controls add between 11,609 cells (middle band) and 11,670 (whole company), of the 11,682 possible.

Each regression is ordinary least squares, and every cell counts once. Besides firm and year effects, it has one indicator for each cohort and event year other than -1, 34 in all, since the 2022 cohort is not observed at event year 3. Every event year of a pledger except -1 has its own indicator, so the pledgers' cells carry no information about the year effects, which come from the controls alone. Each $\beta^b_{g,e}$ is then the change in cohort g's mean Black share between years $g-1$ and $g+e$, minus the change in the year effects between the same two years. This is the cohort-specific effect of Sun and Abraham (2021), with the never-pledged firms as the comparison group. The leads are also differences against year -1, so they trace each cohort's path before its pledge relative to the controls.

Equation (2) averages the cohort coefficients at each event year with weights $\omega_{g,e}$, each cohort's share of the pledgers observed at that event year. With balanced windows these weights are the same in every band. They are the cohort shares of all 115 pledgers at event years -4 to 2, and of the 83 observed at event year 3 (Table D.1). The 2021 cohort, whose earliest numerical race pledges fall between 2 July 2020 and 1 July 2021, carries 55 percent of the weight at event years -4 to 2, 76 percent at event year 3 and 60 percent of τ_b. The 2018 and 2019 cohorts together carry 4 percent of τ_b.

I cluster the standard errors by firm (1,177 firms). They come from the cluster-robust covariance matrix $\hat{V}$ of all the regression's coefficients, with the finite-sample factor $G(N-1)/[(G-1)(N-K)]$ for G firms, N cells and K coefficients. Since K includes the 1,176 firm effects, the standard errors are about 5 percent larger than with a factor that leaves them out. Each event-year estimate is a linear combination $c_e'\hat{b}$ of the coefficient

TABLE D.1: PLEDGE COHORTS AND THEIR WEIGHTS IN THE EVENT STUDY

Cohort g	Pledgers	Years in the window	Weight $\omega_{g,e}$ (percent)		Weight in τ_b (percent)
			$e = -4$ to 2	$e = 3$	
2018	1	2014 to 2021	0.9	1.2	1.0
2019	3	2015 to 2022	2.6	3.6	2.9
2020	16	2016 to 2023	13.9	19.3	15.3
2021	63	2017 to 2024	54.8	75.9	60.1
2022	32	2018 to 2024	27.8	0.0	20.9
All	115		100.0	100.0	100.0

Notes: Cohort g is the year of the first July 1 on or after a firm's earliest numerical race pledge. The window runs from event year -4 to event year 3, or to 2024. $\omega_{g,e}$ is the cohort's weight in the event-year average of Equation (2); the last column is its weight in τ_b, the mean over event years 0 to 3. 115 pledgers of the main estimation sample; the weights are the same in every band. Source: pledge dataset (Section 3.1).

vector $\hat{b}$, with the weights $\omega_{g,e}$ in c_e. The average effect is then $\hat{\tau}_b = \bar{c}'\hat{b}$ with $\bar{c} = \frac{1}{4}\sum_{e=0}^{3} c_e$, and its standard error is $(\bar{c}'\hat{V}\bar{c})^{1/2}$. This standard error uses the covariances between the four post-pledge estimates. They are large and positive (correlations of 0.42 to 0.92 across pairs and bands), because all four are measured against the same year -1 cells and the same year effects. Without these covariances, the standard error at director and above would be 0.042 instead of 0.072. I treat the weights as known, so the standard error conditions on the cohort sizes. For the joint test of the leads, I compare the Wald statistic $(R\hat{b})'(R\hat{V}R')^{-1}(R\hat{b})$, where R stacks c'_{-4}, c'_{-3} and c'_{-2}, with a χ^2 distribution with three degrees of freedom. The p-values of τ_b use the normal distribution.

D.2 Sensitivity to Pre-Trends

Parallel trends after the pledge cannot be tested directly. Rambachan and Roth (2023) replace the assumption with a restriction on the bias δ_e, the gap between the pledgers' and the controls' paths that would have opened without the pledges, and give confidence sets for τ_b that are valid under that restriction. If firms did not anticipate their pledges, the pre-pledge coefficients contain no effect, so they measure the bias before the pledge. I compute the sets with the authors' HonestDiD package. The inputs are the seven event-year coefficients of Appendix D.1 ($e = -4, -3, -2$ and 0 to 3) and their firm-clustered covariance, and the target is τ_b, the mean of the four post-pledge coefficients.

Under the smoothness restriction the bias can bend, but its slope may change by at most M percentage points from one year to the next. With $M = 0$ the bias is a straight line through the pre-pledge coefficients, extended past the pledge. The interval is then

that of an estimate net of a linear pre-pledge trend, close to the trend-adjusted estimates in Panel B of Table 6. A positive M lets the slope drift. By event year e the bias can depart from the line that continues its last pre-pledge slope by up to $M(e+1)(e+2)/2$, that is M, $3M$, $6M$ and $10M$ at event years 0 to 3, or $5M$ on average. With $M = 0.01$, for instance, the average bias over event years 0 to 3 can sit 0.05 points off the line, about a quarter of the estimate at director and above. The relative-magnitudes restriction bounds every year-to-year change in the bias from event year -1 onward by $\bar{M}$ times the largest year-to-year change up to event year -1. With $\bar{M} = 1$, each year after the pledge can see a change as large as the largest pre-pledge one. For each restriction I use the confidence sets that Rambachan and Roth (2023) recommend: fixed-length intervals for smoothness and conditional least-favorable hybrid sets for relative magnitudes.

Under a linear trend the intervals exclude zero at director and above (0.05 to 0.43), at entry (0.08 to 0.55) and across the firm (0.10 to 0.42). They include zero in the middle band (-0.09 to 0.35), whose coefficient three years before the pledge dips to -0.18 (Table D.2). At director and above they still exclude zero with $M = 0.01$ and first include zero at $M = 0.02$; at entry and across the firm they still exclude zero with $M = 0.01$ and first include zero at $M = 0.02$. The pre-pledge coefficients cannot tell these values of M apart: at director and above their slope changes by 0.037 and then by -0.019 points from one pre-pledge year to the next, with standard errors of 0.085 and 0.062. Under relative magnitudes with $\bar{M} = 0.5$, the intervals exclude zero at entry (0.08 to 0.54) and across the firm (0.04 to 0.37) and include it at director and above (-0.01 to 0.40) and in the middle band. With $\bar{M} = 1$ every interval includes zero. The package also has a third restriction, which bounds each change in the slope of the bias from event year -1 onward by $\bar{M}$ times the largest such change up to event year -1. At $\bar{M} = 0.5$ its intervals include zero in every band.

D.3 Trend Adjustment

I report several estimates net of a linear pre-pledge trend. Those listed here all use the same line: Panel B and the slope rows of Table 6, column (4) of Table H.1, the columns beyond the trend in Online Appendix Table OA.15, and the added share net of its pre-trend in Appendix H.2. The pieces of the Asian share by education that Appendix E.3 reports net of their pre-pledge trend are the exception: I adjust them with an ordinary-least-squares line through event years -4 to -1. Let $\hat{\beta}_e$ be the event-time coefficients of an event study or a triple difference with reference year r. The reference year is -1, except for

TABLE D.2: CONFIDENCE SETS ALLOWING FOR VIOLATIONS OF PARALLEL TRENDS

	Director and above	Middle	Entry	Company
ATT (percentage points)	0.199	0.226	0.302	0.209
Conventional 95% interval	[0.058, 0.340]	[0.076, 0.376]	[0.146, 0.459]	[0.099, 0.320]
Panel A: Smoothness, slope of the bias changes by at most M points a year				
$M = 0$ (linear trend)	[0.050, 0.425]	[−0.093, 0.354]	[0.084, 0.554]	[0.099, 0.423]
$M = 0.005$	[0.036, 0.438]	[−0.101, 0.367]	[0.071, 0.567]	[0.071, 0.427]
$M = 0.01$	[0.005, 0.467]	[−0.122, 0.395]	[0.032, 0.591]	[0.017, 0.438]
$M = 0.02$	[−0.064, 0.532]	[−0.175, 0.463]	[−0.050, 0.655]	[−0.091, 0.451]
Panel B: Relative magnitudes, at most $\bar{M}$ times the largest pre-pledge change				
$\bar{M} = 0.5$	[−0.013, 0.396]	[−0.164, 0.590]	[0.085, 0.543]	[0.044, 0.374]
$\bar{M} = 1$	[−0.140, 0.520]	[−0.474, 0.894]	[−0.056, 0.687]	[−0.044, 0.464]

Notes: 95 percent confidence sets of Rambachan and Roth (2023) for the ATT of Table 4 (Black share, mean of event years 0 to 3, percentage points), from the seven event-time coefficients and their firm-clustered covariance. Panel A bounds the yearly change in the slope of the bias by M, and $M = 0$ extrapolates a linear pre-pledge trend (fixed-length intervals). Panel B bounds each yearly change in the bias from event year -1 on by $\bar{M}$ times the largest one up to year -1 (conditional least-favorable hybrid sets). Source: Revelio Labs.

the 24-month outcomes, where it is -3. I fit the line by generalized least squares through zero at r to the three coefficients before r, $\hat{\beta}_{\text{pre}} = (\hat{\beta}_{r-3}, \hat{\beta}_{r-2}, \hat{\beta}_{r-1})'$, with their estimated covariance S:

$$\hat{d} = g'\hat{\beta}_{\text{pre}}, \qquad g = \frac{S^{-1}x}{x'S^{-1}x}, \qquad x = (-3, -2, -1)', \qquad \text{SE}(\hat{d}) = \left(x'S^{-1}x\right)^{-1/2}. \tag{4}$$

Extrapolating the line to event years 0 to 3 and subtracting it gives the adjusted estimate

$$\hat{\tau}^{\text{adj}} = \frac{1}{4}\sum_{e=0}^{3}\left[\hat{\beta}_e - \hat{d}\,(e - r)\right] = \hat{\tau} - \bar{k}\,\hat{d}, \qquad \bar{k} = \frac{1}{4}\sum_{e=0}^{3}(e - r), \tag{5}$$

so $\bar{k} = 2.5$ when $r = -1$ and $\bar{k} = 4.5$ when $r = -3$. The adjusted estimate is a linear combination of the event-time coefficients, with weight $1/4$ on each post-pledge coefficient and $-\bar{k}\,g$ on the three leads. Its standard error comes from their firm-clustered covariance with g held fixed. In the 24-month triple differences, the partly exposed years -2 and -1 are neither leads nor post-pledge years, so the line rests on years -6 to -4. The linear pre-trend tests I report are the t-test of $\hat{d}$.

The Black share at director and above (Table 6) has a slope of -0.014 points a year (SE 0.025). The adjustment moves the estimate from 0.199 to 0.233 and raises its standard error by a third, from 0.072 to 0.096. Promotion within 24 months at entry (Online Appendix

Table OA.15) has a slope of 0.23 (SE 0.12), and subtracting 4.5 times it turns the estimate of 0.56 (SE 0.44) into −0.47 (SE 0.68). For the added share a of Online Appendix OA.C.4, I fit the line to the Poisson coefficients b_{-4}, b_{-3} and b_{-2} of Equation (OA.5), on the log scale, and adjust each post-pledge coefficient before the transformation: $a^{\text{adj}} = \frac{1}{4}\sum_{e=0}^{3}\{1 - \exp\left[-\left(b_e - \hat{d}\,(e+1)\right)\right]\}$. The slope of b_e is 0.020 a year ($p = 0.07$), and a falls from 0.068 to 0.020 (SE 0.043).

The timing of the Asian share (Appendix E.3) uses a different line. The question there is whether the share fell below its trend before the pledge, and a line through zero at event year −1 would rule that out by construction. I fit it by ordinary least squares to event years −4 to −2 alone, which for the 2021 cohort are July 2017 to July 2019: $\ell(e) = \bar{\beta}_{\text{early}} + s\,(e+3)$, with slope $s = (\hat{\beta}_{-2} - \hat{\beta}_{-4})/2$ and $\bar{\beta}_{\text{early}}$ the mean of the three coefficients. The deviation $\hat{\beta}_{-1} - \ell(-1)$ measures a shortfall before the pledge, and the stall in the pledge year is the further change

$$\left[\hat{\beta}_0 - \ell(0)\right] - \left[\hat{\beta}_{-1} - \ell(-1)\right] = \hat{\beta}_0 - \hat{\beta}_{-1} - s, \tag{6}$$

which also equals $\hat{\beta}_0$ minus the ordinary-least-squares line through event years −4 to −1 (with $\hat{\beta}_{-1} = 0$), evaluated at year 0. At director and above the stall is −0.22 (SE 0.09). Across the firm, the 2021 cohort's own coefficients give a deviation of −0.17 (SE 0.08) at event year −1 and a further change of −0.20 (SE 0.09) to year 0. The standard errors come from the clustered covariance of the coefficients.

D.4 Alternative Estimators

TABLE D.3: BLACK SHARE ATT BY ESTIMATOR

	Director and above	Middle	Entry	Company
Panel A: ATT, mean of event years 0 to 3 (percentage points)				
Sun-Abraham (main)	0.199***	0.226***	0.302***	0.209***
	(0.072)	(0.077)	(0.080)	(0.056)
Two-way fixed effects, dynamic	0.195***	0.232***	0.306***	0.214***
	(0.071)	(0.075)	(0.081)	(0.056)
Callaway-Sant'Anna, never-treated	0.199***	0.226***	0.302***	0.209***
	(0.070)	(0.075)	(0.079)	(0.055)
Callaway-Sant'Anna, not-yet-treated	0.199***	0.226***	0.303***	0.210***
	(0.070)	(0.075)	(0.079)	(0.055)
Borusyak-Jaravel-Spiess imputation	0.186**	0.313***	0.310***	0.229***
	(0.079)	(0.082)	(0.096)	(0.067)
Stacked regression	0.193***	0.231***	0.303***	0.212***
	(0.073)	(0.077)	(0.081)	(0.057)
Gardner two-stage	0.186**	0.313***	0.310***	0.229***
	(0.078)	(0.082)	(0.096)	(0.067)
Local-projection DiD	0.199***	0.225***	0.302***	0.209***
	(0.069)	(0.073)	(0.076)	(0.054)
Panel B: Joint test of the three leads (p-value)				
Sun-Abraham (main)	0.92	0.08	0.86	0.35
Two-way fixed effects, dynamic	0.90	0.09	0.84	0.29
Callaway-Sant'Anna, never-treated	0.90	0.06	0.85	0.33
Callaway-Sant'Anna, not-yet-treated	0.90	0.07	0.83	0.27
Borusyak-Jaravel-Spiess imputation	0.90	0.07	0.82	0.26
Stacked regression	0.93	0.10	0.87	0.41
Gardner two-stage	0.90	0.07	0.81	0.25
Local-projection DiD	0.90	0.06	0.82	0.28
Numerical race pledgers	115	115	115	115
Never-pledged controls	1,062	1,062	1,062	1,062

Notes: Black share (education-augmented BIFSG posterior mean), Sun-Abraham row as in the main table. Panel A: mean of event years 0 to 3 of each estimator's event-study coefficients, in percentage points, standard errors in parentheses clustered by firm (bootstrapped for Callaway-Sant'Anna and Borusyak-Jaravel-Spiess). Panel B: joint test of event years -4 to -2. Unweighted firm-band-year cells; the Gardner two-stage estimate drops one control in the middle band. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Source: Revelio Labs.

E Additional Workforce Results

E.1 Persistence through the Backlash

By their 2025 filings, 114 of the 115 pledgers had softened or withdrawn their race targets. Firms chose to retreat, and nearly all of them did, so the designs below compare firms that retreated at different times. They check a mechanism and give no causal estimates. Table E.1 aligns each pledger on the quarter of its first retreat filing. Panel A compares the quarters after it with those before, once against pledgers that had not yet retreated and once against the controls. The Black hire rate did not fall in any comparison. At director level the comparison with pledgers that had not yet retreated fails its lead test, so for that band the result rests on the comparison with the controls (+0.015 per 100 positions a quarter, SE 0.012). The Black share of hires rose in two of the four comparisons, but the rise disappears when the data end in the first quarter of 2025. Panel B shows the Black share of the workforce at the first July 1 after the retreat. No band falls, and the firm-wide interval excludes falls larger than about two fifths of the pledge effect. The data end in mid-2025, so half of the retreating firms are observed for only one or two quarters after their retreat.

E.2 Flow Decomposition

The decomposition of Section 5.4 follows each band's Black share from one July 1 snapshot to the next and assigns every change to four flows: hires from outside the firm, promotions into the band from a lower band of the same firm, separations, and promotions out of the band. A separation is dated to the first snapshot at which the worker is gone from the firm. Moves down to a lower band are left out, and a clear title promotion counts as a promotion (Section 3.3). For each pledge cohort, I compare the pledgers' yearly flow contributions after the pledge with those before it, relative to the same difference at the controls and with the timing of the event study, and then stack the cohorts. The pooled version sums workers across firms, so large firms weigh more. The firm-average version computes the contributions firm by firm and averages them, so each firm counts once, as in the event study. The intervals come from a bootstrap over firms, and Table E.2 reports both versions.

Table 7 splits the hiring channel at director and above by the hire's previous employer. Each row is the origin's contribution to the yearly change in the Black share, pledgers

TABLE E.1: BLACK HIRING AND THE BLACK SHARE AFTER THE FIRST RETREAT

	Director and above		Middle	
	Black hire rate	Black share of hires (pp)	Black hire rate	Black share of hires (pp)
Panel A: Quarters 0 to 4 minus quarters −4 to −1 around the first retreat				
Against not-yet-retreated pledgers	−0.009	0.469	0.022	1.604**
	(0.023)	(0.859)	(0.034)	(0.666)
Joint lead test, p	<0.001	0.025	0.250	0.891
Against never-pledged controls	0.015	1.169**	0.023	0.556
	(0.012)	(0.503)	(0.018)	(0.370)
Joint lead test, p	0.164	0.114	0.683	0.137
Panel B: Black share of the workforce at the first July 1 after the retreat (pp)				
	Director and above	Middle	Entry	Company
Against not-yet-retreated pledgers	0.082	−0.025	−0.045	−0.007
	(0.059)	(0.063)	(0.047)	(0.037)
Joint lead test, p	0.017	0.784	0.458	0.513

Notes: First retreat: the first filing that softens or drops the race target. Panel A: hires dated by start month, against pledgers not yet retreated (Callaway-Sant'Anna) or the controls in the same quarter; Black hire rate per 100 positions a quarter. Data end in 2025 Q2, so for 53 of the 106 retreating firms the post window is one or two quarters. At director level the comparisons with pledgers not yet retreated fail their lead tests. Panel B: event year 0, July 2024 or July 2025. Bootstrap standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

relative to controls, so the rows add up to the hiring channel of Table E.2. The Black hire rate rose significantly from only two small origins: education, nonprofits and health (at the 5 percent level), and a residual group of moves from the firm's own units abroad, other index firms and unidentified employers. From the other origins, the contributions come from a lower volume of hiring at an unchanged or higher Black share. Never-pledged index firms account for 5 percent of the increase, which is the basis of the spillover bound in Section 5.4.

E.3 The Asian Share by Education

Table E.3 splits the Asian share by where each worker was educated, using the schools on her profile. The four pieces are schools only in the United States, schools both in the United States and abroad, schools only abroad, and no school record or none with a country. The pieces add up to the Asian share, and their estimates to its estimate. Asians with some education abroad carry most of the rise at the pledgers: outside the entry band they account for 58 to 72 percent of it. Most of those educated in both places seem to have come as students. Where both start years are recorded, 91 to 93 percent have the foreign

TABLE E.2: FLOW DECOMPOSITION OF THE CHANGE IN THE BLACK SHARE: POOLED AND FIRM-AVERAGE VERSIONS

	Director and above		Middle	
	Pooled	Firm average	Pooled	Firm average
Panel A: Contribution to the yearly change in the Black share (percentage points)				
External hiring	0.079	0.076	0.079	0.093
	[0.031, 0.122]	[0.028, 0.128]	[0.026, 0.131]	[0.026, 0.165]
Promotion into the band	0.021	0.011	0.027	0.002
	[−0.003, 0.044]	[−0.016, 0.039]	[0.007, 0.046]	[−0.032, 0.034]
Separations	−0.005	−0.005	−0.037	−0.031
	[−0.030, 0.023]	[−0.040, 0.032]	[−0.070, 0.008]	[−0.071, 0.009]
Promotion out of the band	0	0	0.001	−0.001
			[−0.015, 0.015]	[−0.019, 0.018]
Net: sum of the four flows	0.095	0.083	0.071	0.063
	[0.034, 0.161]	[0.015, 0.155]	[0.004, 0.141]	[−0.019, 0.148]
Panel B: Share of the net (percent)				
External hiring	83	92	112	146
Promotion into the band	22	14	38	4
Separations	−5	−6	−52	−49
Promotion out of the band	0	0	1	−1

Notes: Contribution of each flow to the yearly change in the band's Black share: change after the pledge (event years −1 to 3) minus change before (−4 to −1), pledgers minus controls, by cohort, with cohorts weighted by their number of pledgers. Pooled sums flows and stocks over firms; firm average weights firms equally. Net sums the four flows; moves down are left out, and nobody is promoted out of the top band. Brackets: 95 percent firm-cluster bootstrap intervals (1,000 draws). 115 numerical race pledgers, 1,063 never-pledged controls; LinkedIn panel.

record first. Across the firm, every piece was already rising before the pledge, the piece educated in both places clearly and the others slowly. Net of its pre-pledge trend, no piece rises detectably (the total: 0.03, SE 0.10; Table 6, Panel B).

At director and above the Asian share stalls in the pledge year and ends 0.22 percentage points (SE 0.09) below its pre-pledge line. Almost all of the stall comes from Asians educated only abroad (−0.10, SE 0.05) and from those educated both abroad and in the United States (−0.09, SE 0.06); Asians educated only in the United States account for almost none of it (−0.02, SE 0.06). With sector-by-year fixed effects the stall is −0.18 (SE 0.10). It coincides with the travel bans and the reduced visa processing of 2020 and 2021, which made hiring from abroad harder and would hit Asians educated abroad most directly. The data cannot tell the two explanations apart. Across the firm, the 2021 cohort was already 0.17 points (SE 0.08) below its 2017 to 2019 trend in July 2020, before any of its firms pledged, and it fell a further 0.20 points (SE 0.09) by July 2021, when its pledges and the restrictions overlapped. Splitting the pledgers by the share of their 2014 to 2017 workers who were educated only abroad does not help either, because that share tracks

the Asian share closely across pledgers (correlation 0.89).

TABLE E.3: THE ASIAN SHARE BY PLACE OF EDUCATION

	Share in 2019 (percent)	Estimate (percentage points)			
		Director and above	Middle	Entry	Company
All Asian	14.0	0.245***	0.420***	0.178	0.309***
		(0.094)	(0.139)	(0.114)	(0.085)
U.S. education only	6.6	0.078	0.124*	0.060	0.081*
		(0.072)	(0.071)	(0.074)	(0.043)
U.S. and foreign education	3.2	0.097*	0.242***	0.059	0.150***
		(0.054)	(0.064)	(0.066)	(0.045)
Foreign education only	2.3	0.046	0.060	−0.014	0.046
		(0.044)	(0.058)	(0.034)	(0.030)
No record or no country	1.9	0.024	−0.006	0.073***	0.032*
		(0.024)	(0.033)	(0.027)	(0.019)
Joint lead p, all Asian		0.008	0.004	0.043	0.001

Notes: Estimates of τ_b (Equation (2)), firm-clustered standard errors in parentheses; 115 numerical race pledgers against 1,062 never-pledged controls. Each piece averages the Asian education-augmented BIFSG probability times an indicator for the countries of the worker's school records (2025 profiles): United States only, United States and abroad, abroad only, or none known. The pieces sum to the Asian share. The share column is the pledgers' mean company-wide share in 2019; the last row tests the first row's leads jointly. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Source: Revelio Labs.

E.4 Headcount Table

This table reports the estimates behind Figure 6 of Section 5.

E.5 Checks against the Firms' EEO-1 Reports

The headcount result of Section 5.3 rests on LinkedIn, so it depends on who keeps a profile. The firms' EEO-1 reports give a second count. I use the reports of federal contractors for 2016 to 2020 (Appendix C.4) and, from 2021 on, the reports that firms published themselves, leaving out the few whose composition jumps implausibly from one year to the next. Each EEO-1 year is matched to the July 1 snapshot of the same year. Of the main sample, 98 pledgers and 612 controls have at least one report, and 60 of the pledgers have reports from both before and after their pledge.

An EEO-1 report counts the employees of the legal entities that file it, so a firm's official count jumps when it buys or sells a business. LinkedIn assigns past jobs to today's parent

TABLE E.4: HEADCOUNT BY BAND AFTER A NUMERICAL RACE PLEDGE

	Director and above	Middle	Entry	Company
Headcount, 100 × log (about percent)	−2.49*	−4.43***	−6.09***	−4.77***
	(1.40)	(1.43)	(1.28)	(1.30)
Headcount without the 2024 snapshot	−2.31	−4.69**	−6.18***	−4.46**
	(1.77)	(1.86)	(1.68)	(1.77)
Band's share of the firm's workforce (pp)	0.38***	0.09	−0.47***	
	(0.12)	(0.13)	(0.18)	
Joint lead p, headcount	0.67	0.25	0.45	0.06

Notes: Estimator: interaction-weighted event study of Sun and Abraham (2021), Equation (1), with firm and year fixed effects, event years −4 to 3, reference year −1. Estimates of τ_b (Equation (2)), the mean of the coefficients for event years 0 to 3, with firm-clustered standard errors in parentheses; 115 numerical race pledgers against 1,063 never-pledged controls. Headcount is 100 times the log of the band's LinkedIn workers on 1 July, or of the whole firm's in the Company column. The band's share is in percentage points (pp). The last row tests jointly that the three pre-pledge coefficients of the first row are zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Source: LinkedIn profiles via Revelio Labs (November 2025 release).

company and has no such jumps. I therefore cut a firm's series wherever the log of its official count moves by more than 0.3 in a year while the log of its LinkedIn count moves by less than 0.1, and at known mergers, and give each stretch its own fixed effect. Table E.5 applies the estimator of Figure 6 to the log of the official count. With the series cut, relative headcount in the four years after the pledge is 1.3 log points lower than in the year before, with a 95 percent interval from −8.0 to 5.7. Keeping every firm-year gives −2.1, with an upper end of 4.8. The reports cannot confirm the decline that LinkedIn shows, which is small on the same firm-years (−0.6) and once the pre-pledge decline is removed (−1.6). They rule out a relative expansion of more than about 6 percent.

TABLE E.5: RELATIVE HEADCOUNT IN THE EEO-1 REPORTS AND ON LINKEDIN

			95% upper end			
	Estimate	SE	Normal	Bootstrap	Pledgers	Controls
EEO-1, cut at mergers	−1.26	3.86	6.31	5.70	98	612
EEO-1, all firm-years	−2.06	3.86	5.50	4.75	98	612
LinkedIn, same firm-years	−0.60	2.64	4.58		98	611
LinkedIn, all firms	−4.77	1.31	−2.22		115	1,062
LinkedIn, net of pre-trend	−1.63	2.18	2.64		115	1,062

Notes: Estimates of Equation (1) for 100 times the log of U.S. headcount, averaged over event years 0 to 3, with standard errors clustered by firm. EEO-1 counts are the federal contractor reports to 2020 and the firms' own published reports after. In the first row each stretch of a firm's series between mergers or divestitures has its own fixed effect. Upper ends use the normal approximation or a wild cluster bootstrap with 9,999 draws. The last row removes a linear trend fitted to event years −4 to −1.

F Economy-Wide Data and New Seats

F.1 Refills of the Movers' Seats

A move is an outside hire into director and above at a pledger, in event years 0 to 3, of a worker labeled Black whose last employee position before the entry was at director level or above at another U.S. employer, the origin. A seat counts as refilled when a different person starts at the origin within 12 months in the same role, one of Revelio's 150 role groups, and in the same band; the first such start is the refill. The benchmark is the origin's Black share among its director-level starts in the 24 months before the departure. Of the 9,348 moves, 77.9 percent were refilled, 43.0 percent by an internal promotion and 34.8 percent by an outside hire, and 7.4 percent by a Black worker (95 percent interval 6.8 to 8.0, bootstrapped over origins). Among refilled seats the Black share was 9.5 percent, against a benchmark of 10.2 percent. The Black refill rate is 7.4 percent for Black departures to pledgers before their pledges, 7.9 percent for Black departures to the controls and 6.0 percent for white-male departures to pledgers, and in each group the Black share among refilled seats is close to the origin's benchmark. With race scored by the posterior the rate is 9.7 percent, with ties between same-day starts broken at random 7.8 percent, and counting a start in any role 9.2 percent. A refill is the origin's next hire into the role, not a person-for-person replacement. The count of new seats in Table 11 is $(1 - L) + L\,r$ times the added seats, where L is the share of the added entries that came from directors who moved (45 in 100) and r the Black refill rate; it ignores later steps of the chain, in which refilled seats open further vacancies. With the posterior refill rate instead of the label rate, 708 of the 1,200 added seats are new. Origins are observed for 91.9 percent of the Black outside hires at pledgers after their pledges; the rest came from abroad (3.8 percent), had no earlier job (3.1 percent) or came from an unresolved employer (0.6 percent).

F.2 Origin Firms and Rivals

For the 38,444 U.S. employers with at least 20 director-level workers in 2019, the pledgers excluded, exposure is the share of their director-level leavers in 2016 to 2019 who moved to a pledger (mean 2.4 percent, standard deviation 5.0 points). In an event study with employer fixed effects and year effects by size and industry, one standard deviation of exposure changes the Black director share by 0.006 points on average over 2020 to 2024 (95 percent interval -0.022 to 0.033), with flat leads ($p = 0.57$). Employers that lost a Black director to a pledger in 2020 to 2022 changed by -0.04 points (-0.13 to 0.07) relative to

matched employers from 2019 to 2023. Their Black director share was already rising faster before 2020, and net of that pre-trend the change is -0.27 (-0.42 to -0.10), about a fifth of the mechanical loss of 1.5 points (28 percent at the end of the interval). The pledgers' rivals show no consistent change. Non-pledging employers in 4-digit industries where the pledgers held more director jobs in 2019 changed by -0.028 points (standard error 0.016) per standard deviation of exposure, and those in metropolitan areas where the pledgers were larger by $+0.054$ (0.023). Only 59 industries and 39 metropolitan areas contain a pledger, and permutation p-values are 0.17 and 0.21, so neither result is reliable.

G Robustness of the Workforce Estimates

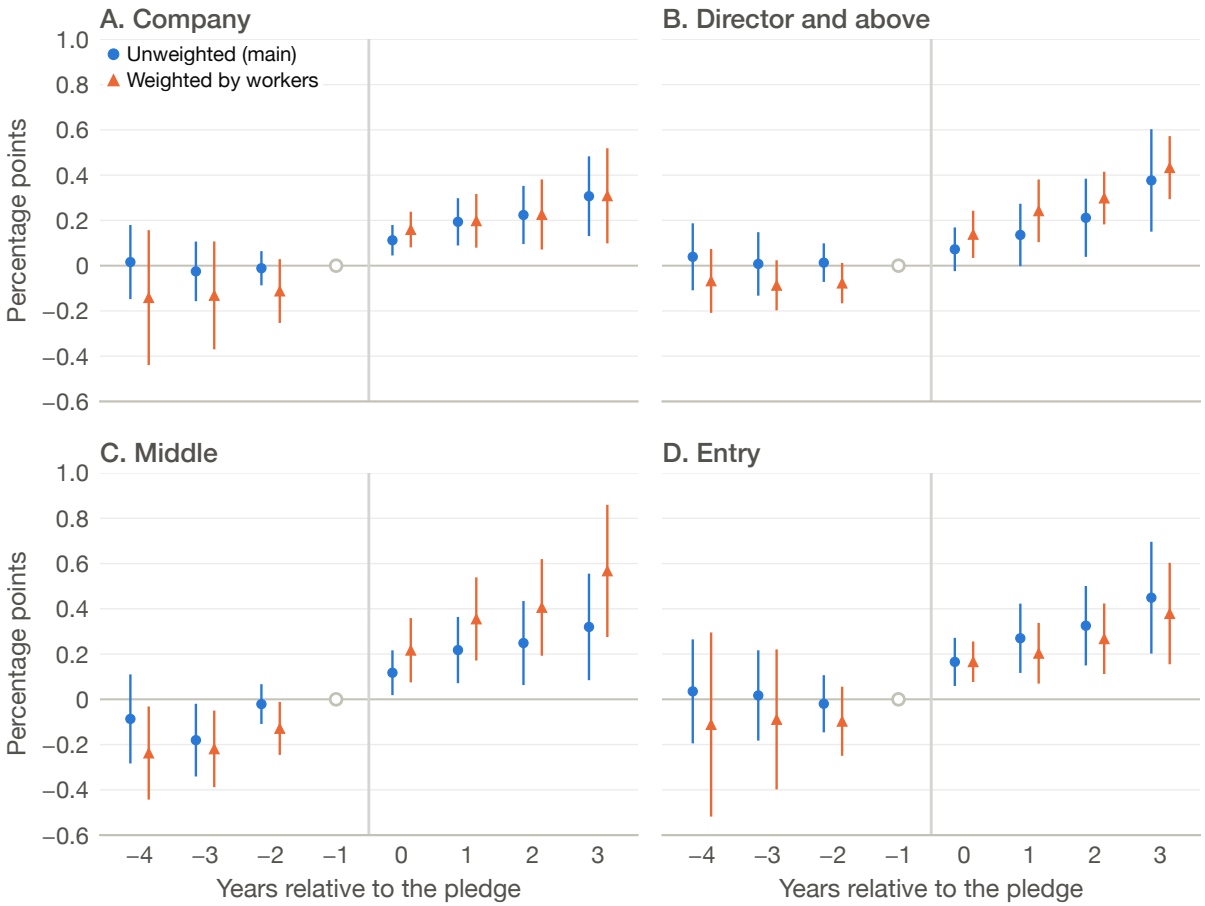
In Table G.1 and Figure G.1 I weight each firm-band-year cell by its number of workers, which moves the estimand toward the effect on the average worker at a pledging firm. The weighted estimates are larger at director level and above (0.278 against 0.199 percentage points) and in the middle band (0.386 against 0.226), smaller at entry (0.254 against 0.302) and close at the company level (0.223 against 0.209). Weighting also moves the leads, which fall below zero in every band: the joint lead test gives $p = 0.05$ in the middle band and $p = 0.06$ at the company level, and when cohorts are also weighted by worker-years, or firms carry fixed pre-2020 weights, it also rejects at the 5 percent level at director level and above ($p = 0.03$). Amazon accounts for part of the gap at director level and above and in the middle band: without it the weighted estimates are 0.249 and 0.351, and the middle-band lead test gives $p = 0.16$. I keep the firm average as the main estimand because each pledge sets a target for that firm's own workforce, so the firm is the unit whose commitment I evaluate.

TABLE G.1: BLACK SHARE ATT WITH CELLS WEIGHTED BY WORKERS

	Director and above	Middle	Entry	Company
Panel A: ATT, mean of event years 0 to 3 (percentage points)				
Unweighted (main)	0.199***	0.226***	0.302***	0.209***
	(0.072)	(0.077)	(0.080)	(0.056)
Weighted by workers	0.278***	0.386***	0.254***	0.223***
	(0.053)	(0.098)	(0.068)	(0.064)
Weighted, without Amazon	0.249***	0.351***	0.230***	0.251***
	(0.054)	(0.120)	(0.076)	(0.071)
Weighted, cohorts by worker-years	0.359***	0.381***	0.254***	0.245***
	(0.062)	(0.089)	(0.073)	(0.068)
Fixed pre-2020 firm weights	0.309***	0.320***	0.199***	0.208***
	(0.062)	(0.094)	(0.076)	(0.069)
Panel B: Joint test of the three leads (p-value)				
Unweighted (main)	0.92	0.08	0.86	0.35
Weighted by workers	0.24	0.05	0.25	0.06
Weighted, without Amazon	0.53	0.16	0.34	0.22
Weighted, cohorts by worker-years	0.03	0.01	0.31	0.09
Fixed pre-2020 firm weights	0.03	0.01	0.52	0.09
Numerical race pledgers	115	115	115	115
Never-pledged controls	1,062	1,062	1,062	1,062

Notes: Black share (education-augmented BIFSG posterior mean), Sun-Abraham, mean of event years 0 to 3 in percentage points, standard errors clustered by firm. The main row counts each firm-band-year cell once. Weighted rows weight cells by workers and cohorts by firms, or by treated worker-years where stated. Fixed weights use each firm's mean pre-2020 cell size for cells and cohorts, so they cannot respond to the pledge (one middle-band control drops). Without Amazon, 114 pledgers. Panel B: event years -4 to -2. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Source: Revelio Labs.

FIGURE G.1: EVENT-STUDY ESTIMATES FOR THE BLACK SHARE WITH CELLS WEIGHTED BY WORKERS



Notes: Sun-Abraham event-study coefficients for the Black share of each band (education-augmented BIFSG posterior mean), in percentage points, with 95 percent confidence intervals clustered by firm. Unweighted firm-band-year cells (main) and the same cells weighted by their workers, offset within each year; the weighted estimates combine cohorts by their number of firms. Year −1 (open marker) is the reference. 115 numerical race pledgers and 1,062 never-pledged controls; cells with at least five workers. Source: Revelio Labs.

H Hire-Level Results

H.1 Alternative Designs

Table H.1 sets the main design of Section 6 against four alternatives. The main design compares Black and white-male hires within the same firm and hire year, and dates each firm's hires relative to its own pledge. The alternatives are the same comparison pooled across firms; a common cutoff in 2020 for every pledger, pooled across firms; the main design net of a linear pre-pledge trend; and, in the pooled version, the general pledgers in place of the never-pledged controls (the within-firm version gives the same picture at director and above). At director and above the designs agree on the outcomes shown, although the human-resources title of Section 6.1 falls only in the aligned designs. Only two gaps move. The gap in the rank of the previous employer narrows in every design, by 0.9 to 2.2 percentile points, and in every design the DEI-title share of Black hires falls relative to white men's, by 0.21 to 0.45 points. Below the top the designs disagree. The pooled designs find entry-level differences in pay, promotion, retention and placement in staff functions; these shrink when hires are compared within firms, and for promotion and retention they disappear once the pre-pledge trend is taken out.

H.2 Bounds on the Added Hires

Online Appendix OA.C.4 derives the bounds. The added share a of the external Black hires into director and above, averaged over event years 0 to 3, is 0.068 (SE 0.024). Table H.3 reports θ, the ratio θ/a with its 95 percent Fieller set, and the within-firm instrumental-variables version with Anderson-Rubin sets (first-stage F of 9.4 for pay).

The sets are wide. Compared with the other Black hires, the added hires entered positions predicted to pay between \$45,400 less and \$63,300 more (Fieller), or between \$36,200 less and \$51,000 more (instrumental variables). For the rank of the previous employer both sets exclude zero, at 6.6 to 65.6 and 5.7 to 52.5 percentile points. Relative to the white men hired alongside them, the added hires therefore came from employers at least as well ranked (a measured among hires from other employers is 0.085, SE 0.027). In the lower tail of pay, below the pre-2020 10th percentile of the firm's white-male hires, the share of added hires is at most about 21 points higher than among the other Black hires. Table H.2 adds retention and promotion within 36 months for hires made by June 2022, and the share of Black hires in the bottom fifth of the firm's pre-2020 hires by pay and

TABLE H.1: BLACK HIRES UNDER ALTERNATIVE DESIGNS

	(1) Aligned, within firm (all starts)	(2) Aligned, pooled	(3) Calendar, pooled	(4) Beyond a linear pre-trend	(5) General pledgers as controls
Panel A: Director and above (levels 5 to 7)					
Top-100 university (pp)	−0.33	−0.53	−0.20	−0.88	−0.45
	(0.61)	(0.64)	(0.58)	(1.22)	(0.64)
Years since bachelor's degree	0.04	−0.05	−0.21	0.25	−0.18
	(0.18)	(0.17)	(0.15)	(0.32)	(0.18)
Previous employer's rank (percentile, movers only)	1.82***	1.51**	0.87*	2.21**	1.17*
	(0.56)	(0.65)	(0.51)	(1.02)	(0.62)
Predicted salary (dollars)	−845	−1,102	−491	−602	−1,764
	(1,092)	(1,217)	(1,162)	(1,891)	(1,259)
Still at the firm after 24 months (pp)	−0.30	−0.07	−0.38	0.40	0.01
	(0.73)	(0.74)	(0.52)	(2.58)	(0.71)
Promoted within 24 months (pp)	−0.28	−0.18	−0.09	0.00	−0.13
	(0.31)	(0.33)	(0.24)	(0.99)	(0.34)
DEI title (pp)	−0.35***	−0.31**	−0.45***	−0.37*	−0.21*
	(0.13)	(0.13)	(0.10)	(0.21)	(0.12)
Panel B: Middle (levels 3 and 4)					
Promoted within 24 months (pp)	0.44	0.93*	0.81**	0.08	0.43
	(0.53)	(0.50)	(0.40)	(1.21)	(0.53)
Panel C: Entry (levels 1 and 2)					
Promoted within 24 months (pp)	0.56	1.45***	1.44***	−0.47	1.06*
	(0.44)	(0.55)	(0.41)	(0.68)	(0.59)
Still at the firm after 24 months (pp)	0.65	1.65*	2.32***	0.10	1.11
	(0.46)	(0.86)	(0.81)	(1.17)	(0.91)
Staff rather than line function (pp)	0.59**	1.34***	1.68**	0.47	1.22**
	(0.29)	(0.44)	(0.67)	(0.47)	(0.54)
Predicted salary (dollars)	277	1,461*	3,419***	134	1,613**
	(361)	(746)	(1,219)	(572)	(774)

Notes: Black minus white-male hires (U.S. position starts, 2016 to 2024), 115 numerical race pledgers minus 1,063 never-pledged controls; firm-clustered standard errors. (1) The design of Table 10 on all position starts, moves inside the firm included (Table 10 keeps outside hires): within destination firm and July 1 hire year, aligned on the pledge: mean of event years 0 to 3 against year −1 (24-month outcomes: hires to 2022, year −3). (2) Pooled across firms. (3) Pooled, hires from 2020 on against earlier ones. (4) Column (1) net of its linear pre-trend. (5) Column (2) against the 193 general pledgers. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Source: Revelio Labs.

previous employer. Section 6.2 discusses them.

The ratio depends on a, and a is fragile. The Black share of the pledgers' external hires was already drifting up before the pledges, by 0.020 a year ($p = 0.07$). Net of a linear pre-trend (Appendix D.3), a is 0.020 (SE 0.043), and at that value the set for pay, like most of the sets, covers the whole line. The interval for θ itself does not need a. It bounds how much the pledges moved the average: −$2,260 to $3,040 for predicted pay.

Mislabeling adds a shift, which I compute on all position starts (Online Appendix OA.C.4). If the pledges raised true Black hiring, the label became purer by about 4 points. That moves θ for pay by $470 to $1,010 toward a lowered bar, between a fifth and nearly half of its 95 percent half-width.

TABLE H.2: RETENTION, PROMOTION AND THE LOWER TAIL OF EXTERNAL BLACK HIRES

	θ	95% interval	Lead p-value	Post years	Firms, last year	Black hires
A. Within 24 months, hires to 2022 (reference year −3)						
Still at the firm	−0.36	[−2.32, 1.59]	0.22	0 to 3	19	21,225
	(1.00)					
Promoted	−0.66	[−1.47, 0.15]	0.22	0 to 3	3	17,073
	(0.41)					
B. Within 24 months, hires to June 2022 (reference year −4)						
Still at the firm	0.00	[−2.29, 2.29]	0.11	0 to 2	16	16,557
	(1.17)					
Promoted	−0.16	[−0.87, 0.54]	0.11	0 to 2	16	15,156
	(0.36)					
C. Within 36 months, hires to June 2022 (reference year −4)						
Still at the firm	0.18	[−2.42, 2.79]	0.05	0 to 2	16	16,557
	(1.33)					
Promoted	0.08	[−0.70, 0.86]	0.25	0 to 2	16	16,557
	(0.40)					
Promoted, July 1 levels, to 2021	−0.11	[−1.37, 1.15]	0.10	0 to 1	16	7,572
	(0.64)					
D. At hire, bottom quintile of the firm's 2016 to 2019 hires (reference year −1)						
Predicted salary	−0.54	[−1.99, 0.92]	0.79	0 to 3	108	26,558
	(0.74)					
Previous employer's rank	0.04	[−1.71, 1.78]	0.06	0 to 3	102	19,472
	(0.89)					

Notes: External hires at director and above, 115 race pledgers and 1,063 controls. Percentage points. θ: mean of the post-pledge event-year coefficients on the Black minus white-male gap, within firm and July 1 hire year; firm-clustered standard errors. Reference years leave out hires whose follow-up overlaps the pledge. Lead p-value: joint test of the earlier years. Firms, last year: pledgers behind the final post year; Black hires: at pledgers, post years. Panel D: below the 20th percentile of the firm's 2016 to 2019 external hires (previous employer: movers). † lead test rejects at 5 percent. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Source: Revelio Labs.

TABLE H.3: BOUNDS ON THE ADDED BLACK HIRES

	Triple difference θ (SE)	Per added hire θ/a [95% set]	Within-firm IV [95% set]
Predicted salary (dollars)	391	5,775	4,933
	(1,352)	[−45,377, 63,272]	[−36,225, 51,033]
Top-100 university (pp)	−0.88	−13.0	−11.1
	(1.09)	[−50.6, 30.6]	[−41.1, 24.4]
Previous employer's rank	1.79***	21.0	17.9
(percentile, movers only)	(0.57)	[6.6, 65.6]	[5.7, 52.5]
Still at the firm after 24 months (pp)	−0.36	−4.5	−2.8
	(1.00)	[−34.5, 25.6]	[−20.3, 13.6]
Promoted within 24 months (pp)	−0.66	−5.9	−3.4
	(0.41)	[−15.4, 1.4]	[−9.2, 0.7]
Lower tails (pp)			
Pay below the firm's pre-2020	−0.10	−1.4	−1.3
white-male 10th percentile	(0.62)	[−27.5, 20.9]	[−22.7, 17.5]
Previous employer below the firm's	0.02	0.2	0.2
pre-2020 white-male 25th percentile	(0.96)	[−32.6, 21.5]	[−29.7, 19.0]

Notes: Black minus white-male gaps among external hires at director and above, 2016 to 2024, within firm and July 1 hire year; 115 numerical race pledgers against 1,063 never-pledged controls, stacked by pledge cohort. θ averages event years 0 to 3 against year -1 (-3 for 24-month outcomes); firm-clustered standard errors. θ/a divides θ by the added share of post-pledge Black hires, $a = 0.068$ (0.024), estimated on each row's sample; Fieller sets. The IV divides θ by the pledges' effect on the log Black share of hires; Anderson-Rubin sets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Source: Revelio Labs.